

DDS-Digital Design Specification

STBIO1

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Table of Contents

Table of Contents.....	1
Table of Tables	3
Table of Figures	3
1. Abstract.....	5
2. Version History	6
3. Reference documents and Ip Repository.....	6
4. Conventions	8
4.1. Abbreviations.....	8
5. STBIO1 toplevel overview	10
6. Symbol.....	14
7. I/O List.....	14
8. Digital Architecture Overview.....	16
8.1. TOP Pinout List	16
9. Power Domains	27
9.1. Introduction.....	27
9.2. Definitions.....	28
9.3. Power Domain Architecture	28
9.4. Power Domain Descriptions.....	29
9.4.1. PD_TOP1V2	29
9.4.2. PD_STREDL.....	29
9.4.3. PD_TOP3V3	29
9.4.4. PD_TOPIO.....	29
10. Interconnect	29
10.1. XBAR Connection Matrix	30
10.2. XBAR Memory Mapping	31
11. STBIO1 Operating Modes	32
12. Modules.....	33
12.1.1. PMU.....	33
12.1.2. PAD Mux.....	45
12.1.3. Main Controller TOP	68
12.2. IPs.....	81
12.2.1. ADSP integration	81
12.2.2. I2C_SPI_AHB integration	93

12.2.3.	SensorHub integration	99
12.2.4.	FIFO_CTRL integration.....	107
12.3.	ISPU integration	111
12.3.1.	Requirements	125
12.4.	STBIO1 Register Map.....	128
12.4.1.	STBIO1 Register Map.....	128
12.4.2.	STBIO1 Register Map Peculiar Requirements.....	130
13.	Soft Reset Procedures	139
13.1.	Main Controller Soft Reset.....	139
14.	Measurement Cycle Examples	139
14.1.	BOOT Phase	140
14.2.	Configuration Phase	141
14.3.	Measurement Cycle.....	142
14.3.1.	ECG Only	143
14.3.2.	ECG and BIA	145
14.3.3.	PPG Only.....	147
15.	SCAN	151
15.1.	Top Level Approach.....	151
16.	BIST	152
16.1.	Reference documents.....	152
16.2.	Memory BIST Architecture.....	152
16.2.1.	MBIST Collar	152
16.2.2.	MBIST Controller	152
16.2.3.	Top Level Approach	153
17.	Digital DFT	154
17.1.	Scan mode	154
17.2.	Debug mode.....	154
17.3.	ADC TEST LOW NOISE.....	155
17.4.	ADC TEST FAST.....	156
17.5.	BIST MODE.....	156

Table of Tables

Table 1: Revision History	6
Table 2: Revision History	7
Table 3: Git Repository Table.....	7
Table 4: Abbreviations Table.....	9
Table 5: I/O List.....	16
Table 6: Digital I/O List.....	27
Table 7: XBAR Connection Matrix.....	30
Table 8: Memory Map	31
Table 9: Operating modes Table.....	32
Table 10: PMU I/O List.....	38
Table 11: Clock Recap Table	42
Table 12: STBIO1 Functional reset Summary table	44
Table 13: PAD MUX I/O List.....	53
Table 14: PAD Configuration Table.....	54
Table 15: Main Controller Parameter	70
Table 16: Main Controller I/O List.....	80
Table 17: ADSP Parameter Table	81
Table 18: ADSP I/O List	82
Table 19: OTP I/O List	83
Table 20 Chosen OTP Timings	90
Table 21 Operation phases timings.....	90
Table 22: I2C/SPI AHB Parameters	94
Table 23: I2C/SPI AHB I/O List	96
Table 24: Sensor Hub Parameter.....	99
Table 25: Sensor Hub I/O List.....	100
Table 26: Fifo Parameter	107
Table 27: FIFO Controller I/O List	107
Table 28: SUB FIFO Memory Address	109
Table 29: Data Tag	110
Table 30: ISPU I/O List	124
Table 31: Address Register Map.....	128
Table 32: Location AFE_BLOCK Registers on the I2C/SPI pages.	129
Table 33: AFE_BLOCK register list.....	135
Table 34: I2C/SPI internal registers.....	138

Table of Figures

Figure 1: Time Slot Definition	11
Figure 2: STBIO1 Device Scheme	12
Figure 3: STBIO1 Reference Application.....	13
Figure 4: STBIO1 - Ball out. TOP View (Ball side down)	14
Figure 5: STBIO1 Digital Engine	16
Figure 6: STBIO1 Digital Scheme	30
Figure 7: Device Start-Up Phase.....	38
Figure 8: Mask Cell for LDO and Fast Ring Oscillator	39

Figure 9: Clocks Scheme	41
Figure 10: STBIO1 Reset Scheme	43
Figure 11: Main Controller Block Diagram	68
Figure 12: Boot Routine	86
Figure 13: Write Routine Start.....	87
Figure 14: WRITE routine and TEST regmap.....	88
Figure 15 : OTP Reading timing diagram	91
Figure 16: OTP Writing timing diagram	93
Figure 17: ISPU turn on phases	125
Figure 18: Timing turn on ISPU.....	126
Figure 19: Device Activity Flowchart	139
Figure 20: Boot Phase Flow	141
Figure 21: Configuration Phase Diagram	142
Figure 22: ECG ONLY FLOWCHART	145
Figure 23: ECG and BIA Measurement Flowchart.....	146
Figure 24: PPG Measurement Flowchart	149
Figure 25 Top Level Controller Configuration.....	153
Figure 26 scan mode activation timing	154
Figure 26 debug mode activation timing.....	155
Figure 27 bist mode activation timing	157

1. Abstract

This document is the specification of the STBIO1 Project.

This document describes the functions, performance characteristics and required verification tests for the STBIO1 Project.

All requirements are considered Reviewed and Accepted once the document is officially approved.

2. Version History

Current Version	Previous Version	Date	Author	Change description
0.1	N.A		Marco Leo	Starting point from VSM01 project
STBIO1_v0.3	0.2	6/17/2025	Marco Leo	Added page numbers, update timing slot definition
STBIO1_v0.4	0.3	8/25/2025	Marco Leo	Fix PMU Peculiar requirements. Added Soft Reset Procedure
STBIO1_v2	1.6.2	1/9/2025	Ugo Garozzo	Add new OTP routines description, Add Soft Reset Procedures
STBIO1_v3	2	3/9/2025	Ugo Garozzo	Add new output on TOP o_BDC_PPG, added Power Domain, Chanced ADSP IP ADSP_v0.3.21.OTP640
STBIO1_v4	3	22/9/2025	Ugo Garozzo, Marco Leo, Aldo Torino, Alessandro Iannuzzi	1. Added reset value TOP; 2. Added Power DomainDescription; 3. Added resets of regmap; 4. Added new A2D input; 5. POR1V2_VDDIO; 6. new ADSP IP 7. new i2c/spi IP 8. Updated Top level Digital Scheme
STBIO1_v5	4	10/10/2025	Aldo Torino, Ugo Garozzo	Added Approvers Data, Fix Tables Size, Fix Clocks and Reset Schemes, removed scan reports section
STBIO1_v6	5	5/11/2025	Ugo Garozzo	Added DCCK, updated DDS_STBIO1_4000 new i2c/spi IP tag documentation, add pad_mux requirements
STBIO1_v7_DOC	6	24/11/2025	Aldo Torino	Added Requirement 00001199 for sub-fifo Offset
STBIO1_v8_DOC	7	25/11/2025	Ugo Garozzo	Updated

Table 1: Revision History

3. Reference documents and Ip Repository

This is a list of reference documents.
[DDS_STBIO1_0002] Definition

Current Version	Document	Author	Change Description
1.0	Main Controller IPOS (stbio1/FE/main_controller_top/Documentation/Specification/)	Aldo Torino, Sonia Andronaco, Giuliana De Milato	
3.0	IPOS I2C SensorHub Quokka uP Documentation Specification FIFO_CTRL_dos	Marco Leo Roberta Priolo	
3.6	Documentation Specification IP Objective Specification ADSP IPOS	Stefano Barbieri Alessandro Alaimo, Luca Gandolfi	
	H9A MEM_OTP_PUMP_85AL05_2 3A User Manual 2.0- 02 https://developer.arm.com/documentation/ih0033/latest/	ARM	
0.2	STRED-LP2 Architecture Reference Manual https://codex.cro.st.com/plugins/document/stbio1/folder/933984	ASMS	

Table 2: Revision History

IP	Git address	Git branch	Git tag
ADSP	git clone ssh://gitolite@codex.cro.st.com/ip-development/ADSP.git	adsp_vecma	ADSP_v0.3.22.OTP640
i2c_spi_slave	git clone ssh://gitolite@codex.cro.st.com/ip-development/i2c_spi_slave.git	vecma	I2C_SPI_AHB_v3.1_061025_DOC
senshub_mst_i2c	ssh://gitolite@codex.cro.st.com/ip-development/senshub_mst_i2c.git	ahb_platf_ip	STBIO1_RTL100_v4.1

Table 3: Git Repository Table

IP external	Description	Author
ISPU	STRED-LP2-SVN_VERSION 9739	SRA
BIST BLOCK	GenericMBIST	STMicroelectronics TRnD
ADSP ROM	H9A_M4_ST_ROM	STMicroelectronics TRnD
ADSP RAM	ST_SPSMALL_512x64m2_bM6_C Umr	STMicroelectronics TRnD
FIFO RAM	ST_SPSMALL_512x64m2_bM6_C Umr	STMicroelectronics TRnD
ISPU CODE	ST_SPUHD_2048x128m4_bM6_C Umr	STMicroelectronics TRnD
ISPU DATA	ST_SPUHD_2048x32m4_bM6_C Umr	STMicroelectronics TRnD
H9A_FU_OT PHV_85AL05_23A	H9A_FU_OTPHV_85AL05_23A_MIM5_6ML_ALUCU revision 2.6.b 640-bit	
XBAR	CoreLink SIE-200 System IP for Embedded - Global Bundle r3p2-00rel0	

Table 1:external IPs

4. Conventions

[DDS_STBIO1_0003] Definition

If not otherwise noted, specification values are valid ...

4.1. Abbreviations

[DDS_STBIO1_00304] Comment

The following abbreviations are used throughout this document:

[DDS_STBIO1_0006] Definition

A	
AGC	Automatic Gain Control
ASIL	Automotive Safety Integrity Level

ASIC	Application Specific Intergrated Circuit
B	
BIST	Built In Self Test
...	
C	
CRM	Clock and Reset Management
F	
FMEA	Failure Mode and Effect Analysis
FMEDA	Failure Mode, Effect and Diagnostic Analysis
FSM	Functional Safety Management
FTA	Fault Tree Analysis
CTRL	Controller
PPG	Photoplethysmography
EDA	Electro-Dermal Activity
GSR	Galvanic Skin Response
PMU	Power Management Unit

Table 4: Abbreviations Table

[DDS_STBIO1_0007] Definition

Category Convention:

Comment:	This category denotes that the content of the object text is a general comment. For example, this may be an explanation why the requirement demands certain items when there would also be other possibilities. A comment should not be necessary to understand the related requirements.
Definition:	This category is used for the definition of terms, wordings, technical expressions etc. It is a documentation of design decisions and general instructions that belong to the document itself, e.g. owner of the document, structure of the document. It is needed to understand a related requirement. It is not linked to any test case.
Assumption	This category is used for requirements this document provide upstream to another document, indicating what this IP/block/system needs to work properly.
Requirement:	This category denotes a requirement that has to be implemented and verified. Accordingly, it is necessary to establish traces from the different requirement levels to the test cases for this category.

5. STBIO1 toplevel overview

Aim of this document is to explain main features of the ASIC STBIO1, electronic interface of ECG, Bio-impedance, EDA and PPG (Photoplethysmography) embedded in a single die.

STBIO1 is an Analog Front End Device with embedded process capabilities.

STBIO1 collects and processes different types of Vital Signals:

- ECG
- PPG
- BIA and GSR

The optical channel (PPG) is designed as an optical transceiver, stimulating up to eight LEDs and measuring the return signal on up to six separate cent inputs. The signal chain rejects signal offsets and corruption from asynchronous modulated interference, typically from ambient light, eliminating the need for optical filters or externally controlled dc cancellation circuitry.

The ECG signal acquisition is designed to support low noise, diagnostic level measurement in the presence of a variety of interferers. The ECG signal chain has several complementary features supporting ECG measurement, such as driven reference for common-mode rejection and lead off detection to identify a fallen electrode.

The BIA signal chain is designed for body impedance and human breathing measurements with a configurable excitation path and measurement path. BIA channel delivers both the real and the imaginary parts of the body and the breathing impedance.

Finally, the last Electro-Dermal Activity (EDA), also called Galvanic Skin Response (GSR) refers to changes in sweat gland activity that are reflective of the intensity of our emotional state, otherwise known as emotional arousal.

STBIO1 is optimized ultra-low-power hardware circuitry for real-time execution of the algorithms is a state-of-the-art feature for any personal electronics, from wearable accessories to high-end applications.

The STBIO1 features an advanced digital engine, composed of multiple IPs able to reduce external host processing workload, reducing bus congestion, providing meta-data useful for final application, both in time and frequency domain.

The main IPs actors in digital domain are:

- **ADSP**
- **Sensor-Hub**
- **ISPU** 32-bit processor
- **Smart FIFO** data collector

ADSP is an embedded IP able to elaborate ultra-low-power custom DSP functions. The main purpose of ADSP is to reduce the computation workload on both **ISPU** and host processor, while saving power through the optimization of both hardware and algorithm.

Sensor hub IP can read external sensors by means of an I2C master protocol, enabling data collection in **FIFO** and embedded elaboration also on external data domain. The data collected by sensor hub is accessible from the Advanced DSP and ISPU, for sensor fusion algorithm elaboration.

The advanced ISPU (32bit proprietary core) is an ultra-low-power, high-performance programmable core which can execute signal processing and AI algorithms in the edge. The

main benefits of the ISPU are C programming and an enhanced ecosystem with libraries and 3rd party tools/IDE. ISPU has 32 Kbyte program RAM, 24 Kbyte Data RAM capability and Floating-point support.

FIFO digital IPs oversees data collecting from multiple sources: internal and external data read by sensor hub, meta-data generated from both ADSP and ISPU.

The internal and external data that can be collected in FIFO are:

- ECG
- BIA
- GSR & 6 channels PPG
- up to 4 external sensors (expressed on 12 bytes each).

All the data are stored inside a **Smart FIFO** that can be configured as:

- **Unique FIFO**: Classic FIFO, the First Data that is written inside the Memory is also the First that is read.
- **Multiple Fifo**: The FIFO is split up to 6 Sub-FIFO (one for each Channels plus one for Sensor Hub and Elaboration results). The depth of each Sub-FIFO is configurable by regmap. Each Sub-FIFO behavior as a Classic FIFO.

[DDS_STBIO1_0101] Definition:

To understand how STBIO1 works, is necessary to define some basic concepts:

- **TIME_SLOT**= represents the interval within measurements are performed. It can be configured by the regmap field TIME_SLOT_SELECTION and its range is from 4096Hz to 64 Hz.
- In each TIME_SLOT we can distinguish two phases:
 - Operative: where all user configured measures are executed and ADSP and/or ISPU eventually runs its algorithms.
 - Sleep: where measures are finished, eventually only ADSP and/or ISPU is terminating its algorithms and writing elaborated data in FIFO.

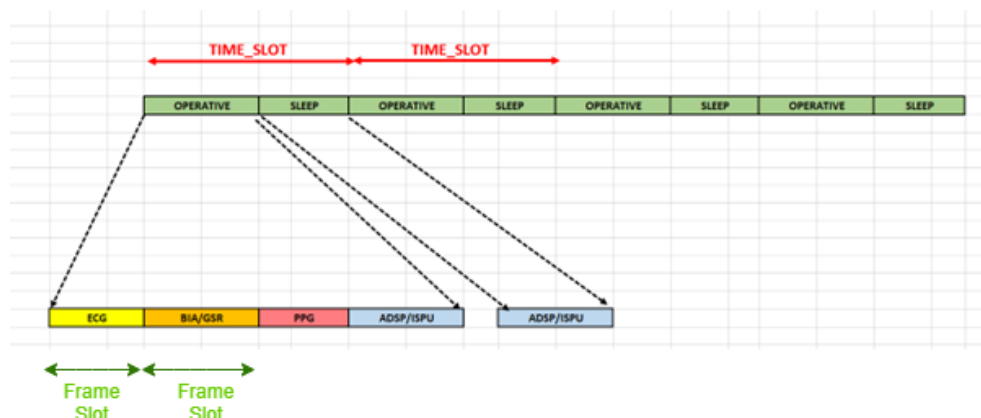


Figure 1: Time Slot Definition

If all the selected operations do not end in a single Time Slot, an error shall be raised.

The Time Slot is composed of different **Frame Slot** representing the time-interval of a single operation (for example, ECG).

We also need to define the other 4 parameters: **Mecg**, **Mbia**, **Mgsr** and **Mppg**, also called **Division Index**.

These parameters are used to define the **Data Rate Interval** of a single Frame Slot.

$$Rate\ Interval = \frac{Time\ Slot}{M_x}$$

With x = ecg,ppg,bia and gsr.

Depending on the value of the Mx parameters, a Time_Slot can be composed by one or more Frame Slot or none. In the last case, the device will remain in SLEEP_OPERATIVE (as described in the next paragraphs). [END]

STBIO1 die size is 3700µm x 4000µm including scribe line of 80µm. Technology used is HCMOS9A (130nm)

PDK_STM_HCMOS9A 11.8.b-03

PDK_STM_PLUS 04.04-02

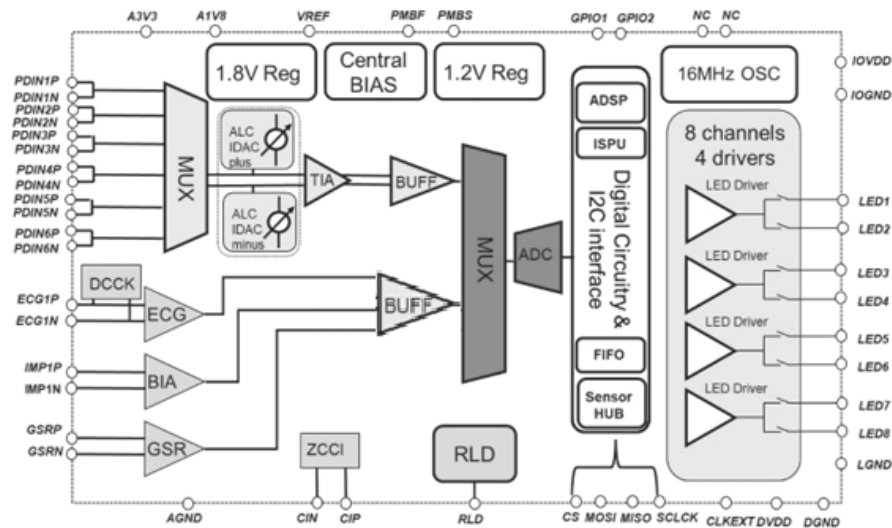


Figure 2: STBIO1 Device Scheme

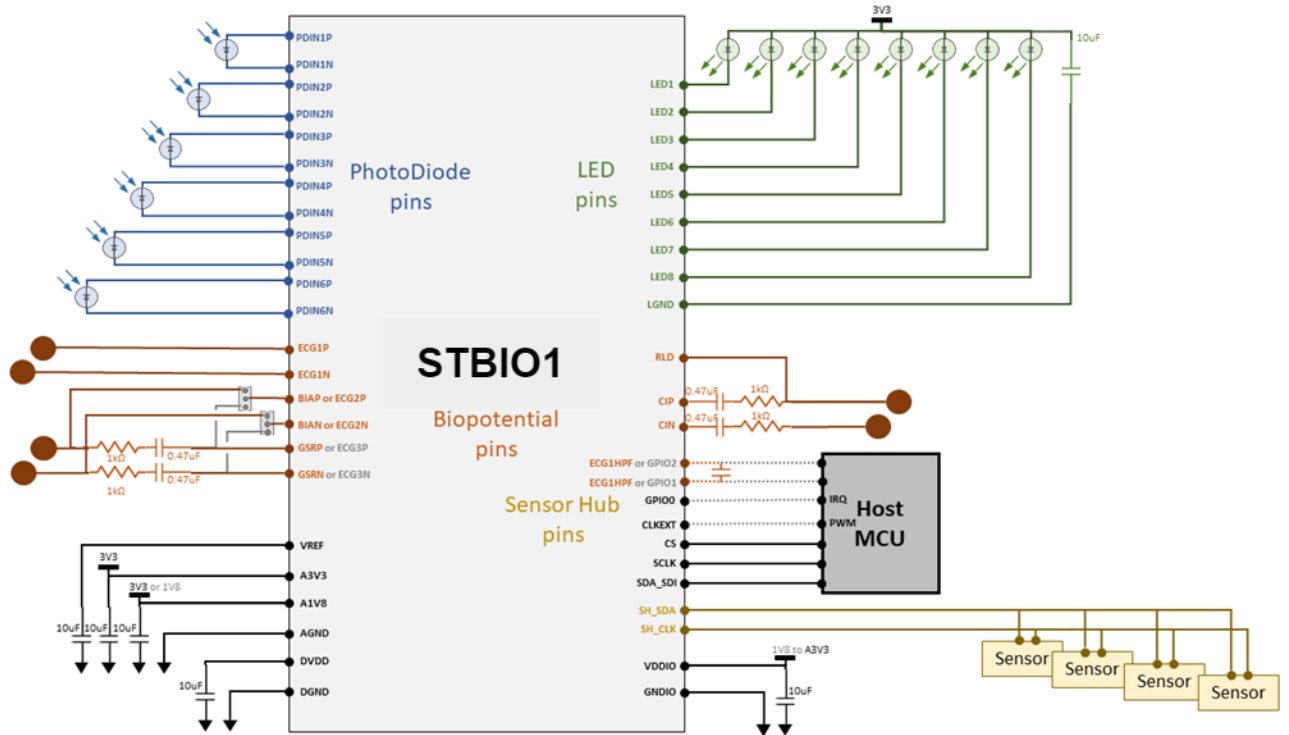


Figure 3: STBIO1 Reference Application

6. Symbol

[DDS_STBIO1_0008] Definition

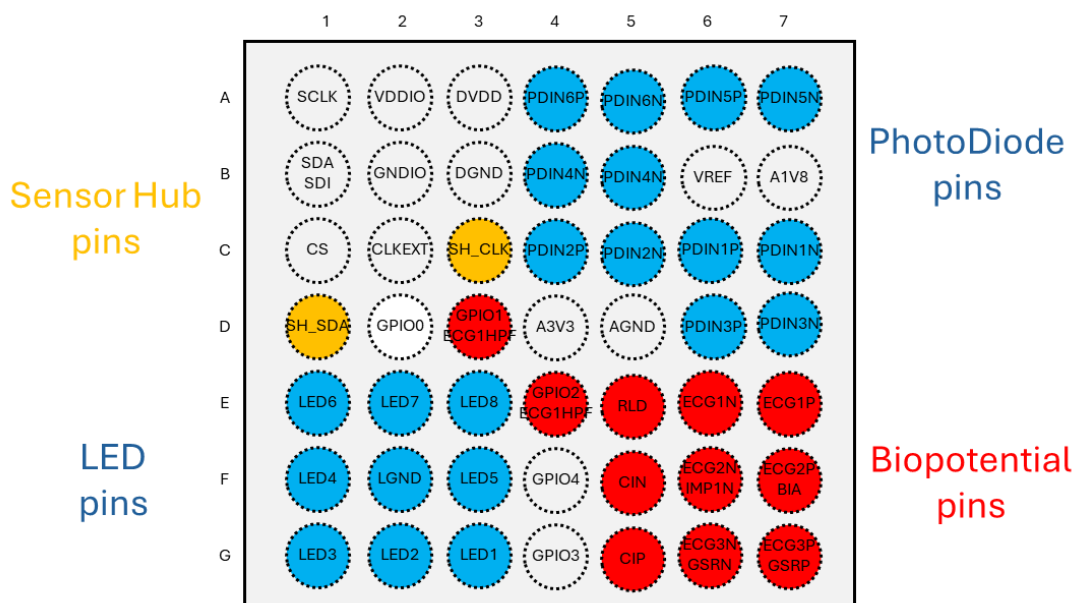


Figure 4: STBIO1 - Ball out. TOP View (Ball side down)

7. I/O List

[DDS_STBIO1_0009] Definition

Nr.	Name	Function	Pin No.	Type
1	AV1V8	1.8 V Analog Power Supply	D5	Power
2	AV3V3	3.3 V Analog Power Supply	D4	Power
3	AGND	Analog Ground	D6	Power
4	DVDD	1.8 V Digital Power Supply	A3	Power
5	GNDD	Digital Ground	B3	Power
6	VDDIO	I/O Power Supply.	A2	Power
7	GNDIO	I/O Ground	B2	Power
8	LGND	LED Ground	F2	Power
9	VREF	Voltage Reference	D7	Analog
10	PDIN5N	Photodiode 5 N	A7	Analog
11	PDIN1P	Photodiode 5 P	A6	Analog
12	PDIN6N	Photodiode 6 N	A5	Analog

Nr.	Name	Function	Pin No.	Type
13	PDIN6P	Photodiode 6 P	A4	Analog
14	PDIN3N	Photodiode 3 N	B7	Analog
15	PDIN3P	Photodiode 3 P	B6	Analog
16	PDIN4N	Photodiode 4 N	B5	Analog
17	PDIN4P	Photodiode 4 P	B4	Analog
18	PDIN2P	Photodiode 2 P	C4	Analog
19	PDIN2N	Photodiode 2 N	C5	Analog
20	PDIN1P	Photodiode 1 P	C6	Analog
21	PDIN1N	Photodiode 1 N	C7	Analog
22	LED1	LED driver 1	G3	Analog
23	LED2	LED driver 2	G2	Analog
24	LED3	LED driver 3	G1	Analog
25	LED4	LED driver 4	F1	Analog
26	LED5	LED driver 5	F3	Analog
27	LED6	LED driver 6	E1	Analog
28	LED7	LED driver 7	E2	Analog
29	LED8	LED driver 8	E3	Analog
30	ECGIP	ECG Positive Input	E7	Analog
31	ECGIN	ECG Negative Input	E6	Analog
32	RLD	Right LEG Driver	E5	Analog
33	IMPIP	BIA Positive Input	F7	Analog
34	IMPIN	BIA Negative Input	F6	Analog
35	GSRP	GSR Positive Input	G7	Analog
36	GSRN	GSR Negative Input	G6	Analog
37	CIP	BIA Injection Positive Output.	G5	Analog
38	CIN	BIA Injection Negative Output.	F	Analog
39	GPIO3	General-Purpose Input/Output 3	G4	Analog
40	GPIO4	General-Purpose Input/Output 4	F4	Analog
41	SCLK	SPI Clock INPUT	A1	Digital
42	CLKext	External Clock	C2	Digital
43	SDA_SDI	I2C / SPI serial data line	B1	Digital
44	SH_SDA	Sensor Hub Data	D1	Digital
45	CS	Chip Select	C1	Digital
46	SH_CK	Sensor Hub Clock	C3	
47	GPIO0	General-Purpose Input/Output 0	D2	Digital
48	GPIO1	General-Purpose Input/Output 1	D3	Digital
49	GPIO2	General-Purpose Input/Output 2	E4	Digital

Table 5: I/O List

8. Digital Architecture Overview

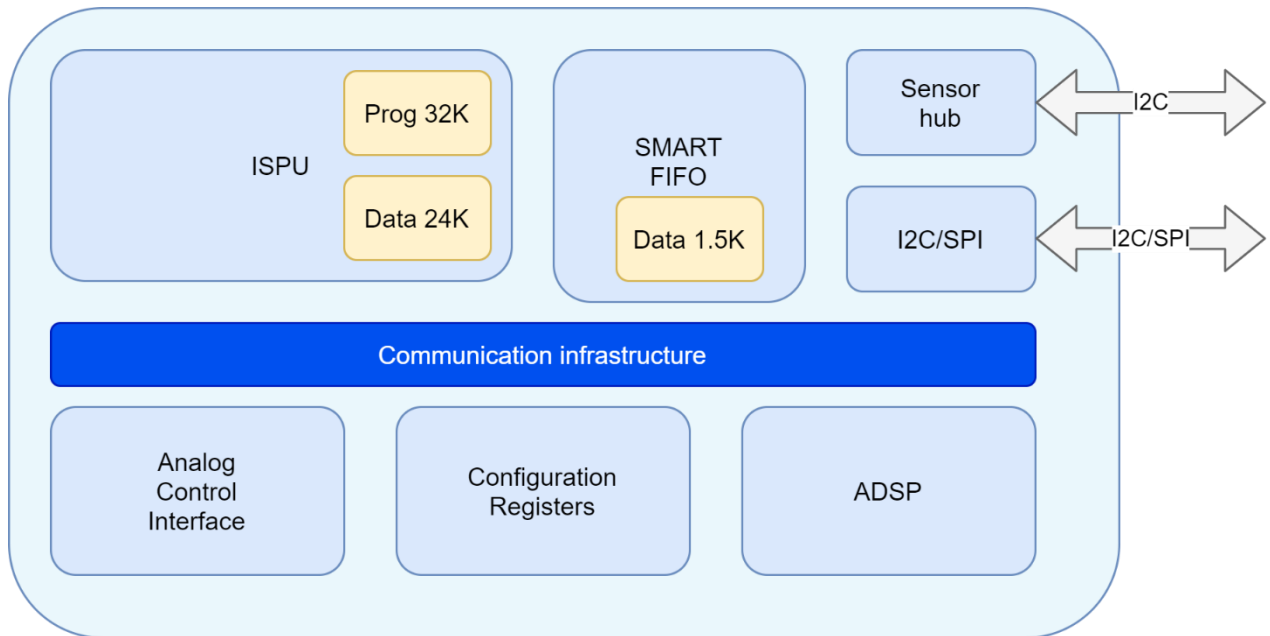


Figure 5: STBIO1 Digital Engine

8.1. TOP Pinout List

Port name	Direction	POR State of Signal	Type	Description
i_comp_rx_out	input	0	wire	to u_main_controller_top, Description: "Inserire LS"
o_BIDAC	output	0	wire [11:0]	from u_scan_out_mux
o_BDC_PPG	output	0	wire [7:0]	from u_scan_out_mux
o_BTIA_gain	output	0	wire [3:0]	from u_scan_out_mux
o_PD_PGA	output	1	wire	from u_scan_out_mux
o_PD_IDAC	output	1	wire	from u_scan_out_mux
o_PD_TIA	output	1	wire	from u_scan_out_mux
o_comp_rx_ck	output	0	wire	from u_scan_out_mux
BT_IDAC_N	output	0	wire [6:0]	from u_scan_out_mux

BT_IDAC_P	output	0	wire [9:0]	from u_scan_out_mux
o_b_cap_idac	output	0	wire [1:0]	from u_scan_out_mux, Description: IDAC Filter Cap
o_rx_ch_frame	output	0	wire [2:0]	from u_scan_out_mux, Description: BMUX_Selection
DIS_LOOP_TIA	output	0	wire	from u_scan_out_mux, Description: Disable TIA loop
T_RX_GPIO1	output	0	wire [1:0]	from u_scan_out_mux, Description: TEST RX signal on
T_RX_GPIO2	output	0	wire [1:0]	from u_scan_out_mux, Description: TEST RX signal on
TM_SEL_1V8	output	0	wire	from u_scan_out_mux, Description: Force PGA input
o_en_tx_1V2	output	0	wire [3:0]	from u_scan_out_mux
o_tx1_outlab_sel_frame_x_1V2	output	0	wire	from u_scan_out_mux
o_tx2_outlab_sel_frame_x_1V2	output	0	wire	from u_scan_out_mux
o_tx3_outlab_sel_frame_x_1V2	output	0	wire	from u_scan_out_mux
o_tx4_outlab_sel_frame_x_1V2	output	0	wire	from u_scan_out_mux
o_tx_preset_1V2	output	0	wire	from u_scan_out_mux
BtrimIREF_TX_1V2	output	0	wire [3:0]	from u_scan_out_mux, Description: TX Ramp Current
DFT_SEL_1V2	output	0	wire	from u_scan_out_mux, Description: VFEED_1V2 Monitor
o_curr_sel_tx_frame_x_1V2	output	0	wire [7:0]	from u_scan_out_mux, Description: Selezione corrente
o_Txi_RAMP_1V2	output	0	wire [3:0]	from u_scan_out_mux, Description: B_RAMP_1V2<3:0>

o_ck_chop_imp	output	0	wire	from u_scan_out_mux
o_ck_hc2_10k	output	0	wire	from u_scan_out_mux
o_ck_m	output	0	wire	from u_scan_out_mux, Description: inserito new bit
o_ck_p	output	0	wire	from u_scan_out_mux, Description: inserito new bit
o_ck_q	output	0	wire	from u_scan_out_mux, Description: inserito new bit
o_short_on	output	0	wire	from u_scan_out_mux, Description: (codex 1417523
o_res_hc2	output	0	wire	from u_scan_out_mux, Description: (codex 1417523
o_resl_on	output	0	wire	from u_scan_out_mux, Description: (codex 1417523
o_pd_hc2	output	1	wire	from u_scan_out_mux
o_pd_imp	output	1	wire	from u_scan_out_mux
o_rst_hc2	output	0	wire	from u_scan_out_mux
o_rst_imp	output	0	wire	from u_scan_out_mux
o_rsti_hc2	output	0	wire	from u_scan_out_mux
o_rsti_imp	output	0	wire	from u_scan_out_mux
o_su_hc2	output	0	wire	from u_scan_out_mux
o_sug_imp	output	0	wire	from u_scan_out_mux
i_ADC_EOC	input	0	wire	to u_main_controller_top/u_pad_mux
i_data_ADC_out	input	0	wire [15:0]	to u_main_controller_top/u_pad_mux
o_adc_mux_bio_channel	output	0	wire [2:0]	from u_scan_out_mux
o_adc_mux_sel	output	0	wire	from u_scan_out_mux
o_gsr_curr_on	output	0	wire	from u_scan_out_mux

o_gsr_on	output	0	wire	from u_scan_out_mux
o_ADC_en	output	0	wire	from u_scan_out_mux
o_ADC_start	output	0	wire	from u_scan_out_mux
o_adc_error_cap8_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap7_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap6_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap5_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap4_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap3_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap2_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap1_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_adc_error_cap0_1V2	output	0	wire [6:0]	from u_scan_out_mux
o_pd_cm_buf	output	1	wire	from u_scan_out_mux
o_pd_rld	output	1	wire	from u_scan_out_mux
o_pd_avg1	output	1	wire	from u_scan_out_mux
o_pd_avg2	output	1	wire	from u_scan_out_mux
POR1V2_1V2	input	0	wire	to u_pad_mux
clk_16m	input	0	wire	to u_pad_mux
clk_64k	input	0	wire	to u_pad_mux
START_CLK_16M	input	0	wire	to u_pad_mux
o_DIS_RC_VIREF	output	0	wire	from u_scan_out_mux
o_EN_LDO1V8	output	0	wire	from u_scan_out_mux
o_PD_CLKF	output	1	wire	from u_scan_out_mux
ADC_CAL_CORR_DIS	output	0	wire	from u_scan_out_mux

ADC_ERROR_CAP9_DEB UG	output	0	wire [7:0]	from u_scan_out_mux
ADC_SR_ENHANCE_DIS	output	0	wire	from u_scan_out_mux
CCK_THRESHN_SEL	output	0	wire [3:0]	from u_scan_out_mux
CCK_THRESHP_SEL	output	0	wire [3:0]	from u_scan_out_mux
CURR_BG_TRIM	output	8	wire [3:0]	from u_scan_out_mux
CURR_PTAT_TRIM	output	8	wire [3:0]	from u_scan_out_mux
DCCK_CURRINJ	output	0	wire [1:0]	from u_scan_out_mux
DCCK_ISINK_IN	output	0	wire [2:0]	from u_scan_out_mux
DCCK_ISOURCE_IN	output	0	wire [2:0]	from u_scan_out_mux
ECG_PSDRES_DOWN	output	0	wire [2:0]	from u_scan_out_mux
ECG_PSDRES_UP	output	0	wire [2:0]	from u_scan_out_mux
EN_DEBUG_ADC_ERROR _CAP9	output	0	wire	from u_scan_out_mux
EN_ECG_50m	output	0	wire	from u_scan_out_mux
EN_GRS_50m	output	0	wire	from u_scan_out_mux
EN_IMP_50m	output	0	wire	from u_scan_out_mux
EN_RLD_CM	output	0	wire	from u_scan_out_mux
EN_SW_IMP	output	1	wire	from u_scan_out_mux, Description: per selezionare
GAIN1_BUF	output	0	wire	from u_scan_out_mux
GSR_IN_TRIM	output	0	wire [7:0]	from u_scan_out_mux
GSR_IP_TRIM	output	0	wire [7:0]	from u_scan_out_mux
IMPP_PSDRES_DOWN	output	0	wire [2:0]	from u_scan_out_mux
IMPP_PSDRES_UP	output	0	wire [2:0]	from u_scan_out_mux
IMPQ_PSDRES_DOWN	output	0	wire [2:0]	from u_scan_out_mux

IMPQ_PSDRES_UP	output	0	wire [2:0]	from u_scan_out_mux
IMP_CURINJN_TRIM	output	0	wire [5:0]	from u_scan_out_mux
IMP_CURINJP_TRIM	output	0	wire [5:0]	from u_scan_out_mux
RLD_CAP_COMP	output	0	wire [3:0]	from u_scan_out_mux, Description: [1/2/3/.../15/16]*16pF
RLD_VREF	output	0	wire [1:0]	from u_scan_out_mux, Description: select 0.9V/ 0.6V
TRIM_LDOD_I	output	0	wire [3:0]	from u_scan_out_mux
TRIM_LDOD_V	output	0	wire [3:0]	from u_scan_out_mux
TRIM_OSC_FAST	output	18	wire [4:0]	from u_scan_out_mux
TRIM_OSC_SLOW	output	0	wire [3:0]	from u_scan_out_mux
TRIM_V1V8	output	7	wire [2:0]	from u_scan_out_mux
VREF_TRIM	output	10	wire [4:0]	from u_scan_out_mux
CAPF_IMP_EN	output	1	wire	from u_scan_out_mux, Description: Enable filtro per
ECG0_GAIN_HC_0	output	1	wire	from u_scan_out_mux
ECG0_GAIN_HC_1	output	1	wire	from u_scan_out_mux
EN_EXT_CAP	output	0	wire	from u_scan_out_mux, Description: Mantenuta opzione
EN_ECG0_0P05HP	output	0	wire	from u_scan_out_mux, Description: aggiunto bit per
EN_ECG12_0P05HP	output	0	wire	from u_scan_out_mux, Description: aggiunto bit per
GAIN_IMP	output	1	wire [1:0]	from u_scan_out_mux
GIMP_DC	output	1	wire	from u_scan_out_mux
o_gsr_switchon	output	0	wire	from u_scan_out_mux, Description: spostato il bit
GSR_GAIN_SEL	output	0	wire [1:0]	from u_scan_out_mux

GSR_GAIN_X1	output	0	wire	from u_scan_out_mux
IGSR_X10	output	0	wire	from u_scan_out_mux
IMP_CURRINJ_CUR	output	2	wire [1:0]	from u_scan_out_mux
IMP_CURRINJ_x10	output	0	wire	from u_scan_out_mux
SEL_GSR_CURR	output	3	wire [1:0]	from u_scan_out_mux
o_ck_bg	output	0	wire	from u_scan_out_mux
o_imp_curinj_pd	output	1	wire	from u_scan_out_mux
o_pd_p	output	1	wire	from u_scan_out_mux
o_pd_n	output	1	wire	from u_scan_out_mux
o_adc_error_cap9	output	0	wire [7:0]	from u_scan_out_mux
o_en_bufxbio	output	0	wire	from u_scan_out_mux
o_ldo_ext	output	0	wire	from u_scan_out_mux
o_clk_ext_div	output	0	wire	from u_scan_out_mux
o_adc_test_cap	output	0	wire [9:0]	from u_scan_out_mux
if_cs_en	output		wire	from u_level_shifter_sel.if_cs_en_ls_out
if_cs_in	input		wire	to u_pad_mux
if_cs_out	output		wire	from u_level_shifter_sel.if_cs_out_ls_out
if_cs_tud	output	1	wire	from u_level_shifter_sel.if_cs_tud_ls_out, Description:
if_gpio0_en	output		wire	from u_level_shifter_sel.if_gpio0_en_ls_out
if_gpio0_in	input		wire	to u_pad_mux/u_pmu.if_sh_ck_in
if_gpio0_out	output		wire	from u_level_shifter_sel.if_gpio0_out_ls_out
if_gpio0_tud	output	1	wire	from u_level_shifter_sel.if_gpio0_tud_ls_out

if_gpio1_en	output		wire	from u_level_shifter_sel.if_gpio1_en_ls_out
if_gpio1_in	input		wire	to u_pad_mux
if_gpio1_out	output		wire	from u_level_shifter_sel.if_gpio1_out_ls_out
if_gpio1_tud	output	1	wire	from u_level_shifter_sel.if_gpio1_tud_ls_out
if_gpio2_in	input		wire	to u_pad_mux
if_gpio2_en	output		wire	from u_level_shifter_sel.if_gpio2_en_ls_out
if_gpio2_out	output	1	wire	from u_level_shifter_sel.if_gpio2_out_ls_out
if_gpio2_tud	output	1	wire	from u_level_shifter_sel.if_gpio2_tud_ls_out
if_gpio3_en	output		wire	from u_level_shifter_sel.if_gpio3_en_ls_out
if_gpio3_in	input		wire	to u_pad_mux, Description: decidere se mantenere l'IN
if_gpio3_out	output	1	wire	from u_level_shifter_sel.if_gpio3_out_ls_out
if_gpio3_tud	output	1	wire	from u_level_shifter_sel.if_gpio3_tud_ls_out
if_gpio4_en	output		wire	from u_level_shifter_sel.if_gpio4_en_ls_out
if_gpio4_in	input		wire	to u_pad_mux, Description: decidere se mantenere l'IN
if_gpio4_out	output	1	wire	from u_level_shifter_sel.if_gpio4_out_ls_out
if_gpio4_tud	output	1	wire	from u_level_shifter_sel.if_gpio4_tud_ls_out
if_sclk_en	output		wire	from u_level_shifter_sel.if_sclk_en_ls_out

if_sclk_in	input		wire	to u_pad_mux
if_sclk_out	output	1	wire	from u_level_shifter_sel.if_sclk_out_ls_out
if_sclk_tud	output	1	wire	from u_level_shifter_sel.if_sclk_tud_ls_out
if_sda_sdi_en	output		wire	from u_level_shifter_sel.if_sda_sdi_en_ls_out
if_sda_sdi_in	input		wire	to u_pad_mux
if_sda_sdi_out	output	1	wire	from u_level_shifter_sel.if_sda_sdi_out_ls_out
if_sda_sdi_tud	output	1	wire	from u_level_shifter_sel.if_sda_sdi_tud_ls_out
if_sh_ck_en	output		wire	from u_level_shifter_sel.if_sh_ck_en_ls_out
if_sh_ck_in	input		wire	to u_pad_mux
if_sh_ck_out	output		wire	from u_level_shifter_sel.if_sh_ck_out_ls_out
if_sh_ck_tud	output	1	wire	from u_level_shifter_sel.if_sh_ck_tud_ls_out
if_sh_sda_en	output		wire	from u_level_shifter_sel.if_sh_sda_en_ls_out
if_sh_sda_in	input		wire	to u_pad_mux
if_sh_sda_out	output	1	wire	from u_level_shifter_sel.if_sh_sda_out_ls_out
if_sh_sda_tud	output	1	wire	from u_level_shifter_sel.if_sh_sda_tud_ls_out
o_trim_ldo_d	output	0	wire [1:0]	from u_level_shifter_sel.o_trim_ldo_d_ls_out
ADC_BUFF_REFRESH_D1 S	output	0	wire	from u_scan_out_mux

ADC_EXTRAPERIOD_DIS	output	0	wire	from u_scan_out_mux
ADC_INPUT_FROM_GPIO	output	0	wire	from u_scan_out_mux
ADC_REF_FROM_GPIO	output	0	wire	from u_scan_out_mux
AFE_SPARE_0_1V2	output	0	wire [7:0]	from u_scan_out_mux
AFE_SPARE_1	output	0	wire [7:0]	from u_scan_out_mux, Description: modificato l'utilizzo
DIS_LDOD_HP	output	0	wire	from u_scan_out_mux
DIS_LDOD_IP	output	0	wire	from u_scan_out_mux
EN_LDOD_FIFO	output	0	wire	from u_scan_out_mux
EN_LDOD_STREDL	output	0	wire	from u_scan_out_mux
OTP_SPARE	output	00007F00	wire [31:0]	from u_scan_out_mux, Description: i PMB_EXT sono collegati
OVERCURR_N_IN_i	input	0	wire	to u_regmap_otp
OVERCURR_P_IN_i	input	0	wire	to u_regmap_otp
O_DCCK_N_i	input	0	wire	to u_pad_mux/u_regmap_otp/u_regmap_otp.DCCK_N_IRQ_i/u_N_not_logic.A,
O_DCCK_P_i	input	0	wire	to u_pad_mux/u_regmap_otp/u_regmap_otp.DCCK_P_IRQ_i/u_P_not_logic.A,
t_en_gpio1	output	0	wire	from u_scan_out_mux, Description: spostato il bit
t_en_gpio2	output	0	wire	from u_scan_out_mux, Description: spostato il bit
t_en_gpio3	output	0	wire	from u_scan_out_mux
t_en_gpio4	output	0	wire	from u_scan_out_mux
t_ecg0_out_pre_buf	output	0	wire	from u_scan_out_mux

t_offset_gpio12	output	0	wire	from u_scan_out_mux
t_in_bio_buf34	output	0	wire	from u_scan_out_mux
t_ecg1_out_pre_bufp_3	output	0	wire	from u_scan_out_mux
t_ecg1_out_pre_bufn_4	output	0	wire	from u_scan_out_mux
t_ecg2_out_pre_bufn_3	output	0	wire	from u_scan_out_mux
t_ecg2_out_pre_bufp_4	output	0	wire	from u_scan_out_mux
T_CM_HLT_2	output	0	wire	from u_scan_out_mux
T_CM_RLD_1	output	0	wire	from u_scan_out_mux
T_ECG0_AC_1	output	0	wire	from u_scan_out_mux
T_ECG0_DC_2	output	0	wire	from u_scan_out_mux
T_EXT_HPF_ECG0	output	0	wire	from u_scan_out_mux
T_FORCE_1V2	output	0	wire	from u_pad_mux.o_t_force_1v2
T_IBG_1u_2	output	0	wire	from u_scan_out_mux
T_IMPP_AC_1	output	0	wire	from u_scan_out_mux
T_IMPP_DC_2	output	0	wire	from u_scan_out_mux
T_IMPQ_AC_2	output	0	wire	from u_scan_out_mux
T_IMPQ_DC_1	output	0	wire	from u_scan_out_mux
T_IN_BIO_BUF	output	0	wire	from u_scan_out_mux
T_IPTAT_1u_1	output	0	wire	from u_scan_out_mux
T_OP55_1	output	0	wire	from u_scan_out_mux
T_OP90_2	output	0	wire	from u_scan_out_mux
T_OUT_BIO_BUF	output	0	wire	from u_scan_out_mux
T_OUT_GSR	output	0	wire	from u_scan_out_mux
T_OUT_PPG_BUF	output	0	wire	from u_scan_out_mux

T_SENSE_1V2	output	0	wire	from u_scan_out_mux
T_SENSEIBG_1V2	output	0	wire	from u_scan_out_mux
T_CM_HLT_1	output	0	wire	from u_scan_out_mux
T_SENSEVBG_1V2	output	0	wire	from u_scan_out_mux
o_end_calibration_1V2	output	0	wire	from u_scan_out_mux
pd_osc_clk32k	output	0	wire	from u_scan_out_mux, Description: se si seleziona
o_pd_buf_gpio1	output	1	wire	from u_scan_out_mux
o_pd_buf_gpio2	output	1	wire	from u_scan_out_mux
EN_CLK_EXT	output	0	wire	from u_scan_out_mux
WHO_AM_I	input	D7	wire [7:0]	to u_regmap_otp.DEV_ID_i, Description: ? OK ?
if_sclk_i3c_in	input		wire	to u_pad_mux
if_sda_sdi_i3c_in	input		wire	to u_pad_mux
VREF	input		wire	to u_otp
IREF	input		wire	to u_otp
OTP_IREF_WAKEUP	output	0	wire	from u_scan_out_mux, Description: Enable Mirror OTP

Table 6: Digital I/O List

9. Power Domains

9.1. Introduction

This document paragraph describes the Power Domain architecture implemented in the STBIO1. The goal is to provide a detailed overview of the power domain partitioning, their functionalities, control modes, and interactions between the various functional blocks.

9.2. Definitions

- **Power Domain (PD):** A set of logic blocks sharing the same power supply, which can be powered on/off independently.
- **Always-On Domain (AON):** Domain that is always powered, even in low-power modes.
- **Retention Domain:** Domain that retains register state even when the main logic is powered down.
- **Switchable Domain:** Domain that can be completely powered off to save energy.

9.3. Power Domain Architecture

Domain Name	Included Block(s)	Domain Type	Voltage [V]	Control Mode	Notes
PD_TOP1V2	ADSP, ca_fifo_bist_c, ca_qk_bist_c, ca_qk_rom_bist, multi_channel_fifo, i2c_spi_slave, main_controller_top, senshub_mst_i2c, pad_mux, pmu, stbio1_regmap, scan_out_mux, ta_stbio1_bist_ctrl, xbar_afe	Always-On	1.2	HW	Always active
PD_STREDL	STREDL	Switchable	1.2	SW	Can be powered down in idle
PD_TOP3V3	H9A_MEM_OTP_PUMP_85AL05_2 3A	Always-On	2.7÷4.7V	HW	Always active
PD_TOPIO	level_shifter_sel	Always-On	1.8V÷3.6V	HW	Always active

9.4. Power Domain Descriptions

9.4.1. PD_TOP1V2

- **Function:** Ensures continuity of critical functions (Data Storage Mode, Sensor-Hub acquisition, FIFO read and write data, Elaboration Mode with ADSP only).
- **Characteristics:** Never powered down.
- **Control:** Managed by hardware.

9.4.2. PD_STREDL

- **Function:** Powers the STREDL core and associated cache.
- **Characteristics:** Can be powered down when STREDL is not in use.
- **Control:** Managed via software (power management driver)(see [12.3.1](#))

9.4.3. PD_TOP3V3

- **Function:** power supply to the OTP memory block.
- **Characteristics:** Never powered down. It provides 4.7V of power supply to the OTP memory block during programming (write) operations
- **Control:** Managed by hardware.

9.4.4. PD_TOPIO

- **Function:** Powers communication PAD
- **Characteristics:** Never powered down.
- **Control:** Managed by hardware.

10. Interconnect

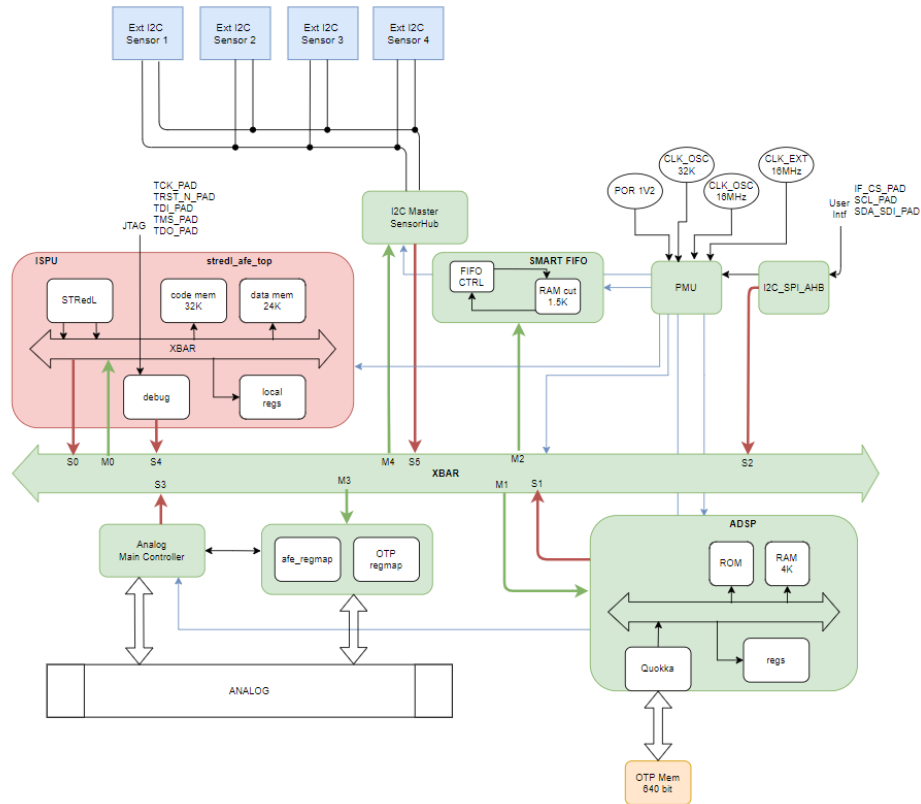


Figure 6: STBIO1 Digital Scheme

10.1. XBAR Connection Matrix

Destination Source	Regmap	SENSOR HUB	ISPU	FIFO	ADSP
Main Controller	[DDS_STBIO1_0201]			[DDS_STBIO1_0202]	[DDS_STBIO1_0219]
ADSP	[DDS_STBIO1_0203]	[DDS_STBIO1_0204]		[DDS_STBIO1_0206]	
I2C_SPI_AHB	[DDS_STBIO1_0207]	[DDS_STBIO1_0208]	[DDS_STBIO1_0209]	[DDS_STBIO1_0210]	[DDS_STBIO1_0221]
SENSOR HUB				[DDS_STBIO1_0211]	
ISPU	[DDS_STBIO1_0212]	[DDS_STBIO1_0213]		[DDS_STBIO1_0215]	[DDS_STBIO1_0222]
ISPU debug	[DDS_STBIO1_0216]	[DDS_STBIO1_0217]	[DDS_STBIO1_0218]	[DDS_STBIO1_0220]	[DDS_STBIO1_0223]

Table 7: XBAR Connection Matrix

[DDS_STBIO1_XBAR_01] Assumption

Every master and slave supports AHBLite and none supports burst communication

[End]

10.2. XBAR Memory Mapping

AHB base start address	AHB base end address	Address block	Direct Access	Req_ID
0x0_0000	0x0_7FFF	ISPU CODE RAM 32KB	N.A.	[DDS_STBIO1_10100]
0x0_8000	0x3_FFFF	RESERVED	N.A.	[DDS_STBIO1_10101]
0x4_0000	0x4_5FFF	ISPU DATA RAM 24KB	N.A.	[DDS_STBIO1_10102]
0x4_6000	0x7_FFFF	RESERVED	N.A.	[DDS_STBIO1_10103]
0x8_0000	0x8_010F	ISPU SYSTEM REGISTERS	N.A.	[DDS_STBIO1_10104]
0x8_0110	0xB_FFFF	RESERVED		[DDS_STBIO1_20000]
0xC_0000	0xC_0800	ADSP PROG ROM	N.A.	[DDS_STBIO1_10105]
0xC_0804	0xC_0FFF	RESERVED		[DDS_STBIO1_20001]
0xC_1000	0xC_1DFF	ADSP DATA MEMORY	N.A.	[DDS_STBIO1_10106]
0xC_1E00	0xC_1E20	ADSP REGISTERS	N.A.	[DDS_STBIO1_10107]
0xC_1E24	0xC_3FFF	NOT USED		
0xC_4000	0xC_4FFF	FIFO RAM MEMORY	N.A.	[DDS_STBIO1_10108]
0xC_5000	0xC_5060	FIFO REGISTERS	N.A.	[DDS_STBIO1_10109]
0xC_5064	0xC_5FFF	NOT USED		
0xC_6000	0xC_6FFF	SENSOR HUB	N.A.	[DDS_STBIO1_10110]
0xC_7000	0xC_7FFF	NOT USED		
0xC_8000	0xC_DFFF	AFE SYSTEM REGISTERS	N.A. with exception (see 12.4)	[DDS_STBIO1_10111]
0xE_0000	0xF_FFFF	STOP CLOCK (Reserved)	N.A.	[DDS_STBIO1_10112]

Table 8: Memory Map

11. STBIO1 Operating Modes

Digital blocks status for different operating modes is reported in table, where ON indicates the clock is running and OFF that it is gated:

	PMU	ADSP	Regmap	FIFO_CTRL	Sensor hub	Main CTRL	ISPU	Req_ID
Power Down	On	Off	Off	Off	Off	On	Off	DDS_STBIO1_10200
Boot	On	On	On	Off	Off	On	Off	DDS_STBIO1_10201
Configuration Phase	On	Off	On	Off	Off	On	Off	DDS_STBIO1_10202
Operative data storage mode	On	Off	Off	On	On	On	On (optionally)	DDS_STBIO1_10203
Operative elaboration mode	On	On	Off	On	On	On	On (optionally)	DDS_STBIO1_10204
Sleep	On	Off	Off	On (Optionally)	Off	On	On (optionally)	DDS_STBIO1_10205

Table 9: Operating modes Table

12. Modules

Instantiations

- u_pmu: pmu
- u_pad_mux: pad_mux
- u_regmap_otp: stbio1_regmap
- u_masterblaze_ahb: senshub_mst_i2c
- u_xbar_afe: xbar_afe: interconnect
- u_ADSP: ADSP
- u_fifo: Fifo Controller
- u_ispu: ispu
- u_i2c_spi_slave: i2c_spi_slave
- u_ca_fifo_bist_c: FIFO RAM (Hard Macro)
- u_ca_qk_bist_c: ADSP RAM (Hard Macro)
- u_ca_qk_rom_bist: ADSP ROM (Hard Macro)
- u_psw_ctrl: ISPU Power Island
- u_ta_afe_bist_ctrl: STBIO1 Bist Controller (Hard Macro)
- u_otp: otp (Hard Macro)
- u_main_controller_top: main_controller_top

12.1.1. PMU

12.1.1.1. PMU TOP Description

PMU oversees the Device Start-Up phase, the clock generation to the Digital and Analog Domain and Reset Sync to the different Clock Domain.

12.1.1.2. PMU TOP I/O List

Port name	Direction	Type	Description
clk_16m	input	wire	Clock 16 MHz from analog ring oscillators
clk_64k	input	wire	Clock 64kHz from analog ring oscillators
POR1V2_1V2	input	wire	POR Signal
START_CLK_16M	input	wire	Clock ready for 16MHz
PDClockF	input	wire	Power Down for 16MHz Ring Oscillator from main controller
i_bio_divider	input	wire	Enable 32kHz geneartion from 16MHz when a Bio-Channel is activated
EN_LDO	input	wire	enable LDO from main controller
PD_CLK_REG	input	wire	power down 16MHz from Regmap

EN_LDO1V8_REG	input	wire	Enable LDO from regmap
bist_mode	input	wire	enable bist mode
debug_mode	input	wire	enable debug
scan_mode	input	wire	enable scan mode
scan_enable	input	wire	start scan mode
scan_clk	input	wire	scan clk
scan_rstn	input	wire	scan rst
clk_bistctlr	input	wire	bist controller and collar clock
rst_bistctlr_n	input	wire	bist controller and collar resetn
clk_ext_en	input	wire	enable external clock from pad
clk_ext_slow	input	wire	external clock 32kHz
i_en_sh_clk	input	wire	enable sensor hub clock
i_senshub_clk_sel	input	wire [2:0]	selector for sensor hub clock
i_test_clk_fast	input	wire	enable test and trimming for 16MHz clock
i_half_charge_ldo	input	wire	enable the low power mode for ldo when only ppg is selected
mask_en_ldo_pdclkf	input	wire	mask the control of pdf and ldo control during boot
clk_en_s0	input	wire	clock gating logic inside stredl_afe_top
clk_en_s1	input	wire	if present should be HSEL STMC AHB Master
clk_en_s2_i2c	input	wire	if present should be HSEL STMC AHB Master
clk_en_s2_on	input	wire	clk gating ON from AHB I2C/SPI Interface
clk_en_s2_off	input	wire	clk gating OFF from AHB I2C/SPI Interface
clk_en_s3	input	wire	clk gating from AHB Master Main Controller

clk_en_s4	input	wire	clock gating logic inside stredl_afe_top
clk_en_s5	input	wire	clock gating logic inside masterblaze_ahb
clk_en_m0	input	wire	clock gating logic inside stredl_afe_top
clk_en_m1	input	wire	HSEL M1 from xbar to STMC AHB Slave
clk_en_m2	input	wire	HSEL M2 from xbar to FIFO
clk_en_m22	input	wire	HSEL M2 from xbar to MASTERBLAZE
clk_en_m3	input	wire	HSEL M3 from xbar to Regmap
iso_i	input	wire	from register
vdden_i	input	wire	enable ISPU Power Island
i_start_operative_16MHz	input	wire	enable the lead of check
main_ctrl_sw_resetrn	input	wire	soft reset for main controller
stmc_sw_resetrn	input	wire	adsp soft reset
fifo_sw_resetrn	input	wire	fifo soft reset
masterblaze_sw_resetrn	input	wire	sensor hub soft reset
regmap_sw_resetrn	input	wire	regmap soft reset
adsp_run	input	wire	adsp is running
adsp_xbar_sw_en	input	wire	adsp clk ahb
i_en_clk_gsr_16M	input	wire	enable the clock for gsr digital circuit
i_en_ck_m_p_q	input	wire	enable quadrature and phase clock
i_en_clk_mod_debug	input	wire	enable the clock modulation debug
i_en_chop_ref	input	wire	enable chopper clock
i_en_ck_chop_imp	input	wire	enable chopper clock for bia
i_en_ck_hc2_10k	input	wire	enable health channel clock
i_freq_bia_m_p_q	input	wire [2:0]	frequency selector for phase and quadrature

i_en_clk_16MHz_sleep	input	wire	enable clock in SLEEP mode when a bio-channel is selected
EN_ADC_TEST_LOW_NOISE	input	wire	enable adc test low noise
EN_ADC_TEST_FAST	input	wire	enable adc test fast
if_sh_ck_in	input	wire	sensor hub clock interface
o_EN_LDO1V8	output	wire	enable ldo to analog domain
o_PD_CLKF	output	wire	power down 16MHz clock to analog domain
o_CLK_16M_READY	output	wire	clock ready to the digital domain
o_DIS_RC_VIREF	output	wire	dis rc viref to the analog domain
pd_osc_clk32k	output	wire	power down for clk 64kHz
tst_clk_32k	output	wire	clock slow 32k to main_controller
tst_clk_16m	output	wire	clock fast 16M
tst_clk_bistctrl	output	wire	test clock for bist ctrl
tst_rst_bistctrl_n	output	wire	test reset for bist ctrl
clk_stredl	output	wire	clock gating logic inside stredl_afe_top
clk_xbar	output	wire	xbar clock
clk_intf	output	wire	interface clock
clk_main_ctrl	output	wire	clock for main controller
clk_stmc	output	wire	adsp clock
clk_stmc_ahb	output	wire	adsp clk elaboration
clk_fifo	output	wire	fifo clock
clk_masterblaze_m	output	wire	sensor hub master clock
clk_masterblaze_s	output	wire	sensor hub slave clock
clk_regmap	output	wire	regmap clock
o_clk_ext_en	output	wire	enable external clock

resetrn_stredl	output	wire	reset mngmt inside stredl_afe_top
resetrn_xbar	output	wire	xbar reset
resetrn_intf	output	wire	reset interface
resetrn_main_ctrl	output	wire	reset main controller
resetrn_32k_main_ctrl	output	wire	reset 32kHz domain main controller
resetrn_stmc	output	wire	reset adsp
resetrn_fifo	output	wire	reset FIFO
resetrn_masterblaze	output	wire	reset sensor hub
resetrn_regmap	output	wire	reset regmap
o_trimming_clk_fast	output	wire	trimming output clock fast
iso_o	output	wire	left open . to be connected in synthesis
vdden_o	output	wire	to be connected to psw_ctrl.VDDIEN
o_ck_bg	output	wire	bg clock to analog domain
o_sh_clk	output	wire	sensor hub clock
o_ck_chop_imp	output	wire	chopper bia clock to analog domain
o_ck_m	output	wire	modulation clock to analog domain
o_ck_hc2_10k	output	wire	healt channel 10kHz clock to analog domain
o_ck_p	output	wire	phase clock to analog domain
o_ck_q	output	wire	quadrature clock to analog domain
o_clk_16M_mc	output	wire	16MHz clock for Main Controller
o_clk_dft_adc	output	wire	16MHz clock for ADC DFT
o_clk_gsr_16M	output	wire	gsr clock to digital domain
o_ADC_clk_serial_dft	output	wire	adc serial clock in dft
o_sh_clk_gate	output	wire	status clk reg clk sh 16Mhz

Table 10: PMU I/O List

12.1.1.3. PMU Peculiar Requirements

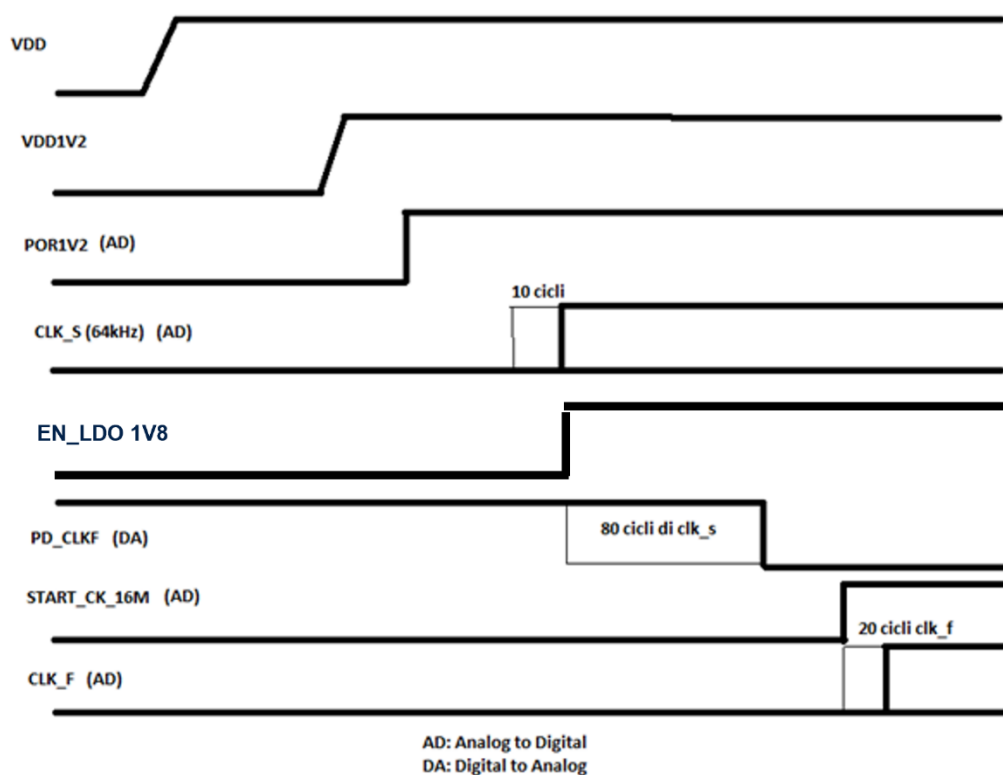


Figure 7: Device Start-Up Phase

[DDS_STBIO1_0102] Definition:

w_EN_LDO and w_PDClockF are as follows:

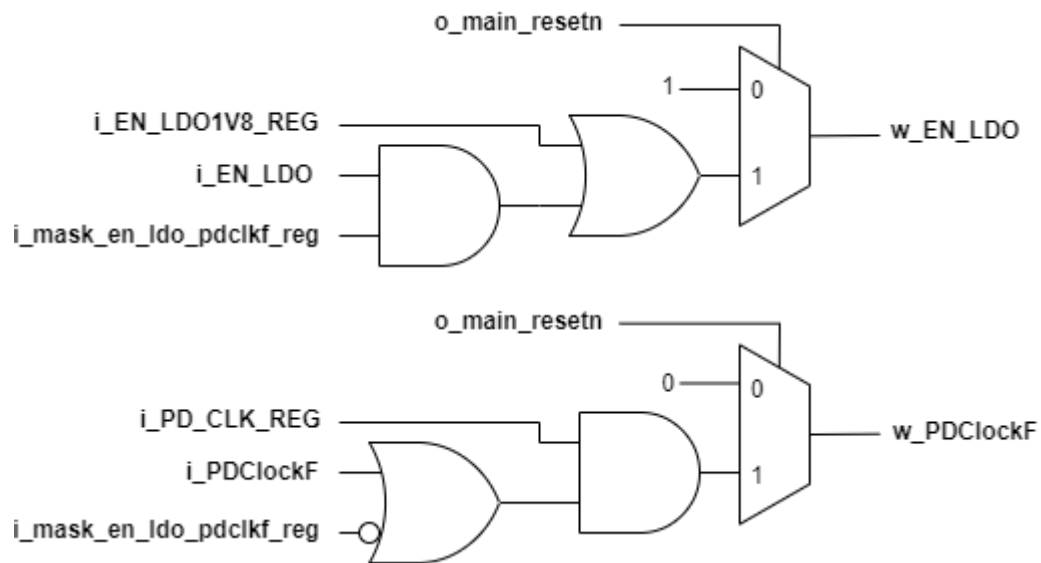


Figure 8: Mask Cell for LDO and Fast Ring Oscillator

w_EN_LDO_sync32 and w_PDClockF_sync32 are the relative 32kHz synchronized versions. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0103] Definition:

clk_64k_divided (32kHz) is i_clk_64k (64kHz) divided by two. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0104] Requirement:

When the device is turned on, i_POR1V2_1V2 shall be high. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0105] Requirement:

When i_POR1V2_1V2 is low, o_EN_LDO1V8 shall be low. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0106] Requirement:

When i_POR1V2_1V2 is low, o_PD_CLKF shall be high. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0107] Requirement:

When i_POR1V2_1V2 is low, o_CLK_16M_READY shall be low. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0108] Requirement:

When i_POR1V2_1V2 is low, o_DIS_RC_VIREF shall be high. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0109] Requirement:

When i_POR1V2_1V2 rises, after at least 10 cycles of i_clk_64k (64kHz), o_EN_LDO1V8 shall be high and the clk_64k_divided is considered stable. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0110] Requirement:

When clk_64k_divided is stable, after at least 352 cycles of clk_64k_divided (32kHz), o_DIS_RC_VIREF shall be low. [TO: IPOS_PMU] [END]

[DDS_STBIO1_0111] Requirement:

When i_mask_en_ldo_pdclkf_reg and o_EN_LDO1V8 are set to 1, o_PD_CLKF shall be low after at least 40 cycles of the clk_64k_divided. [TO: IPOS_PMU] [\[END\]](#)

[DDS_STBIO1_0112] Requirement:

When i_START_CLK_16M is high, after at least 20 cycles of i_clk_16M (16MHz), o_CLK_16M_READY shall be high. [TO: IPOS_PMU] [\[END\]](#)

[DDS_STBIO1_0113] Requirement:

When i_START_CLK_16M is high for the first time, after at least 20 cycles of i_clk_16M (16MHz), o_main_resetrn shall be high. [TO: IPOS_PMU] [\[END\]](#)

12.1.1.4. Clocks and Reset

Clocks scheme is shown in the below figure

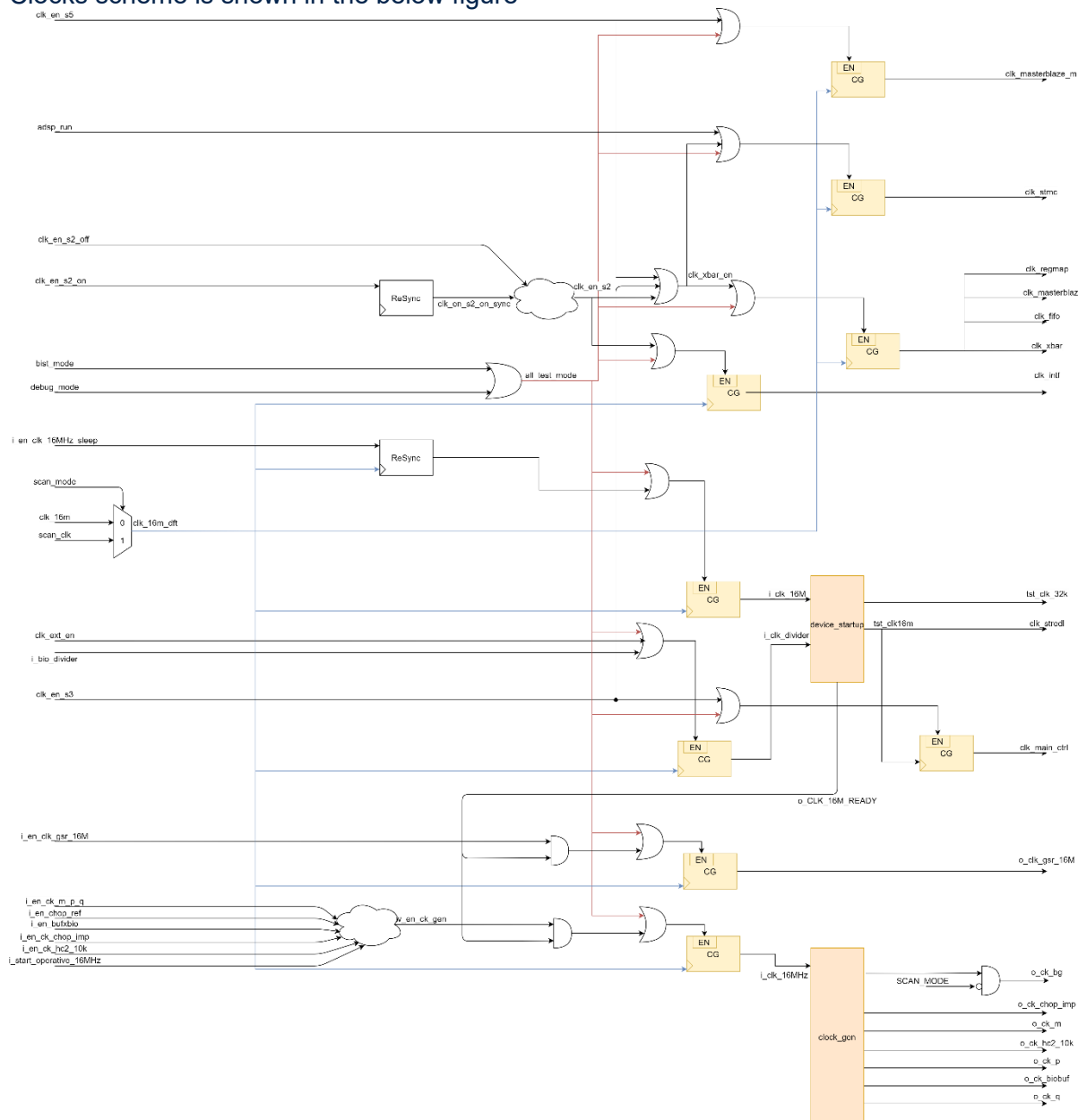


Figure 9: Clocks Scheme

Clocks scheme is shown in the above figure

Req_ID	Source	Target	Path Delay	Clock Gating	Analog Control and Description	F Max
Requirements					Definition	
DDS_STBIO1_1020	stbio1_top.clk_16m	stbio1_top.dig_wrapper_LV.u_pmu.clk_16m		No	Pwr_down controlled by pmu	16 MHz
DDS_STBIO1_1021	stbio1_top.clk_64k	stbio1_top.dig_wrapper_LV.u_pmu.clk_64k		No	No	64 kHz

DDS_STBIO1_1027	stbio1_top.dig_wrapper_LV.u.pmu.tst_clk_32k	stbio1_top.dig_wrapper_LV.u.main_controller_top.i_clk_32kHz		No	No	32 kHz
DDS_STBIO1_1022	stbio1_top.dig_wrapper_LV.u.pmu.clk_16m_df	stbio1_top.dig_wrapper_LV.u.ADSP.s_tmc_clk	1	stbio1_top.dig_wrapper_LV.u.pmu.adsp_run stbio1_top.dig_wrapper_LV.u.pmu.all_test_mode ~stbio1_top.dig_wrapper_LV.u.pmu.ad_mux_scan_enable	No	16 MHz
DDS_STBIO1_1023	stbio1_top.dig_wrapper_LV.u.pmu.w_sh_clk	stbio1_top.dig_wrapper_LV.u.master_blaze_ahb.sys_clk	1	stbio1_top.dig_wrapper_LV.u.pmu.i_en_sh_clk stbio1_top.dig_wrapper_LV.u.pmu.all_test_mode ~stbio1_top.dig_wrapper_LV.u.pmu.ad_mux_scan_enable	No	16 MHz
DDS_STBIO1_1024	stbio1_top.dig_wrapper_LV.u.pmu.clk_16m_df	stbio1_top.dig_wrapper_LV.u.xbar_ahbclk	1	stbio1_top.dig_wrapper_LV.u.pmu.clk_xbar_en stbio1_top.dig_wrapper_LV.u.pmu.all_test_mode ~stbio1_top.dig_wrapper_LV.u.pmu.ad_mux_scan_enable	No	16 MHz
DDS_STBIO1_2024	stbio1_top.dig_wrapper_LV.u.pmu.clk_xbar	stbio1_top.dig_wrapper_LV.u.fifo.HCLK	1	stbio1_top.dig_wrapper_LV.u.pmu.clk_en_m2 stbio1_top.dig_wrapper_LV.u.pmu.all_test_mode stbio1_top.dig_wrapper_LV.u.pmu.clk_en_m2_reg ~stbio1_top.dig_wrapper_LV.u.pmu.ad_mux_scan_enable	No	16 MHz
DDS_STBIO1_2025	stbio1_top.dig_wrapper_LV.u.pmu.clk_stredl	stbio1_top.dig_wrapper_LV.u.ispu.clk		No	Pwr_down controlled by pmu	16 MHz
DDS_STBIO1_2026	stbio1_top.dig_wrapper_LV.u.pmu.clk_xbar	stbio1_top.dig_wrapper_LV.u.regmap_otp.hclk	1	stbio1_top.dig_wrapper_LV.u.pmu.clk_en_m3 stbio1_top.dig_wrapper_LV.u.pmu.all_test_mode stbio1_top.dig_wrapper_LV.u.pmu.clk_en_m3_reg stbio1_top.dig_wrapper_LV.u.pmu.clk_en_s3 ~stbio1_top.dig_wrapper_LV.u.pmu.ad_mux_scan_enable	No	16 MHz
DDS_STBIO1_1025	stbio1_top.u.pmu.u.device.startup.tst_clk_32k	stbio1_top.dig_wrapper_LV.tst_clk_32k		No	No	32 kHz
DDS_STBIO1_1026	stbio1_top.dig_wrapper_LV.u.pmu.tst_clk_16m	stbio1_top.dig_wrapper_LV.u.main_controller_top.i_clk_16MHz	1	stbio1_top.dig_wrapper_LV.u.pmu.i_en_clk_16MHz_sleep stbio1_top.dig_wrapper_LV.u.pmu.scan_mode ~stbio1_top.dig_wrapper_LV.u.pmu.ad_mux_scan_enable	No	16 MHz

Table 11: Clock Recap Table

Reset scheme is shown in the below figure

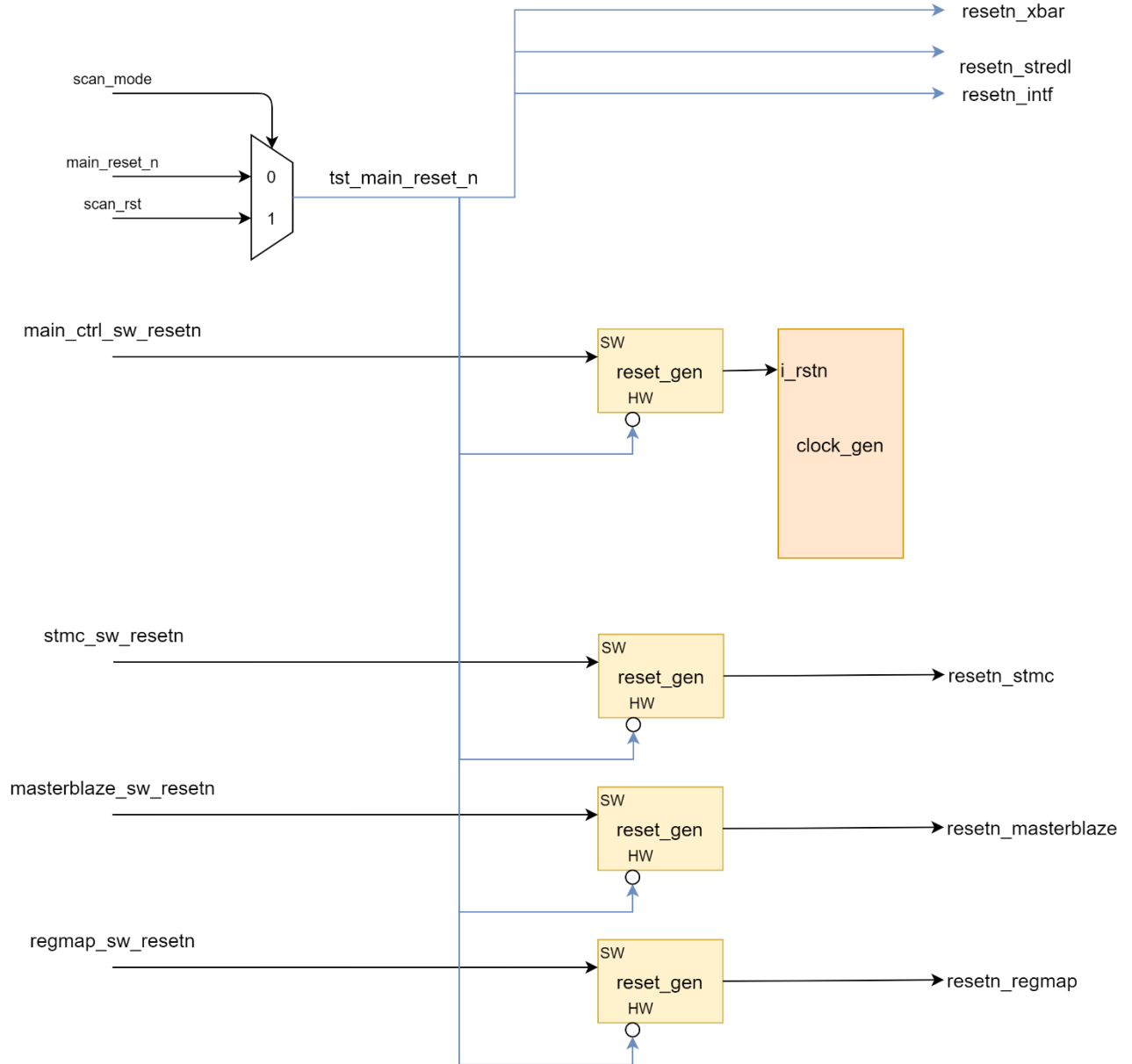


Figure 10: STBIO1 Reset Scheme

Req_ID	Source	Target
DDS_STBIO1_1050	u_stbio1_top.dig_wrapper_LV.u_pmu.resetn_stmc	u_stbio1_top.dig_wrapper_LV.u_ADSP.HRESETn
DDS_STBIO1_1051	u_stbio1_top.dig_wrapper_LV.u_pmu.resetn_masterblaze	u_stbio1_top.dig_wrapper_LV.u_masterblaze_ahb.HRESETn
DDS_STBIO1_1052	u_stbio1_top.dig_wrapper_LV.u_pmu.resetn_regmap	u_stbio1_top.dig_wrapper_LV.u_regmap_top.hresetn
DDS_STBIO1_2052	stbio1_top.dig_wrapper_LV.u_pmu.u_device_startup.u_por_main_resetn.RST_N_FF[0]	u_stbio1_top.dig_wrapper_LV.u_xbar_afe.hresetn
DDS_STBIO1_2053	u_stbio1_top.dig_wrapper_LV.u_pmu.resetn_regmap	u_stbio1_top.dig_wrapper_LV.u_ispu.main_rst_n
DDS_STBIO1_1053	u_stbio1_top.dig_wrapper_LV.u_pmu.resetn_fifo	u_stbio1_top.dig_wrapper_LV.u_multi_channel_fifo_top.HRESETn

DDS_STBIO1_1054	u_stbio1_top.dig_wrapper_LV.u_p mu.resetn_main_ctrl	u_stbio1_top.dig_wra pper_LV.u_main_cont roller_top.i_rstn
DDS_STBIO1_2054	u_stbio1_top.dig_wrapper_LV.u_p mu.resetn_32k_main_ctrl	u_stbio1_top.dig_wra pper_LV.u_main_cont roller_top.i_rstn_sync 32

Table 12: STBIO1 Functional reset Summary table

12.1.2. PAD Mux

In the following table are reported the Pad Configuration for each operative mode of the device.
The
BIST Mode, Scan Mode and Debug Mode are reserved for ST Testing Purpose

12.1.2.1. PAD MUX TOP I/O List

Port name	Direction	Type	Description
i_write_protection_en	input		write_protection_en OTP regmap enable from regmap
o_write_protection_en	output		write_protection_en OTP regmap enable to regmap
i_tap_bist_mode	input		bist_mode signal used retun in functional mode
i_EN_OP05HP_ECG0	input		bit enable select PDRES 05 ECG0 channel
i_EN_OP05HP_ECG12	input		bit enable select PDRES 05 ECG12 channel
IMPQ_PSDRES_UP_02	input	[2:0]	OTP value for Analog block
IMPQ_PSDRES_DOWN_02	input	[2:0]	OTP value for Analog block
IMPP_PSDRES_UP_02	input	[2:0]	OTP value for Analog block
IMPP_PSDRES_DOWN_02	input	[2:0]	OTP value for Analog block
ECG_PSDRES_DOWN_02	input	[2:0]	OTP value for Analog block
ECG_PSDRES_UP_02	input	[2:0]	OTP value for Analog block
IMPQ_PSDRES_UP_05	input	[2:0]	OTP value for Analog block
IMPQ_PSDRES_DOWN_05	input	[2:0]	OTP value for Analog block
IMPP_PSDRES_UP_05	input	[2:0]	OTP value for Analog block
IMPP_PSDRES_DOWN_05	input	[2:0]	OTP value for Analog block
ECG_PSDRES_DOWN_05	input	[2:0]	OTP value for Analog block
ECG_PSDRES_UP_05	input	[2:0]	OTP value for Analog block

IMPQ_PSDRES_UP	output	[2:0]	OTP value for Analog block
IMPQ_PSDRES_DOWN	output	[2:0]	OTP value for Analog block
IMPP_PSDRES_UP	output	[2:0]	OTP value for Analog block
IMPP_PSDRES_DOWN	output	[2:0]	OTP value for Analog block
ECG_PSDRES_DOWN	output	[2:0]	OTP value for Analog block
ECG_PSDRES_UP	output	[2:0]	OTP value for Analog block
i_TS_IRQ_ST	input		time slot irq
i_DCCK_IRQ_ST	input		DCCK irq change status
o_en_tx_1V2	input		input to LED pilot
o_tx_preset_1V2	input		input to LED pilot
o_tx1_outlab_sel_frame_x_1V2	input		input to LED pilot
sel_irq_gpio1	input	[3:0]	sel irq gpi1 {000:NC, 001:i_fifo_irq, 010:i_LED_Pilot, 011:i_DCCK_irq, 100:i_error_time_slot, 101:i_time_slot_irq}
sel_irq_gpio2	input	[3:0]	sel irq gpi2 {000:NC, 001:i_fifo_irq, 010:i_LED_Pilot, 011:i_DCCK_irq, 100:i_error_time_slot, 101:i_time_slot_irq}
sel_irq_gpio3	input	[3:0]	sel irq gpi3 {000:NC, 001:i_fifo_irq, 010:i_LED_Pilot, 011:i_DCCK_irq, 100:i_error_time_slot, 101:i_time_slot_irq}
sel_irq_gpio4	input	[3:0]	sel irq gpi4 {000:NC, 001:i_fifo_irq, 010:i_LED_Pilot, 011:i_DCCK_irq, 100:i_error_time_slot, 101:i_time_slot_irq}
DCCK_P_en	input		Manages lead off mode DCCK_P, Source: Regmap
DCCK_N_en	input		Manages lead off mode DCCK_P, Source: Regmap
MODE_OPERATION	input		Start Divice operations status

o_pd_p	output		enable lead off mode DCCK_P, Destination:Analog block
o_pd_n	output		enable lead off mode DCCK_N, Destination:Analog block
O_DCCK_P_i	input		status lead off mode DCCK_P {0:low impedance,1:floating electrode}, Source: Analog block
O_DCCK_N_i	input		status lead off mode DCCK_N {0:low impedance,1:floating electrode}, Source: Analog block
if_gpio1_in_reg	output		ZI value from regmap
if_gpio1_en_reg	input		EN value from regmap
if_gpio1_out_reg	input		OUT value from regmap
if_gpio2_in_reg	output		ZI value from regmap// ZI
if_gpio2_en_reg	input		EN value from regmap
if_gpio2_out_reg	input		OUT value from regmap
if_gpio3_in_reg	output		ZI value from regmap
if_gpio3_en_reg	input		EN value from regmap
if_gpio3_out_reg	input		OUT value from regmap
if_gpio4_in_reg	output		ZI value from regmap
if_gpio4_en_reg	input		EN value from regmap
if_gpio4_out_reg	input		OUT value from regmap
o_ext_reg_dt_afe	input	[7:0]	i2c/spi read data u_regmap_otp
o_ext_reg_dt	output	[7:0]	i2c/spi read data output
i_ext_reg_add	input	[7:0]	i2c/spi address
i_ext_reg_dt	input	[7:0]	i2c/spi write data
i_ext_reg_ck_wr	input		i2c/spi clk

i_ext_reg_we	input		i2c/spi regmap direct write enable
if_sclk_in	input		ZI_I2C
if_sclk_i3c_in	input		ZI_I3C
if_sclk_out	output		A
if_sclk_en	output		EN
if_sclk_tud	output		TUD
if_sda_sdi_in	input		ZI_I2C
if_sda_sdi_i3c_in	input		ZI_I3C
if_sda_sdi_out	output		A
if_sda_sdi_en	output		EN
if_sda_sdi_tud	output		TUD
if_cs_in	input		ZI_CS
if_cs_out	output		A
if_cs_en	output		EN
if_cs_tud	output		TUD
if_gpio0_in	input		ZI_GPIO0
if_gpio0_out	output		A
if_gpio0_en	output		EN
if_gpio0_tud	output		TUD
if_gpio1_in	input		ZI_GPIO1
if_gpio1_out	output		A
if_gpio1_en	output		EN
if_gpio1_tud	output		TUD
if_gpio2_in	input		ZI_GPIO2

if_gpio2_out	output		A
if_gpio2_en	output		EN
if_gpio2_tud	output		TUD
if_gpio3_in	input		ZI GPIO3
if_gpio3_out	output		A
if_gpio3_en	output		EN
if_gpio3_tud	output		TUD
if_gpio4_in	input		ZI GPIO4
if_gpio4_out	output		A
if_gpio4_en	output		EN
if_gpio4_tud	output		TUD
if_sh_ck_in	input		ZI sensor hub ck
if_sh_ck_out	output		A
if_sh_ck_en	output		EN
if_sh_ck_tud	output		TUD
if_sh_sda_in	input		ZI sensor hub SDA
if_sh_sda_out	output		A
if_sh_sda_en	output		EN
if_sh_sda_tud	output		TUD
POR1V2_1V2	input		POR
ispu_tck	output		jtag clk port ispu in ispu debug mode
ispu_tdi	output		jtag data in port ispu in ispu debug mode
ispu_tdo	input		jtag data out port ispu in ispu debug mode
ispu_tdo_en	input		jtag en port ispu in ispu debug mode

ispu_tms	output		jtag state in port ispu in ispu debug mode
ispu_trst_n	output		jtag reset in port ispu in ispu debug mode
i_if_ck	output		i2c/spi clk
i_if_cs	output		i2c/spi cs
i_if_sda_sdi_in	output		i2c/spi data
o_if_cs_en_n	input		i2c/spi cs en from i2c/spi ip
o_if_cs_out_n	input		i2c/spi cs out from i2c/spi ip
o_if_cs_tud	input		i2c/spi cs tud from i2c/spi ip
o_if_sda_sdi_en_n	input		i2c/spi data en from i2c/spi ip
o_if_sda_sdi_out	input		i2c/spi data out from i2c/spi ip
scl_master_in	output		from sensorhub
scl_master_out	input		from sensorhub
sda_master_in	output		to sensorhub
sda_master_out	input		from sensorhub
afe_bist_ctrl_tdi	output		bist jtag port
afe_bist_ctrl_trstn	output		bist jtag port
afe_bist_ctrl_tms	output		bist jtag port
afe_bist_ctrl_tdo	input		bist jtag port
afe_bist_ctrl_tdo_en_n	input		bist jtag port
afe_bist_ctrl_tck	output		bist jtag port
ck_16m	input		clk16MHz trimming in debug mode
ck_2m	input		clk2MHz trimming in debug mode
ck_64k	input		clk64MHz trimming in debug mode
scan_rstn	output		scan reset

scan_enable	output		scan enable
scan_clk	output		scan clock
ext_cap_en	input		external cap in debug mode
irq_fifo	input		fifo irq from main controller - regmap
error_time_slot	input		error timeslot from error time slot
gpio_in	output		gpio0 in in functional mode from regmap
gpio_out	input		gpio0 out in functional mode from regmap
gpio_en	input		gpio0 en in functional mode from regmap
irq_SENSHUB_STS	input		sensorhub status
irq_ISPU_MC_INTR_STS	input		irq main controller data-valid
scan_mode	output		stbio1 scan mode
bist_mode	output		stbio1 bist mode
debug_mode	output		stbio1 debug mode
memsafe	output		memsafe for bist
test_clk_fast	input		enable test clk 16MHz trimming
test_clk_slow	input		enable test clk 64kHz trimming
if_sclk_tud_reg	input		if sclk_tud from reg
if_sda_sdi_tud_reg	input		if sda_sdi_tud from re
if_sh_sda_tud_reg	input		if sh_sda_tud from reg
if_gpio0_tud_reg	input		if gpio0_tud from reg
if_gpio1_tud_reg	input		if gpio1_tud from reg
if_gpio2_tud_reg	input		if gpio2_tud from reg
if_gpio3_tud_reg	input		if gpio3_tud from reg

if_gpio4_tud_reg	input		if gpio4_tud from reg
if_sh_ck_tud_reg	input		if sh_ck_tud from reg
clk_dft_adc	input		16MHz adc dft clk
i_ADC_clk_serial_dft	input		serial clk for adc dft 16Mhz:Fast mode, if_sh_ck_in:Low noise mode
EN_ADC_TEST_FAST	input		enable adc test fast
EN_ADC_TEST_LOW_NOISE	input		enable adc test low noise
i_ADC_EOC	input		adc eoc input
i_ADC_start	input		adc star from main controller
i_data_ADC_out	input	[15:0]	adc data input
o_ADC_start	output		adc star output
i_N_clk	input	[6-1:0]	n clk start time high
i_saturation_irq	input		interrupt saturation from main controller
i_end_boot	input		end boot signal from adsp
quokka_run	input		status quokka elaboration 1:run 0:not elaboration
sensor_hub_status	input		status sensor_hob from regmap
SH_CKL_EN	input		sensorhub clk enable from regmap (master_end_op_clr or sleep_master_clr or ahb_master_on_clr)
i_t_force_1v2	input		input force signal to IDDQ test
o_t_force_1v2	output		force signal to IDDQ test
i_bfail_fifo	input		bist fail fifo ram
i_bfail_qk	input		bist fail adsp ram
i_bfail_ispu1	input		bist fail ispu ram instruction mem
i_bfail_ispu2	input		bist fail ispu ram data mem 1

i_bfail_ispu3	input		bist fail ispu ram data mem 2
i_bfail_ispu4	input		bist fail ispu ram data mem 3
i_bend	input		bist end signal
i_bbad	input		signal fail bist test
o_bypass	output		bypass memory signal

Table 1313: PAD MUX I/O List

12.1.2.2. Requirements

[DDS_STBIO1_300] Definition:

The operative mode is selected by a i2c/spi write on register

MODE_CONFIG

MODE_CONFIG_0 register

7	6	5	4	3	2	1	0
RESERVED	bypass	memsafe	ispu_debug_mode	debug_mode	bist_mode	SCAN_COMPR_EN	scan_mode
	RW	RW	RW	RW	W	W	W

Address: 0xFE

Type: RW

Reset: 0x00

[6]	bypass: SPIRIT parameter _resetPort_:reset_no_scan SPIRIT parameter _clockPort_:clk_no_scan
[5]	memsafe: SPIRIT parameter _resetPort_:reset_no_scan SPIRIT parameter _clockPort_:clk_no_scan
[4]	ispu_debug_mode:
[3]	debug_mode:
[2]	bist_mode:
[1]	SCAN_COMPR_EN: SPIRIT parameter _resetPort_:reset_no_scan SPIRIT parameter _customType_:W1S SPIRIT parameter _clockPort_:clk_no_scan
[0]	scan_mode: SPIRIT parameter _customType_:W1S SPIRIT parameter _resetPort_:reset_no_scan SPIRIT parameter _clockPort_:clk_no_scan

[Vpriority High] [End]

PAD	Type	Functional Mode (MODE_CONFIG_0=0x00)				SCAN Mode (MODE_CONFIG_0=0x01)	BIST Mode (MODE_CONFIG_0=0x02)	Debug Mode (MODE_CONFIG_0=0x04)				ISPU Debug Mode (MODE_CONFIG_0=0x08)
		operative_mode				scan_in2	scan_in1	scan_in0	scan_in3	scan_in4	scan_in5	scan_in6
SDA	Analog	SPIRIT Clock				scan_in2	scan_in1	scan_in0	scan_in3	scan_in4	scan_in5	scan_in6
SDA	Analog	SPIRIT Slave Inout / OCSDA				scan_in2	scan_in1	scan_in0	scan_in3	scan_in4	scan_in5	scan_in6
CS	Analog	SPIRIT Chip Select				scan_in2	scan_in1	scan_in0	scan_in3	scan_in4	scan_in5	scan_in6
GPIO0	Analog	General Purpose				scan_in2	scan_in1	scan_in0	scan_in3	scan_in4	scan_in5	scan_in6
GPIO1	Analog	General Purpose	Start Time start (RQ_SEL1_gpio1_in_ssel=1)	LED_Pwr (RQ_SEL1_gpio1_in_ssel=2)	DOCK_in (RQ_SEL1_gpio1_in_ssel=3)	error_time_start (RQ_SEL1_gpio1_in_ssel=4)	input	sense Internal V1V2 (sen=1)	sense Internal V1V2 (sen=1)	ck_16m	ck_32k	ck_16m
GPIO2	Analog	General Purpose	Start Time start (RQ_SEL1_gpio2_in_ssel=1)	LED_Pwr (RQ_SEL1_gpio2_in_ssel=2)	DOCK_in (RQ_SEL1_gpio2_in_ssel=3)	error_time_start (RQ_SEL1_gpio2_in_ssel=4)	input	sense Internal V1V2 (sen=1)	sense Internal V1V2 (sen=1)	ck_16m	ck_32k	ck_16m
GPIO3/PM5F	Analog	General Purpose	Start Time start (RQ_SEL1_gpio3_in_ssel=1)	LED_Pwr (RQ_SEL1_gpio3_in_ssel=2)	DOCK_in (RQ_SEL1_gpio3_in_ssel=3)	error_time_start (RQ_SEL1_gpio3_in_ssel=4)	input	sense Internal V1V2 (sen=1)	sense Internal V1V2 (sen=1)	ck_16m	ck_32k	ck_16m
GPIO4/PM5B	Analog	General Purpose	Start Time start (RQ_SEL1_gpio4_in_ssel=1)	LED_Pwr (RQ_SEL1_gpio4_in_ssel=2)	DOCK_in (RQ_SEL1_gpio4_in_ssel=3)	error_time_start (RQ_SEL1_gpio4_in_ssel=4)	input	sense Internal V1V2 (sen=1)	sense Internal V1V2 (sen=1)	ck_16m	ck_32k	ck_16m
SH_OK	Analog	General Purpose	Start Time start (RQ_SEL1_gpio5_in_ssel=1)	LED_Pwr (RQ_SEL1_gpio5_in_ssel=2)	DOCK_in (RQ_SEL1_gpio5_in_ssel=3)	error_time_start (RQ_SEL1_gpio5_in_ssel=4)	input	sense Internal V1V2 (sen=1)	sense Internal V1V2 (sen=1)	ck_16m	ck_32k	ck_16m
SH_SDA	Analog	Sensor HUB Data				scan_in2	scan_in1	scan_in0	scan_in3	scan_in4	scan_in5	scan_in6

Table 1414: PAD Configuration Table

[DDS_STBIO1_301] Assumption

It will be enabled by two consecutive writes, pag. 3 to address 0x40 to set the page and 0x00 to address 0x3E

[End]

[DDS_STBIO1_302] Requirement

If MODE_CONFIG register is 0x00 the operative mode shall be functional.

[Vpriority High] [End]

[DDS_STBIO1_303] Requirement

If MODE_CONFIG register is 0x01 the operative mode shall be SCAN.

[End]

[DDS_STBIO1_304] Requirement

If MODE_CONFIG register is 0x04 the operative mode shall be BIST.

[End]

[DDS_STBIO1_305] Requirement

If MODE_CONFIG register is 0x08 the operative mode shall be DEBUG.

[End]

[DDS_STBIO1_306] Requirement

If MODE_CONFIG register is 0x10 the operative mode shall be ISPU_DEBUG.

[Vpriority High] [End]

[DDS_STBIO1_307] Requirement

If MODE_CONFIG register is 0x20 the operative mode shall be Bist MEMSAFE mode.

[Vpriority High] [End]

[DDS_STBIO1_308] Requirement

If MODE_CONFIG register is 0x40 the operative mode shall be Bist Bypass mode.

[Vpriority High] [End]

[DDS_STBIO1_314] Requirement

In function mode, the GPIO1,2,3,4_out shall be General-Purpose if sel irq gpi1,2,3,4 is 0x00 and ext_cap_en is "0"

[Vpriority High][End]

[DDS_STBIO1_309] Requirement

In function mode, the GPIO1,2,3,4_out shall be irq_fifo if sel irq gpio1,2,3,4 is 0x01 and ext_cap_en is "0"

[Vpriority High] [End]

[DDS_STBIO1_310] Requirement

In function mode, the GPIO1,2,3,4_out shall be i_LED_Pilot if sel irq gpi1,2,3,4 is 0x02 and ext_cap_en is "0"

[Vpriority High] [End]

[DDS_STBIO1_311] Requirement

In function mode, the GPIO1,2,3,4_out shall be i_DCCK_irq if sel_irq_gpio1,2,3,4 is 0x03 and ext_cap_en is "0"

[Vpriority High] [End]

[DDS_STBIO1_312] Requirement

In function mode, the GPIO1,2,3,4 output shall be i_error_time_slot if sel_irq_gpio1,2,3,4 is 0x04 and ext_cap_en is "0"

[Vpriority High] [End]

[DDS_STBIO1_313] Requirement

In function mode, the GPIO1,2,3,4 output shall be i_time_slot_irq if sel_irq_gpio1,2,3,4 is 0x05 and ext_cap_en is "0"

[Vpriority High] [End]

Table 15 functional mode Static Connectivity Checks

ENABLE	SOURCE	DESTINATION	Req ID
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_fifo_irq_FIFO_U_M_IRQ_STATUS	stbio1_top.if_gpio1_out	DDS_STBIO1_GPIO1_309
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_fifo_irq_FIFO_U_M_IRQ_STATUS	stbio1_top.if_gpio2_out	DDS_STBIO1_GPIO2_309
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	stbio1_top.dig_wrapper_LV.u_fifo_irq_FIFO_U_M_IRQ_STATUS	stbio1_top.if_gpio3_out	DDS_STBIO1_GPIO3_309
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	stbio1_top.dig_wrapper_LV.u_fifo_irq_FIFO_U_M_IRQ_STATUS	stbio1_top.if_gpio4_out	DDS_STBIO1_GPIO4_309
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 1) }	1'b0	stbio1_top.if_gpio1_en	DDS_STBIO1_GPIO1_309

<pre> == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en } </pre>			1_EN_309
<pre> {(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio2_irq_sel == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en } </pre>	1'b0	stbio1_top.if_gpio2_en	DDS_STB IO1_GPIO 2_EN_309
<pre> {(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio3_irq_sel == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) } </pre>	1'b0	stbio1_top.if_gpio3_en	DDS_STB IO1_GPIO 3_EN_309
<pre> {(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio4_irq_sel == 1) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)} </pre>	1'b0	stbio1_top.if_gpio4_en	DDS_STB IO1_GPIO 4_EN_309
<pre> {(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio1_irq_sel == 5) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en } </pre>	stbio1_top.dig_wrapper_LV.u_regmap_otp.irq_TS_IRQ_ST	stbio1_top.if_gpio1_out	DDS_STB IO1_GPIO 1_313
<pre> {(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio1_irq_sel == 5) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en } </pre>	1'b0	stbio1_top.if_gpio1_en	DDS_STB IO1_GPIO 1_EN_313
<pre> {(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio2_irq_sel == 5) && !pad_mux__path.scan_mode </pre>	stbio1_top.dig_wrapper_LV.u_regmap_otp.irq_TS_IRQ_ST	stbio1_top.if_gpio2_out	DDS_STB IO1_GPIO 2_313

&& (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }			
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 5) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }}	1'b0	stbio1_top.if_gpio2_en	DDS_STB IO1_GPIO 2_EN_313
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 5) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}}	stbio1_top.dig_wrapper_LV.u_regmap_otp.irq_TS_IRQ_ST	stbio1_top.if_gpio3_out	DDS_STB IO1_GPIO 3_313
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 5) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}}	1'b0	stbio1_top.if_gpio3_en	DDS_STB IO1_GPIO 3_EN_313
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 5) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}}	stbio1_top.dig_wrapper_LV.u_regmap_otp.irq_TS_IRQ_ST	stbio1_top.if_gpio4_out	DDS_STB IO1_GPIO 4_313
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 5) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}}	1'b0	stbio1_top.if_gpio4_en	DDS_STB IO1_GPIO 4_EN_313
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 4) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }}	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_error_time_slot	stbio1_top.if_gpio1_out	DDS_STB IO1_GPIO 1_312

{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 4) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	1'b0	stbio1_top.if_gpio1_en	DDS_STB IO1_GPIO 1_EN_31 2
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 4) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_error_time_slot	stbio1_top.if_gpio2_out	DDS_STB IO1_GPIO 2_312
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 4) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	1'b0	stbio1_top.if_gpio2_en	DDS_STB IO1_GPIO 2_EN_31 2
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 4) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_error_time_slot	stbio1_top.if_gpio3_out	DDS_STB IO1_GPIO 3_312
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 4) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	1'b0	stbio1_top.if_gpio3_en	DDS_STB IO1_GPIO 3_EN_31 2
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 4) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_error_time_slot	stbio1_top.if_gpio4_out	DDS_STB IO1_GPIO 4_312
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 4) && !pad_mux__path.scan_mode }	1'b0	stbio1_top.if_gpio4_en	DDS_STB IO1_GPIO 4_EN_31 2

&& (pad_mux__path.mission_mode == 1)}			
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 2) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en}}	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_en_tx & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx_preset & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx1_outlab_sel_frame_x	stbio1_top.if_gpio1_out	DDS_STB IO1_GPIO 1_310
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 2) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en}}	1'b0	stbio1_top.if_gpio1_en	DDS_STB IO1_GPIO 1_EN_310
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 2) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en}}	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_en_tx & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx_preset & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx1_outlab_sel_frame_x	stbio1_top.if_gpio2_out	DDS_STB IO1_GPIO 2_310
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 2) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en}}	1'b0	stbio1_top.if_gpio2_en	DDS_STB IO1_GPIO 2_EN_310
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 2) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}}	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_en_tx & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx_preset & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx1_outlab_sel_frame_x	stbio1_top.if_gpio3_out	DDS_STB IO1_GPIO 3_310
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 2) && !pad_mux__path.scan_mode &&	1'b0	stbio1_top.if_gpio3_en	DDS_STB IO1_GPIO 3_EN_310

(pad_mux__path.mission_mode == 1) }			
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 2) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}	stbio1_top.dig_wrapper_LV.u_main_controller_top.o_en_tx & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx_preset & stbio1_top.dig_wrapper_LV.u_main_controller_top.o_tx1_outlab_sel frame_x	stbio1_top.if_gpio4_out	DDS_STB IO1_GPIO 4_310
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 2) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}	1'b0	stbio1_top.if_gpio4_en	DDS_STB IO1_GPIO 4_EN_310
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_regmap_otp.irq_DCCK_IRQ_ST	stbio1_top.if_gpio1_out	DDS_STB IO1_GPIO 1_311
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio1_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	1'b0	stbio1_top.if_gpio1_en	DDS_STB IO1_GPIO 1_EN_311
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_regmap_otp.irq_DCCK_IRQ_ST	stbio1_top.if_gpio2_out	DDS_STB IO1_GPIO 2_311
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) &&	1'b0	stbio1_top.if_gpio2_en	DDS_STB IO1_GPIO 2_EN_311

!pad_mux__path.ext_cap_en }			
{{(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio3_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1)}}	stbio1_top.dig_wrapper_LV.u_r egmap_otp.irq_DCCK_IRQ_ST	stbio1_top.if_g pio3_out	DDS_STB IO1_GPIO 3_311
{{(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio3_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1)}}	1'b0	stbio1_top.if_g pio3_en	DDS_STB IO1_GPIO 3_EN_31 1
{{(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio4_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1)}}	stbio1_top.dig_wrapper_LV.u_r egmap_otp.irq_DCCK_IRQ_ST	stbio1_top.if_g pio4_out	DDS_STB IO1_GPIO 4_311
{{(stbio1_top.dig_wrapper_LV .u_regmap_otp.gpio4_irq_sel == 3) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1)}}	1'b0	stbio1_top.if_g pio4_en	DDS_STB IO1_GPIO 4_EN_31 1
{ !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1) && pad_mux__path.ext_cap_en}	stbio1_top.dig_wrapper_LV.u_fi fo.irq_FIFO_U_M_IRQ_STATU S stbio1_top.dig_wrapper_LV.u_ main_controller_top.o_error_tim e_slot	stbio1_top.if_g pio0_out	DDS_STB IO1_GPIO 0_EXT_C AP_315
{ !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1) && pad_mux__path.ext_cap_en}	1'b0	stbio1_top.if_g pio0_en	DDS_STB IO1_GPIO 0_EN_EX T_CAP_3 15
{ !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1) && pad_mux__path.ext_cap_en}	1'b1	stbio1_top.if_g pio1_en	DDS_STB IO1_GPIO 1_EN_EX T_CAP_3 15
{ !pad_mux__path.scan_mode && (pad_mux__path.mission_mo	1'b1	stbio1_top.if_g pio2_en	DDS_STB IO1_GPIO 2_EN_EX

de == 1) && pad_mux_path.ext_cap_en}			T_CAP_3 15
{ !pad_mux__path.scan_mode && ((pad_mux__path.mission_m ode == 1) pad_mux__path.debug_mod e pad_mux__path.ispu_debug _mode) }	stbio1_top.if_cs_in	stbio1_top.dig_ wrapper_LV.u_ i2c_spi_slave.i _if_cs	DDS_STB IO1_CS_3 16
{ !pad_mux__path.scan_mode && ((pad_mux__path.mission_m ode == 1) pad_mux__path.debug_mod e pad_mux__path.ispu_debug _mode) }	!stbio1_top.dig_wrapper_LV.u_i 2c_spi_slave.o_if_cs_en_n	stbio1_top.if_c s_en	DDS_STB IO1_CS_ EN_316
{ !pad_mux__path.scan_mode && ((pad_mux__path.mission_m ode == 1) pad_mux__path.debug_mod e pad_mux__path.ispu_debug _mode) }	!stbio1_top.dig_wrapper_LV.u_i 2c_spi_slave.o_if_cs_out_n	stbio1_top.if_c s_out	DDS_STB IO1_CS_ OUT_316
{ !pad_mux__path.scan_mode && ((pad_mux__path.mission_m ode == 1) pad_mux__path.ispu_debug _mode) && stbio1_top.dig_wrapper_LV.u _regmap_otp.SH_CKL_EN }	stbio1_top.dig_wrapper_LV.u_ masterblaze_ahb.sda_master_o ut	stbio1_top.if_s h_sda_en	DDS_STB IO1_SH_ SDA_EN_ 317
{ !pad_mux__path.scan_mode && !pad_mux__path.w_bist_mod e && !pad_mux__path.debug_mod e}	1'b0	stbio1_top.if_s h_sda_out	DDS_STB IO1_SH_ SDA_OU T_317
{ !pad_mux__path.scan_mode && ((pad_mux__path.mission_m ode == 1) pad_mux__path.ispu_debug _mode)}}	stbio1_top.if_sh_sda_in	stbio1_top.dig_ wrapper_LV.u_ masterblaze_a hb.sda_master _in	DDS_STB IO1_SH_ SDA_IN_ 317

{ !pad_mux__path.scan_mode && ((pad_mux__path.mission_m ode == 1) pad_mux__path.ispu_debug _mode) && stbio1_top.dig_wrapper_LV.u _regmap_otp.SH CKL_EN }	stbio1_top.dig_wrapper_LV.u _masterblaze_ahb.scl_master_o ut	stbio1_top.if_s h_ck_en	DDS_STB IO1_SH_ SCL_EN_ 317
{ !pad_mux__path.scan_mode && !pad_mux__path.w_bist_mod e && !pad_mux__path.debug_mod e}	1'b0	stbio1_top.if_s h_ck_out	DDS_STB IO1_SH_ SCL_OUT _317
{ !pad_mux__path.scan_mode && ((pad_mux__path.mission_m ode == 1) pad_mux__path.ispu_debug _mode)}	stbio1_top.if_sh_ck_in	stbio1_top.dig_ wrapper_LV.u_ _masterblaze_a hb.scl_master_ in	DDS_STB IO1_SH_ SCL_IN_3 17
{{(stbio1_top.dig_wrapper_LV _u_regmap_otp.gpio1_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_r egmap_otp.gpio1_out	stbio1_top.if_g pio1_out	DDS_STB IO1_GPIO 1_314
{{(stbio1_top.dig_wrapper_LV _u_regmap_otp.gpio1_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_r egmap_otp.gpio1_en	stbio1_top.if_g pio1_en	DDS_STB IO1_GPIO 1_EN_31 4
{{(stbio1_top.dig_wrapper_LV _u_regmap_otp.gpio1_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mo de == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.if_gpio1_in	stbio1_top.dig_ wrapper_LV.u_ _regmap_otp.gp io1_in_i	DDS_STB IO1_GPIO 1_IN_314

{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_out	stbio1_top.if_gpio2_out	DDS_STB IO1_GPIO 2_314
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_en	stbio1_top.if_gpio2_en	DDS_STB IO1_GPIO 2_EN_314
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.if_gpio2_in	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio2_in_i	DDS_STB IO1_GPIO 2_IN_314
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_out	stbio1_top.if_gpio3_out	DDS_STB IO1_GPIO 3_314
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_en	stbio1_top.if_gpio3_en	DDS_STB IO1_GPIO 3_EN_314
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) }	stbio1_top.if_gpio3_in	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio3_in_i	DDS_STB IO1_GPIO 3_IN_314
{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 0) && !pad_mux__path.scan_mode	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_out	stbio1_top.if_gpio4_out	DDS_STB IO1_GPIO 4_314

&& (pad_mux__path.mission_mode == 1))			
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}}	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_en	stbio1_top.if_gpio4_en	DDS_STB IO1_GPIO 4_EN_314
{{(stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_irq_sel == 0) && !pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1)}}	stbio1_top.if_gpio4_in	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio4_in_i	DDS_STB IO1_GPIO 4_IN_314
{!pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio0_out	stbio1_top.if_gpio0_out	DDS_STB IO1_GPIO 0_318
{!pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio0_en	stbio1_top.if_gpio0_en	DDS_STB IO1_GPIO 0_EN_318
{!pad_mux__path.scan_mode && (pad_mux__path.mission_mode == 1) && !pad_mux__path.ext_cap_en }	stbio1_top.if_gpio0_in	stbio1_top.dig_wrapper_LV.u_regmap_otp.gpio0_in_i	DDS_STB IO1_GPIO 0_IN_318

DCCK IP

Pin Configuration: The diagram shows the DCCK pin configuration. The DCCK_P pin is connected to the positive supply (1.8V), and the DCCK_N pin is connected to the negative supply (0V). The DCCK pin is also connected to the DCCK_P and DCCK_N pins.

DCCK_0 (0x0011) :

DCCK_CURRINJ:	Value
00	→ 25nA
01	→ 50nA
10	→ 100nA
11	→ 200nA

DCCK_1 (0x0012) :

DCCK_ISINK_IN:	Value
000	→ NC
001	→ INN_ECG
001	→ INP_ECG
001	→ INN_IMP
001	→ INP_IMP
001	→ INN_GSR
001	→ INP_GSR

DCCK_ISOURCE_IN:

Value	Source
000	→ NC
001	→ INN_ECG
001	→ INP_ECG
001	→ INN_IMP
001	→ INP_IMP
001	→ INN_GSR
001	→ INP_GSR

DCCK_P and DCCK_N Internal Circuitry: The diagram shows the internal structure of the DCCK_P and DCCK_N pins. The DCCK_P pin is connected to the positive supply (1.8V) and the DCCK_N pin is connected to the negative supply (0V). The DCCK_P pin is also connected to the DCCK_N pin. The DCCK_P pin is connected to the DCCK_N pin. The DCCK_P pin is connected to the DCCK_N pin.

If O_DCCK_P_i changes its value from '1' to '0' and DCCK_IRQ_MASK is set to '1', then irq_DCCK_IRQ_ST, output of u_regmap_otp, shall be set to '1'
[End]

12.1.3. Main Controller TOP

Instantiations

- inst_main_controller_fsm: main_controller_fsm
- inst_configuration_register_update: configuration_register_update
- inst_ECG_S_U: ECG_S_U
- inst_BIA_S_U: BIA_S_U
- inst_ppg_fsm: ppg_fsm
- inst_Digital_ramp_case: Digital_ramp_case
- inst_ALC_compensation: ALC_compensation
- inst_ADC_phases: ADC_phases
- inst_average_block: average_block
- inst_mainctrl_ahb_master: mainctrl_ahb_master
- inst_main_controller_clock_gatin_unit: main_controller_clock_gatin_unit

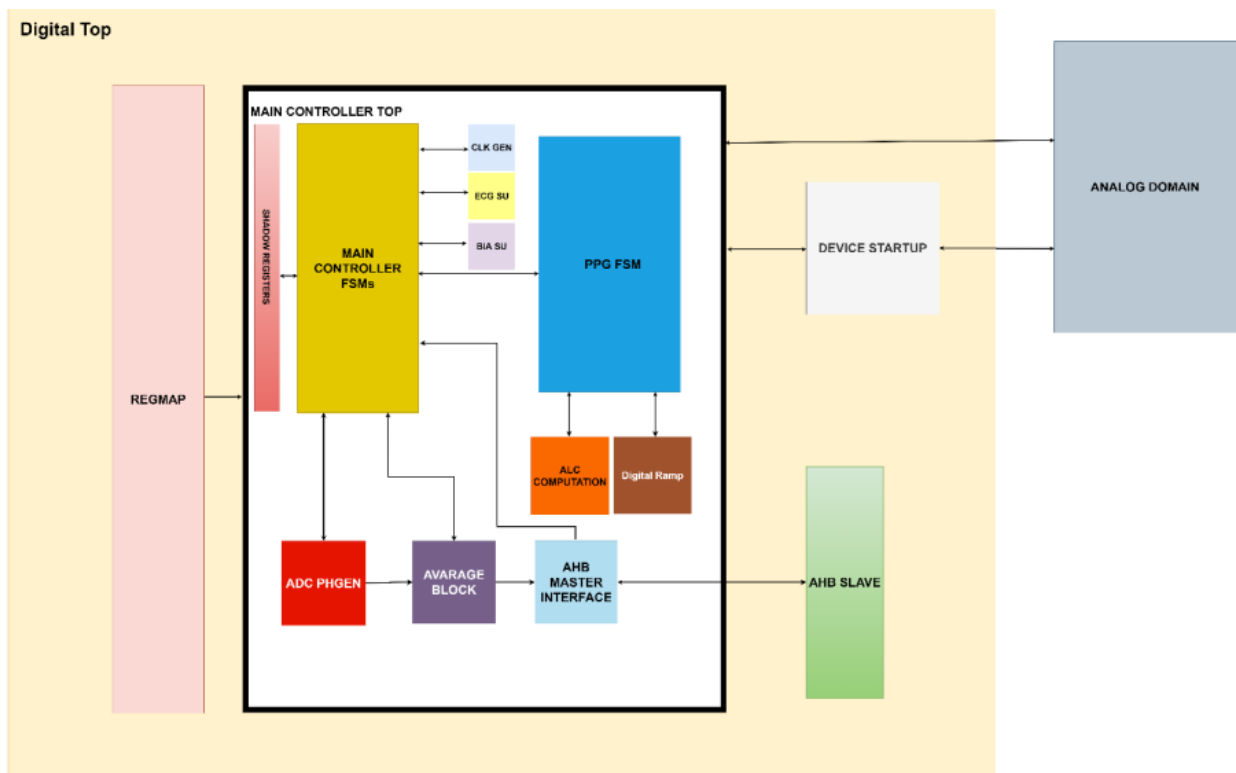


Figure 11: Main Controller Block Diagram

12.1.3.1. Main Controller TOP Description

The Main Controller is in charge of generating and receiving all the control signals for the Analog Domain, furthermore, performs DSP elaboration on ADC Output and writes it in different Address.

The IP is divided into different sub-blocks, all at the same hierarchy level and described in the next chapter. Together with the Analog Domain, this IP is interfaced with the FIFO Controller, the ADSP and the PMU, these last two will not be described in this document. The IP is divided into different sub-blocks, all at the same hierarchy level and described in the next chapter.

12.1.3.2. Main Controller TOP I/O List

Generic name	Type	Value	Description
ppg_frame_bit_top		4	
n_max_frame		12	
ppg_rx_channel_bit_top		3	
T_tx_led_set_bit_top		9	
T_tx_led_off_max_bit_top		11	
m_odr_max_bit_top		12	
mode_operation_n_bit_top		2	
idac_dc_ppg_frame_bit_top		6	
time_slot_sel_bit_top		3	
T_tx_startup_bit_top		8	
T_rx_set_bit_top		9	
Nmax_bit_top		4	
base_address_bit_top		12	
delta_bit_top		4	
gain_max_bit_top		4	
T1_max_bit_top		9	
T2_max_bit_top		5	
N_clk_max_bit_top		6	
n_average_ppg_max_bit_top		3	
rx_ch_gain_max_bit_top		4	
BTIA_ALC_register_bit_top		9	
n_channel_bit_top		4	
n_channel_adc_bit		4	

quokka_routine_bit_top		16	
vcm_bit_top		14	
gnc_cmm_top		15	
id_frame_max_bit_top		4	
data_in_size_bit_top		16	
ppg_start_up_bit_top		9	
data_out_size_bit_top		20	
n_average_ecg_bia_gsr_max_bit_top		8	
rx_ch_frame_n_bit_top		3	
curr_sel_bit_top		8	
bit_outab_sel_top		1	
n_average_ppg_max_value		128	
n_average_ecg_bia_gsr_max_value		16	
max_frame_repetition_top		2	
n_tx_channel		4	
ahb_base_address	[31:0]	32'h000C4000	
time_buffer_const_bit_top		10	
T1_max_bit_from_decoder_top		4	
T2_max_bit_from_decoder_top		2	
ppg_frame_bit_ALC_BTIA_top		12	
ecg_channel_bit		2	

Table 16: Main Controller Parameter

Port name	Direction	Type	Output Reset Value	Description
<i>i_clk_32kHz</i>	input			clk 32kHz from pmu
<i>i_clk_16MHz</i>	input			clk 16MHz from pmu
<i>i_clk_gsr_16M</i>	input			clk for gsr signal
<i>i_clk_ahb_master</i>	input			clock for ahb master
<i>i_rstn</i>	input			reset in 16MHz domain
<i>i_rstn_sync_32</i>	input			reset in 32kHz domain
<i>i_wave_velocity_mode</i>	input			enable of the wave velocity mode
<i>i_ispu_end_elab</i>	input			
<i>i_n_ppg_frame</i>	input	[ppg_frame_bit_top-1:0]		number of frame selected
<i>i_const_value_ioff</i>	input	[data_in_size_bit_top-1:0]		constant value from regmap used for ppg ioff calibration
<i>i_rx_ch_frame</i>	input	[n_max_frame-1:0]		ppg rx channel frame from regamp
<i>i_T_tx_led_set</i>	input	[n_max_frame-1:0]		tx led set time from regmap
<i>i_T_tx_led_off</i>	input	[n_max_frame-1:0]		tx led off time from regmap
<i>i_curr_sel_tx_frame_x</i>	input	[n_max_frame-1:0]		current injection
<i>i_tx1_outlab_sel_frame_x</i>	input	[n_max_frame-1:0]		led output selected from regmap for each frame
<i>i_tx2_outlab_sel_frame_x</i>	input	[n_max_frame-1:0]		led output selected from regmap for each frame
<i>i_tx3_outlab_sel_frame_x</i>	input	[n_max_frame-1:0]		led output selected from regmap for each frame
<i>i_tx4_outlab_sel_frame_x</i>	input	[n_max_frame-1:0]		led output selected from regmap for each frame
<i>i_idac_dc_ppg_frame</i>	input	[n_max_frame-1:0]		idac value for each frame selected from regmap
<i>i_average_ppg</i>	input	[n_max_frame-1:0]		average ppg for each frame selected from regmap
<i>i_gain_alc_start</i>	input	[gain_max_bit_top-1:0]		btia gain for alc from regmap
<i>i_gain_start</i>	input	[n_max_frame-1:0]		btia gain start for compensation from regmap
<i>i_Tx_idac_Vref_debug</i>	input	[Nmax_bit_top-1:0]		Idac value for debug from regmap

Port name	Direction	Type	Output Reset Value	Description
<i>i_ppg_start_up_time</i>	input	[ppg_start_up_decod_bit_top-1:0]		start up time for ppg from regmap
<i>i_en_txi</i>	input	[n_max_frame-1:0]		tx enabling for each frame from regmap
<i>i_frame_repetition</i>	input	[n_max_frame-1:0]		i frame repetition for each frame from regmap
<i>i_T_preset</i>	input	[1:0]		Preset time for digital ramp from regmap
<i>i_m_ecg</i>	input	[m_odr_max_bit_top-1:0]		m parameter for ecg from regmap
<i>i_m_bia</i>	input	[m_odr_max_bit_top-1:0]		m parameter for bia from regmap
<i>i_m_gsr</i>	input	[m_odr_max_bit_top-1:0]		m parameter for gsr from regmap
<i>i_m_ppg</i>	input	[m_odr_max_bit_top-1:0]		m parameter for ppg from regmap
<i>i_debug_adc_error_cap8_0_bypass</i>	input			error cap bypass for adc from regmap
<i>i_base_address_ecg0</i>	input	[base_address_bit_top-1:0]		base address for ecg0 in multiple fifo
<i>i_base_address_bia</i>	input	[base_address_bit_top-1:0]		base address for bia/ecg1-2 in multiple fifo
<i>i_base_address_gsr</i>	input	[base_address_bit_top-1:0]		base address for gsr in multiple fifo
<i>i_base_address_ppg</i>	input	[base_address_bit_top-1:0]		base address for ppg in multiple fifo
<i>i_frame_ppg_debug</i>	input	[ppg_frame_bit_top-1:0]		frame selection for ppg data debug
<i>i_en_buf_gpio1</i>	input			enable buffer on gpio1
<i>i_en_buf_gpio2</i>	input			enable buffer on gpio2
<i>i_en_biobuf</i>	input			enable biobuffer
<i>i_do_calibration</i>	input			enabling for adc calibration
<i>i_mode_operation</i>	input	[mode_operation_n_bit_top-1:0]		mode operation selection
<i>i_T_rx_set</i>	input	[T1_max_bit_from_decoder_top-1:0]		rx channel time set
<i>i_en_clkf</i>	input			en 16 MHz clock
<i>i_quokka_boot_end</i>	input			from quokka
<i>i_quokka_run</i>	input			from quokka

Port name	Direction	Type	Output Reset Value	Description
<i>i_clk_16MHz_ready</i>	input			(from analog circuit)
<i>i_imp_curinj_en</i>	input			enabling current injection for bia
<i>i_en_ldo_1v8</i>	input			enabling ldo 1v8
<i>i_debug_comp_rx_ck</i>	input			enabling debug for comp rx ck
<i>i_debug_gsr_on</i>	input			gsr on debug mode
<i>i_debug_gsr_curr_on</i>	input			gsr curr on debug mode
<i>i_alc_disabled</i>	input	[n_max_frame-1:0]		disabling alc bit for each frame
<i>i_en_cm_buf</i>	input			enabling common buffer
<i>i_en_rld</i>	input			enabling rld
<i>i_en_avg1</i>	input			enabling avg1
<i>i_en_avg2</i>	input			enabling avg1
<i>i_hc_en</i>	input			enabling health channel
<i>i_debug</i>	input			enabling debug mode
<i>i_imp_en</i>	input			enabling bia channel
<i>i_debug_en_tia</i>	input			enabling debug tia
<i>i_debug_en_buffer</i>	input			enabling debug for buffer
<i>i_debug_en_idac</i>	input			enabling debug for IDAC
<i>i_en_clk_chop10k</i>	input			enabling chopper clock
<i>i_debug_tx_preset</i>	input			enabling tx preset
<i>i_en_ldo_1v8_ext</i>	input			enabling external ldo 1v8
<i>i_clk_imp_10k_en</i>	input			enabling clock chopper for bia
<i>i_b_cap_idac</i>	input	[1:0]		enabling cap for idac
<i>i_adc_mux_bio_channel_debug</i>	input	[2:0]		enabling debug adc mux bio channel
<i>i_adc_mux_sel_debug</i>	input			enabling adc mux

Port name	Direction	Type	Output Reset Value	Description
<i>i_ispu_core_sw_rstn</i>	input			ispu core enabling
<i>i_T_TIAON</i>	input	[T1_max_bit_from_decoder_top-1:0]		TIA Time
<i>i_scan_mode</i>	input			scan mode enabling
<i>i_reset_debug_adc_data</i>	input			reset adc data debug
<i>i_start_adc_debug</i>	input			pilot adc start om debug
<i>i_en_ADC_debug</i>	input			enabling adc debug
<i>i_debug_adc_readout_bypass</i>	input			enabling adc readout bypass
<i>i_su_hlt_dly</i>	input			enabling start up delay for ecg
<i>i_multiple_fifo_sel</i>	input			enabling multiple fifo
<i>i_debug_calibration</i>	input			enabling debug calibraiton
<i>i_en_gsr_dc_cds</i>	input	[1:0]		enabling gsr cds mode
<i>i_ADC_EOC</i>	input			end of conversion from adc
<i>i_ecg0_sel</i>	input			select ecg0 channel
<i>i_bia_sel</i>	input			select bia channel
<i>i_gsr_sel</i>	input			select gsr channel
<i>i_ppg_sel</i>	input			select ppg channel
<i>i_ppg_ioff_calibration</i>	input			enabling ppg ioff calibration
<i>i_ecg1_ecg2_sel</i>	input			enabling ecg1 and ecg2 channel
<i>i_rst_async_ecg</i>	input			reset async for ecg
<i>i_rst_async_bia</i>	input			reset async for bia
<i>i_ppg_quokka_elab_sel</i>	input			enabling adsp elab for ppg
<i>i_data_ADC_out</i>	input	[data_in_size_bit_top-1:0]		(from analog circuit)
<i>i_time_slot_selection</i>	input	[time_slot_sel_bit_top-1:0]		odr selection from regmap

Port name	Direction	Type	Output Reset Value	Description
<i>i_Nmax_to_reg</i>	input	[Nmax_bit_top-1:0]		Max value for digital ramp
<i>i_nstart_case</i>	input	[Nmax_bit_top-1:0]		n_start value for frist sterp
<i>i_debug_rx_ch</i>	input	[rx_ch_frame_n_bit_top-1:0]		enabling debug for rx channel
<i>i_T_delta1_to_reg</i>	input	[delta_bit_top-1:0]		delta 1 time for digital ramp
<i>i_gain_max_set</i>	input	[gain_max_bit_top-1:0]		gain max for alc
<i>i_BTIA_ALC_debug</i>	input	[BTIA_ALC_register_bit_top-1:0]		BTIA ALC Debug enable
<i>i_comp_rx_out</i>	input			input from analog comparator
<i>i_adc_error_cap8_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap7_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap6_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap5_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap4_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap3_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap2_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap1_debug</i>	input	[6:0]		adc cap error debug
<i>i_adc_error_cap0_debug</i>	input	[6:0]		adc cap error debug
<i>i_BIDAC_debug_register</i>	input	[IDAC_offset-1:0]		
<i>i_sleep_master</i>	input			
<i>i_lpm</i>	input	[1:0]		low power mode for ldo
<i>i_mpm</i>	input	[1:0]		medium power mode for ldo
<i>i_hpm</i>	input	[1:0]		high power mode for ldo
<i>i_dis_rc_viref</i>	input			dis rc viref
<i>i_T1</i>	input	[T1_max_bit_from_decoder_top-1:0]		4 to 10 us=160nclk

Port name	Direction	Type	Output Reset Value	Description
<i>i_T2</i>	input	[T2_max_bit_from_decoder_top-1:0]		0.5 to 2 us~32
<i>i_N_clk_ppg</i>	input	[N_clk_max_bit_top-1:0]		Sample time for ppg
<i>i_N_clk_ecg</i>	input	[N_clk_max_bit_top-1:0]		Sample time for ecg
<i>i_N_clk_cal</i>	input	[N_clk_max_bit_top-1:0]		Sample time for adc calibration
<i>scan_enable</i>	input			scan enable for scan mode
<i>i_timer_buffer_config</i>	input	[1:0]		timer buffer configuration
<i>i_ppg_alc_frame_save</i>	input	[ppg_frame_bit_top-1:0]		alc data frame to save
<i>i_reset_ppg_ioff_calib</i>	input			
HRDATA	input	[31:0]		AHB Protocol signals
HREADY	input			AHB Protocol signals
HRESP	input	[1:0]		AHB Protocol signals
HGRANT	input			AHB Protocol signals
SCANENABLE	input			Scan Test Mode Enbl
SCANINHCLK	input			Scan Chain Input
SCANOUTHCLK	output			Scan Chain Output
HSEL	output		0	AHB Protocol signals
HADDR	output	[31:0]	0	AHB Protocol signals
HTRANS	output	[1:0]	0	AHB Protocol signals
HWRITE	output		0	AHB Protocol signals
HSIZE	output	[2:0]	0	AHB Protocol signals
HBURST	output	[2:0]	0	AHB Protocol signals
HPROT	output	[3:0]	0	AHB Protocol signals
HWDATA	output	[31:0]	0	AHB Protocol signals

Port name	Direction	Type	Output Reset Value	Description
HBUSREQ	output		0	AHB Protocol signals
HLOCK	output		0	AHB Protocol signals
o_bio_clock_div	output		0	output for clock chopper
o_BIDAC	output	[IDAC_offset-1:0]	0	IDAC for top
o_rst_imp	output		1	bia start up rst signal
o_rsti_imp	output		1	bia start up rsti signal
o_sug_imp	output		1	bia start up sug channel
o_pd_imp	output		1	bia start up pd channel
o_start_operative_16MHz	output		0	start operative 16MHz
o_su_hc2	output		1	ecg start up hc2 signal
o_rst_hc2	output		1	ecg start up rst signal
o_rsti_hc2	output		1	ecg start up rsti signal
o_pd_hc2	output		1	ecg start up pd signal
o_reset_reg_rst	output		1	end of reset async phase
o_reset_reg_rst_imp_async	output		1	end of reset async phase
o_imp_curinj_pd	output		0	bia current injection output
o_Txi_IDAC_Vref	output	[Nmax_bit_top-1:0]	0	TX IDAC VREF TO analog domain
o_comp_rx_ck	output		0	output for comparator debug
o_BTIA_gain	output	[gain_max_bit_top-1:0]	0	BTIA gain output to analog domain
o_data_out_debug_adc	output	[15:0]	0	o data output for adc debug
o_en_mask_ldo_pdclkf_reg	output		1	mask for en_ldo and pdf
o_pd_clkf	output		1	pd_clkf
o_trim_ldo_d	output	[1:0]	3	trim ldo value
o_BDC_PPG	output	[idac_dc_ppg_frame_bit_top-1:0]	0	idac dc ppg frame output

Port name	Direction	Type	Output Reset Value	Description
o_adc_mux_bio_channel	output	[2:0]	0	adc mux bio channel
o_adc_mux_sel	output		0	adc mux sel output
o_error_time_slot	output		0	error time slot irq
o_curr_sel_tx_frame_x	output	[curr_sel_bit_top-1:0]	0	current selection for tx channel output
o_tx1_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	0	tx outlab sel frame x
o_tx2_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	0	tx outlab sel frame x
o_tx3_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	0	tx outlab sel frame x
o_tx4_outlab_sel_frame_x	output	[bit_outab_sel_top-1:0]	0	tx outlab sel frame x
o_tx_preset	output		0	tx prese to analog domain
o_en_tx	output	[n_tx_channel-1:0]	0	tx enable to analog domain
o_PD_TIA	output		1	enabling tia to analog domain
o_PD_PGA	output		1	enabling buffer to analog domain
o_PD_IDAC	output		1	enabling idac to analog domain
o_gsr_curr_on	output		0	gsr curr on pilot to analog domain
o_gsr_on	output		0	grs on on pilot to analog domain
o_ADC_en	output		0	ADC enabling
o_half_charge_idc	output		0	
o_ADC_start	output		0	ADC Start sampling
o_N_step2_to_reg	output	[rx_ch_gain_limit-1:0]	0	N_Step2 calibration value
o_adc_test_cap	output	[9:0]	0	adc test cap
o_adc_error_cap9	output	[7:0]	0	adc calibration result
o_adc_error_cap8	output	[6:0]	0	adc calibration result
o_adc_error_cap7	output	[6:0]	0	adc calibration result
o_adc_error_cap6	output	[6:0]	0	adc calibration result

Port name	Direction	Type	Output Reset Value	Description
o_adc_error_cap5	output	[6:0]	0	adc calibration result
o_adc_error_cap4	output	[6:0]	0	adc calibration result
o_adc_error_cap3	output	[6:0]	0	adc calibration result
o_adc_error_cap2	output	[6:0]	0	adc calibration result
o_adc_error_cap1	output	[6:0]	0	adc calibration result
o_adc_error_cap0	output	[6:0]	0	adc calibration result
o_pd_cm_buf	output		1	power down common buffer
o_pd_rld	output		1	power down rld
o_pd_avg1	output		1	power down avg1
o_pd_avg2	output		1	power down avg2
o_mc_ahb_clk_en	output		0	enable for master ahb clock
o_main_controller_device_status_register	output	[2:0]	0	device status register
o_main_controller_elab_status_register	output	[3:0]	0	device status register
o_en_ido_1v8	output		0	enable ldo
o_en_bufxbio	output		0	enable biobuffer to analog domain
o_rx_ch_frame	output	[ppg_rx_channel_bit_top-1:0]	0	selected rx ch frame to analog domain
o_ido_ext	output		0	enable external ldo
o_ppg_status_register	output	[n_state_bit-1:0]	0	ppg state status register
o_b_cap_idac	output	[1:0]	0	b cap idac to analog domain
o_ispu_irq_ctrl	output	[3:0]	0	ispu irq
o_irq_start_measurement	output		0	irq start measurement to GPIO
o_end_calibration	output		1	end calibration to analog domain
o_alc_data	output	[data_out_size_bit_top+2:0]	0	alc data to regmap
o_ppg_data_raw	output	[data_out_size_bit_top+2:0]	0	ppg raw data to regmap

Port name	Direction	Type	Output Reset Value	Description
o_pd_buf_gpio1	output		1	power down for buffer gpio1
o_pd_buf_gpio2	output		1	power down for buffer gpio2
o_adc_data_debug_alc_1	output	[frame_ppg_debug_alc_top-1:0]	0	alc 1 registers data debug
o_adc_data_debug_alc_2	output	[frame_ppg_debug_alc_top-1:0]	0	alc 2 registers data debug
o_adc_data_debug_ppg_raw	output	[frame_ppg_debug_ppg_raw_top-1:0]	0	ppg raw registers data debug
o_en_lead_off	output		0	mask lead off
o_mc_busy	output		0	main controller is working
o_en_ck_m_p_q	output		0	clock m and q bia
o_en_ck_chop_imp	output		0	clock chopper bia
o_saturation_flag	output	[n_max_frame-1:0]	0	ppg saturation flag
o_en_ck_hc2_10k	output		0	clock chipper ecg
o_en_clk_16MHz_sleep	output		1	sleep signal for pmu
o_gsr_clk_en_16M	output		0	enable clock for gsr channel

Table 17: Main Controller I/O List

12.1.3.3. Main Controller Peculiar Requirements

Main Controller manage all the Analog domain, so all its requirements shall be considered as High Priority. To check correctly the requirements, it's important to cover all the different phase described, partially, in [chapter 12](#) and its IPOS.

12.2. IPs

12.2.1. ADSP integration

Generic name	Type	Value	TOP CONNECTION
ADDRWIDTH		12	

Table 18: ADSP Parameter Table

Port name	Direction	Type	TOP CONNECTION
stmc_clk	input	wire	u_pmu.tst_clk_16m
HCLK	input	wire	u_pmu.clk_stmc
HRESETn	input	wire	u_pmu.resetn_stmc
HSELS	input	wire	u_xbar_afe.
HADDRS	input	wire [ADDRWIDTH-1:0]	u_xbar_afe.haddr_M1
HTRANS	input	wire [1:0]	u_xbar_afe.htrans_M1
HSIZES	input	wire [2:0]	u_xbar_afe.
HWRITES	input	wire	u_xbar_afe.
HREADY	input	wire	u_xbar_afe.
HWDATAS	input	wire [31:0]	u_xbar_afe.
HREADYOUTS	output	wire	u_xbar_afe.
HRESPS	output	wire	u_xbar_afe.
HRDATAS	output	wire [31:0]	u_xbar_afe.
HRDATAM	input	wire [31:0]	
HREADYM	input	wire	
HRESPM	input	wire [1:0]	
HGRANTM	input	wire	
HSELM	output	wire	
HADDRM	output	wire [31:0]	
HTRANS	output	wire [1:0]	
HWRITE	output	wire	
HSIZE	output	wire [2:0]	

HBURSTM	output	wire [2:0]	
HPROTM	output	wire [3:0]	
HWDATAM	output	wire [31:0]	
HBUSREQM	output	wire	
HLOCKM	output	wire	
stmc_data_CSN	output	wire	
stmc_data_WEN	output	wire	
stmc_data_A	output	wire [8:0]	
stmc_data_D	output	wire [63:0]	
stmc_data_M	output	wire [63:0]	
stmc_data_Q	input	wire [63:0]	
stmc_rom_CLK	output	wire	
stmc_rom_CSN	output	wire	
stmc_rom_A	output	wire [10:0]	
stmc_rom_Q	input	wire [22:0]	
quokka_run	output	wire	u_pmu.adsp_run main_controller_top.i_quokka_r un
quokka_boot_end	output	wire	main_controller_top.i_quokka_b oot_end
ahb_master_clk_en	output	wire	U_pmu.clk_en_s1
scan_mode	input	wire	
scan_en	input	wire	
scan_clk	input	wire	
scan_rst_n	input	wire	

Table 19: ADSP I/O List

12.2.1.1. ADSP Peculiar Requirements

[DDS_STBIO1_0010] Definition:

ADSP is responsible for managing signals for the OTP

memory H9A_FU_OTPHV_85AL05_23A_MIM5_6ML_ALUCU/2.6.b-00, which are required for its reading, programming and testing procedures.

The selected OTP memory has a data bit cut size of 640 bits with an 8-bit word size.

Each data bit is physically redundant to optimize memory yield. Consequently, the OTP memory is organized into 2 banks, each containing 80 addresses of 8-bit size.

[End]

12.2.1.1.1. OTP I/O List

[DDS_STBIO1_0011] Definition:

Pin Name	Function
PDN	Power down command. Memory is in low power mode when PDN is low.
A<6:0>	A<6:0> is the word address for read or programming operation
BANKEN<1:0>	Bank selection bits. Used to program and read each bank separately. Or to activate redundancy when reading (2 banks read simultaneously, correction enabled)
BITSEL<2:0>	To define the bit to be programmed in the 8 bit word. Decimal value of (BITSEL<2:0>)
SELECTWLBL	To select antifuse block to be programmed
PRG	To select programming mode (active high)
FUSE	Fuse command to program bit cell.
TREAD<3:0>	For read condition trimming (sense current and read voltage).
VREF	Analog 1.25V+/-5% bandgap reference voltage
IREF	Pin requiring external current sink of 10uA+/-30%

Table 20: OTP I/O List

[End]

12.2.1.1.2. Requirements

[DDS_STBIO1_0012] Definition:

There are three routine that the ADSP use to interact with the OTP memory, which are the Write, Boot and Test routine. In ADSP registers map there are 3 specific bits used to trigger these OTP routines:

AHB ADDRES S	AHB REGISTER	7	6	5	4	3	2	1	0
C1E02	CONTROL_ A	/	/	/	/	OTP_TES T	OTP_WRI T E	OTP_BOO T	SOFT_RESE T

- OTP BOOT Bit used to trigger Boot Routine.
- OTP WRITE Bit used to trigger Write routine.
- OTP TEST Bit used to trigger Test routine.

The same register also contain a SOFT_RESET bit used to trigger the corresponding soft reset routine.

[End]

[DDS_STBIO1_0013] Definition:

To move OTP signal, ADSP shall writes new value in a regbank by AHB protocol.

Register Name	Register Title	Address Offset	Reset value	Bits	Field name	Access	Req ID
OTP_PDN	OTP_PDN register	0xc90000	0x0000	0	OTP_PDN	RW	DDS_STBIO1_1000
OTP_A	OTP_A register	0xc9004	0x0000	6:0	OTP_A	RW	DDS_STBIO1_1001
OTP_BANKEN	OTP_BANKEN register	0xc9008	0x0000	1:0	OTP_BANKEN	RW	DDS_STBIO1_1002
OTP_BITSEL	OTP_BITSEL register	0xc900C	0x0000	2:0	OTP_BITSEL	RW	DDS_STBIO1_1003
OTP_SELECT_WLBL	OTP_SELECT_WLBL register	0xc9010	0x0000	0	OTP_SELECT_WLBL	RW	DDS_STBIO1_1004
OTP_PRG	OTP_PRG register	0xc9014	0x0000	0	OTP_PRG	RW	DDS_STBIO1_1005
OTP_FUSE	OTP_FUSE register	0xc9018	0x0000	0	OTP_FUSE	RW	DDS_STBIO1_1006
OTP_TREAD	OTP_TREAD register	0xc9020	0x0800	3:0	OTP_TREAD	RW	DDS_STBIO1_1007
OTP_D	OTP_D register	0xc9024	0x0000	7:0	OTP_D	R	DDS_STBIO1_1008
OTP_IREF_WAKEUP	OTP_IREF_WAKEUP register	0xc9028	0x0000	0	OTP_IREF_WAKEUP	RW	DDS_STBIO1_1009

OTP_D_TEST	OTP_D_TEST register	0xc902C	0x0000	7:0	OTP_D_TEST	RW	DDS_STBIO1_1010
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[Vpriority High] [End]

[DDS_STBIO1_0014] Requirement:

The BOOT routine shall start when the OTP_BOOT bit rises and shall complete when quokka_boot_end rises.

[Vpriority High] [End]

[DDS_STBIO1_0015] Requirement:

During OTP boot operation, the ADSP shall copy the content of the OTP memory into the OTP registers, following the procedure to access OTP memory

[Vpriority High] [End]

[DDS_STBIO1_0016] Requirement:

During BOOT routine, ADSP shall first read the last location of the OTP memory, at address 0x7F. If the content is 0xFF, called "lifecycle" byte, then the READ operation proceeds from address 0 to address 0x2F; otherwise, ADSP shall not perform any further reads.

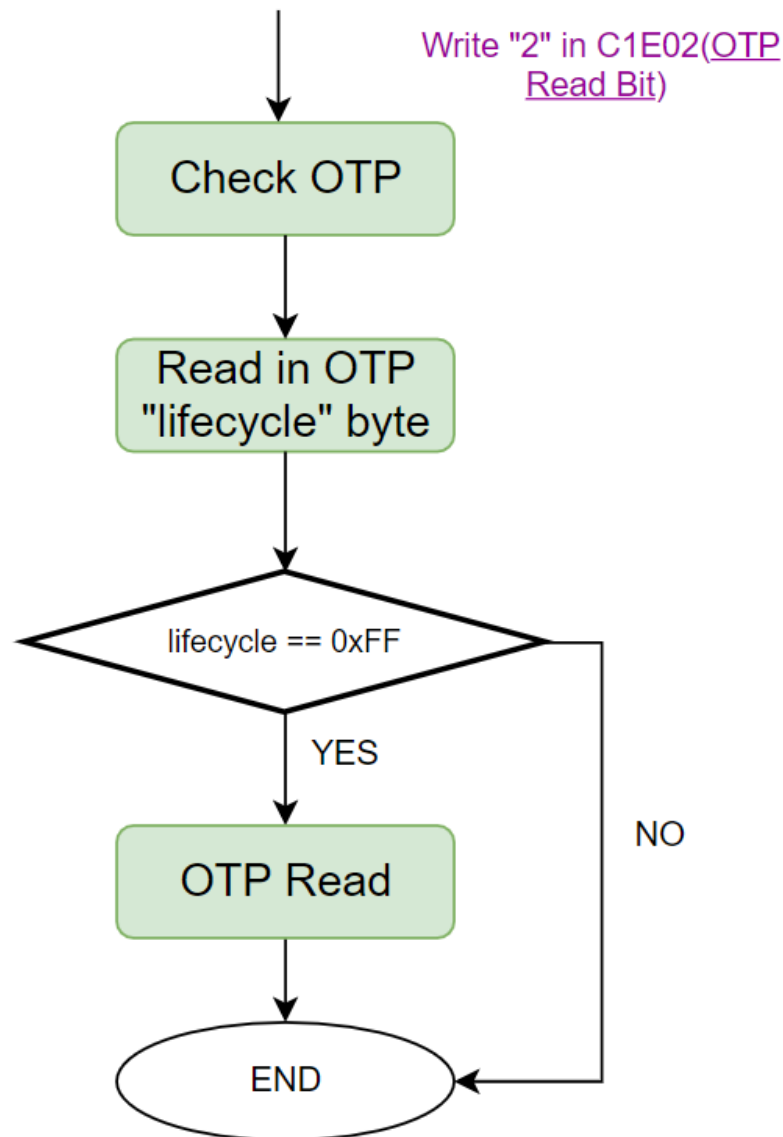


Figure 12: Boot Routine

[Vpriority High] [End]

[DDS_STBIO1_0017] Requirement:
To perform a BOOT routine, OTP_BOOT bit shall be written when quokka_run is low.
[Vpriority High] [End]

[DDS_STBIO1_0018] Requirement:
The WRITE routine shall start when OTP Write Bit rises and completes when quokka_run falls.
[Vpriority High] [End]

[DDS_STBIO1_0019] Requirement:

During WRITE routine ADSP shall copy the content of the OTP regmap into the OTP memory, following the procedure to burn the fuses as expressed in the OTP datasheet

[Vpriority High] [End]

[DDS_STBIO1_0020] Requirement:

During WRITE routine, ADSP shall allow copying all OTP Registers into the OTP Memory only if the location 0x24 of OTP Memory (OTP[24] in the following diagram) is not set to 0xFF.

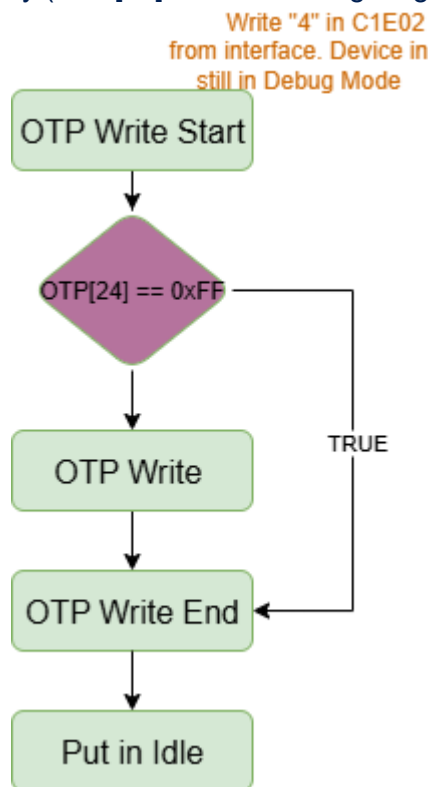


Figure 13: Write Routine Start

[Vpriority High] [End]

[DDS_STBIO1_0021] Requirement:

During the WRITE routine and after copying all OTP Registers into the OTP Memory, ADSP shall compare each value inside the OTP memory with the corresponding value in OTP Registers to check if the WRITE routine has been succesfull. In that case ADSP shall write 0xFF into the last location of the OTP memory.

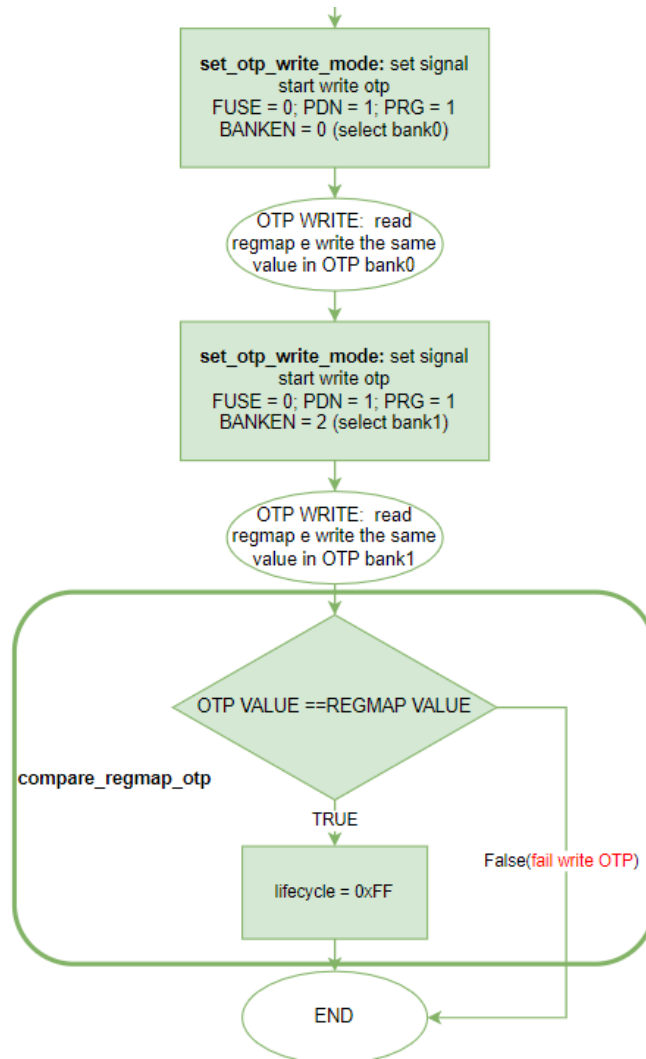


Figure 14: WRITE routine and TEST regmap

[Vpriority High] [End]

[DDS_STBIO1_0022] Requirement:
OTP_WRITE, OTP_BOOT and OTP_TEST bits shall be cleared by ADSP at the end of their respective operation

[Vpriority High] [End]

[DDS_STBIO1_0023] Requirement:
A new operation shall start only when quokka_run is low, and one between OTP_TEST bit, OTP_WRITE bit or OTP_BOOT bit is written to 1

[Vpriority High] [End]

[DDS_STBIO1_0024] Requirement:

OTP_IREF_WAKEUP shall be set to one by an external I2C/SPI to AHB write before initiating a new OTP_TEST or OTP_WRITE operation

[Vpriority High] [End]

[DDS_STBIO1_0025] Requirement:

OTP_IREF_WAKEUP shall be set to zero by an external I2C/SPI to AHB write at the end of OTP_TEST or OTP_WRITE operation

[Vpriority High] [End]

[DDS_STBIO1_0026] Requirement:

After startup phase, once started with an operation, the IP shall ignore any further start incoming transactions.

[Vpriority High] [End]

[DDS_STBIO1_0027] Requirement:

The TEST routine shall start when OTP Test Bit rises and completes when quokka_run falls.

[Vpriority High] [End]

[DDS_STBIO1_0028] Requirement:

The TEST routines shall allow reading an 8-bit word on OTP of a specific address written in OTP_A register, and write this value in OTP_D_TEST register.

[Vpriority High] [End]

[DDS_STBIO1_0029] Definition:

Timing parameter	Value
ts_pdnr_tread	3 μ s
ts_pdnr_fuse	3 μ s
th_pdnr_selectwbl	8 μ s
ts_selectwblr_a	6 μ s
ts_selectwblr_prg	5 μ s
ts_fuser_selectwbl	5 μ s
tfuse_high	100 μ s
th_fusef_selectwbl	5 μ s
t_selectwbl_high_read	10 μ s
td_selectwblr_dout	8 μ s
tselectwbl_low	7 μ s
th_selectwblf_a	2*clk_period

ts_pdnf_selectwbl	500 ns
th_pdnf_tread	150 ns
th_pdnf_fuse	150 ns
th_selectwblf_prg	2*clk_period
clk_period	31ns
n_mirror_reg	6

Table 21 Chosen OTP Timings

[End]

[DDS_STBIO1_0030] Definition:

Timing parameter	Value
t_state1_cycles	ts_pdnr_tread_fuse_cycles + th_pdnr_selectwbl_cycles
t_s1_int1_cycles	ts_pdnr_tread_fuse_cycles
t_s1_int2_cycles	t_state1_cycles - ts_selectwblr_a_cycles
t_s1_int3_cycles	t_state1_cycles - ts_selectwblr_prg_cycles
t_state2_write_cycles	ts_fuser_selectwbl_cycles + tfuse_high_cycles + th_fusef_selectwbl_cycles
t_s2_int1_write_cycles	ts_fuser_selectwbl_cycles
t_s2_int2_write_cycles	t_state2_write_cycles - th_fusef_selectwbl_cycles
t_state2_read_cycles	t_selectwbl_high_read_cycles
t_s2_int1_read_cycles	td_selectwblr_dout_cycles
t_state3_cycles	tselectwbl_low_cycles
t_s3_int1_cycles	th_selectwblf_a_cycles
t_s3_int2_cycles	tselectwbl_low_cycles - ts_selectwblr_a_cycles
t_state4_cycles	ts_pdnf_selectwbl_cycles + th_pdnf_tread_fuse_cycles
t_s4_int1_cycles	th_selectwblf_a_cycles
t_s4_int2_cycles	th_selectwblf_prg_cycles
t_s4_int3_cycles	ts_pdnf_selectwbl_cycles

Table 22 Operation phases timings

[End]

[DDS_STBIO1_0031] Definition:



Figure 15 : OTP Reading timing diagram

[End]

[DDS_STBIO1_0032] Requirement:

After each Boot routine performed by the ADSP, a value of 0x04 shall be written once to address 0x0CA00C to ensure correct operation of the AHB transactions.

[Vpriority High] [End]

[DDS_STBIO1_0033] Requirement:

The Soft Reset routine shall start when SOFT_RESET Bit (in CONTROL_A register) rises and completes when quokka_run falls.

[Vpriority High] [End]

[DDS_STBIO1_0034] Requirement:

During the Soft Reset routine, ADSP shall write the ADSP RAM with the value 0. Only the first 3.5Kbyte of ADSP RAM shall be cleared, from address 0x000 to address 0xDFF

[Vpriority High] [End]

[DDS_STBIO1_0035] Requirement:

If during the start of a new operation, more than one bit is written into CONTROL_A reg in the same write operation, ADSP shall execute only the routine at higher priority and discard the others.

Priority is listed below, with first routine at higher priority:

1. SOFT RESET routine
2. OTP WRITE routine
3. OTP TEST routine
4. OTP BOOT routine

[Vpriority High] [End]

[DDS_STBIO1_0036] Requirement:

If bit OTP_PRG10[31] in OTP Registers, called write_protection_en, is set to 1, and STBIO1 is not in debug_mode, register map shall refuse any incoming write request if the transaction address is within the OTP Registers space (from 0xCD000 to 0xCD02F).

[End]

[DDS_STBIO1_0037] Definition:

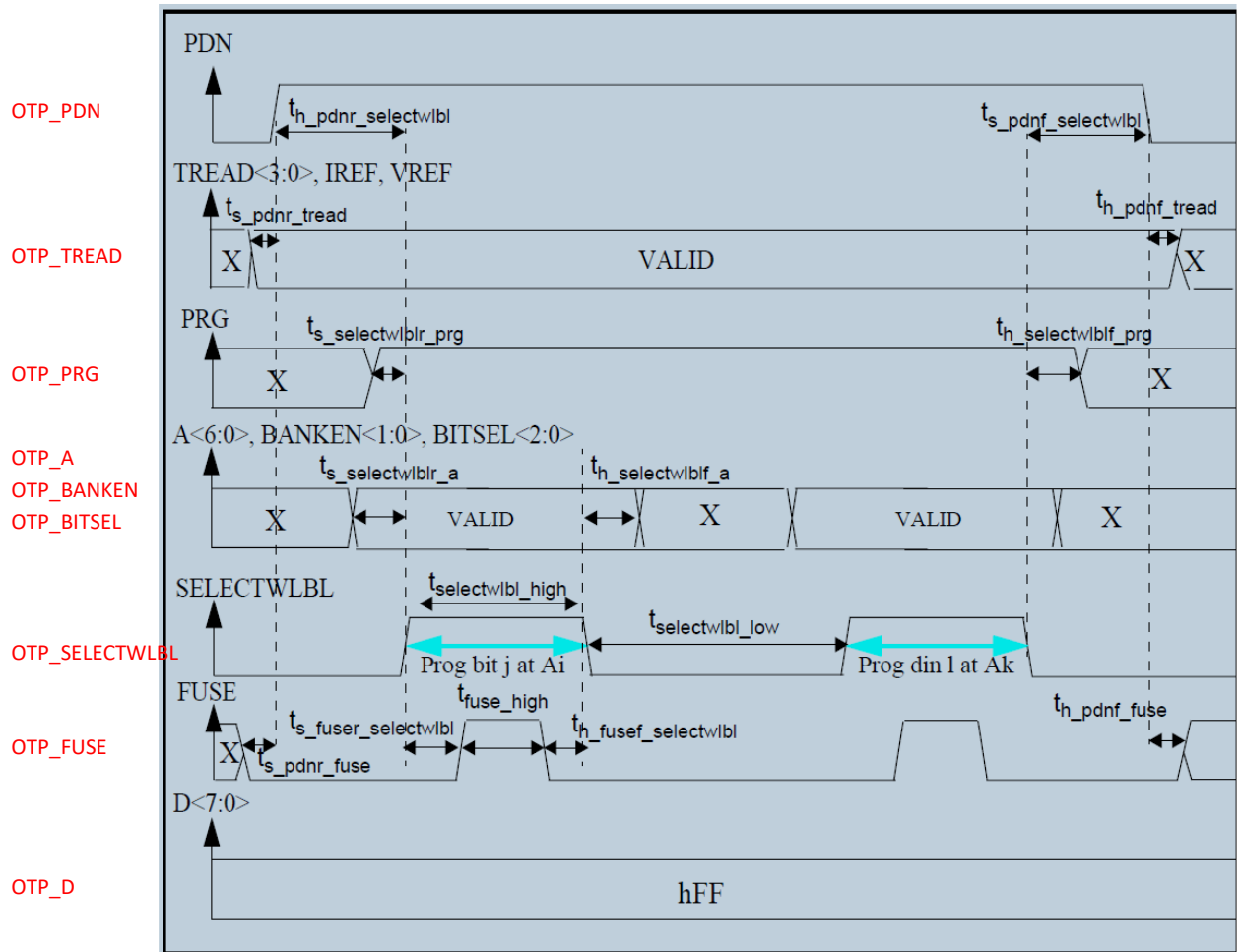


Figure 16: OTP Writing timing diagram

[End]

12.2.2. I2C_SPI_AHB integration

Generic name	Type	Value	TOP CONNECTION	
fifo_dpt		8	FIFO depth. Suggested val 8	[DDS_STBIO1_8100]
rd_to_max_val		6000	Time available between double byte write to communicate the number of bytes to be read and the following read. If timer expires the operation is aborted and proper flag is activated. Suggested val. 1200	[DDS_STBIO1_8101]
i2c_hs_mode		1'b0	To activate HS I2C	[DDS_STBIO1_8102]
i2c_master_code		8'h00	Master code for HS I2C	[DDS_STBIO1_8103]
ser_mode		"i2c_spi"	To choose if the interface is SPI only, I2C only or both. Suggested : i2c_spi	[DDS_STBIO1_8104]
ip_version		8'haa	IP version	[DDS_STBIO1_8105]
i2c_dev_id		7'h5f	I2C device address	[DDS_STBIO1_8106]

Table 23: I2C/SPI AHB Parameters

Port name	Direction	Type	TOP CONNECTION
rst_n	input		POR
i_if_cs	input		CS for SPI interface
o_if_cs_tud	output		Control for CS PAD. 1 in scan mode 0 otherwise
o_if_cs_en_n	output		Set to 0 for CS pad id bidir pad is used. Could be inverted depending on the PAD
o_if_cs_out_n	output		Set to 0 for CS pad id bidir pad is used. Could be inverted depending on the PAD
i_if_ck	input		Interface clock

i_if_sda_sdi_in	input		SDA/SDI from pad
o_if_sda_sdi_en_n	output		SDA/SDI pad enable
o_if_sda_sdi_out	output		SDA/SDI to pad
o_ctrl_reg_7	output	[7:0]	8 configuration bit set by local register (addr. 0x5e)
i_ext_reg_dt	input	[7:0]	Data from external register block (if present)
o_ext_reg_wr_add	output	[7:0]	Write address to external register block (if present)
o_ext_reg_rd_add	output	[7:0]	Read address to external register block (if present)
o_ext_reg_dt	output	[7:0]	Data to external register block (if present)
o_ext_reg_ck_wr	output		Write enable for external register block (if present)
o_ext_reg_we	output		Write direct enable for external register block (if present)
rst_16m_n			Sw reset for AHB domain
sys_16m_clk	input		AHB clock
o_hselm	output		AHB_M
o_haddrm	output	[31:0]	AHB_M
o_htransm	output	[1:0]	AHB_M
o_hmasterm	output	[3:0]	AHB_M
o_hwritem	output		AHB_M
o_hsizem	output	[2:0]	AHB_M
o_hmastlockm	output		AHB_M
o_hwdatam	output	[31:0]	AHB_M
o_hburstm	output	[2:0]	AHB_M
o_hprotm	output	[3:0]	AHB_M
i_hrdatam	input	[31:0]	AHB_M
i_hreadym	input		AHB_M
i_hresp	input		AHB_M
o_gt_ck_on	output		Pulse on interface clock to inform system that AHB clock must be switched on

o_gt_ck_off	output		Pulse on AHB clock to inform system that AHB clock can be switched off
o_ahb_error	output		Error on AHB side. Active if bit 1 or of register at address 0x4f are active.
o_rd_byte_nr_err	output		Active when
o_req_error	output		Active when number of byte is missing for read or timeout between double yte write and read is expired
scan_mode	input		Scan mode
scan_enable	input		Scan enable
scan_clk	input		Scan clk
scan_rst	input		Scan reset

Table 24: I2C/SPI AHB I/O List

12.2.2.1. I2C_SPI_AHB Peculiar Requirements

[DDS_STBIO1_8000] Requirement: I2C interface shall support all the specified modalities (Standard, FM, FM+) and, if configured, HS is also supported.

[End]

[DDS_STBIO1_8001] Requirement: Clock frequency ratio between system (AHB) and I2C/SPI interfaces domains shall be at least 2 times.

[TO: IPOS_I2C_SPI_AHB_0006]

[Vpriority High]

[End]

[DDS_STBIO1_8002] Requirement: The IP has a serial interface (I2C or SPI) and a parallel bus interface AHB.

[TO: IPOS_I2C_SPI_AHB_0007]

[End]

[DDS_STBIO1_8003] Requirement: An I2C/SPI write access to the address 0x60-0x67 shall be translated in an AHB write access to the configured AHB address (contained in registers SAM_ADDR_BYTE0/1/2/3, address 0x50/0x51/0x52/0x53).

[TO: IPOS_I2C_SPI_AHB_0009]

[Vpriority High]

[End]

[DDS_STBIO1_8004] Requirement: A double byte write access to the address 0x68-0x6f (with data that is the number of bytes that the master wants to read) shall start the data preload from AHB in order to have the internal fifo not empty when the read request will be done by the master.

[TO: IPOS_I2C_SPI_AHB_0010]

[Vpriority High]

[End]

[DDS_STBIO1_8005] Requirement: An I2C/SPI read access to the address 0x70-0x7f shall be translated in an AHB read access to the configured AHB address (contained in registers SAM_ADDR_BYTE0/1/2/3, address 0x50/0x51/0x52/0x53).

[TO: IPOS_I2C_SPI_AHB_0011]

[Vpriority High]

[End]

[DDS_STBIO1_8006] Requirement: On AHB interface, data size shall be limited to BYTE and WORD.

[TO: IPOS_I2C_SPI_AHB_0361]

[End]

[DDS_STBIO1_8007] Requirement: On AHB interface, BURST shall not be supported as well as LOCKED transfers.

[TO: IPOS_I2C_SPI_AHB_0362]
[End]

12.2.3. SensorHub integration

Generic name	Type	Value	TOP CONNECTION
ADDRWIDTH			
EBMwriteAddr			
TAGid			

Table 25: Sensor Hub Parameter

Port name	Direction	Type	TOP CONNECTION
scan_mode	input		
scan_enable	input		
scan_rstn	input		
HCLKM	input		
HCLKS	input		
sys_clk	input		
HRESETn	input		
DEBUG_MODE_i	input		
scl_master_in	input		
sda_master_in	input		
scl_master_out	output		
sda_master_out	output		
dvalid_masterblaze_ext	input		
dvalid_masterblaze_ext_enable	input		
sleep_master	output		
master_end_op_pulse	output		
SH_SMARTFIFO_OCCUPATION	Input		
fifo_mode	input		
fifo_offset	input	[11:0]	
HSELS	input		
HADDRS	input	[ADDRWIDTH-1:0]	

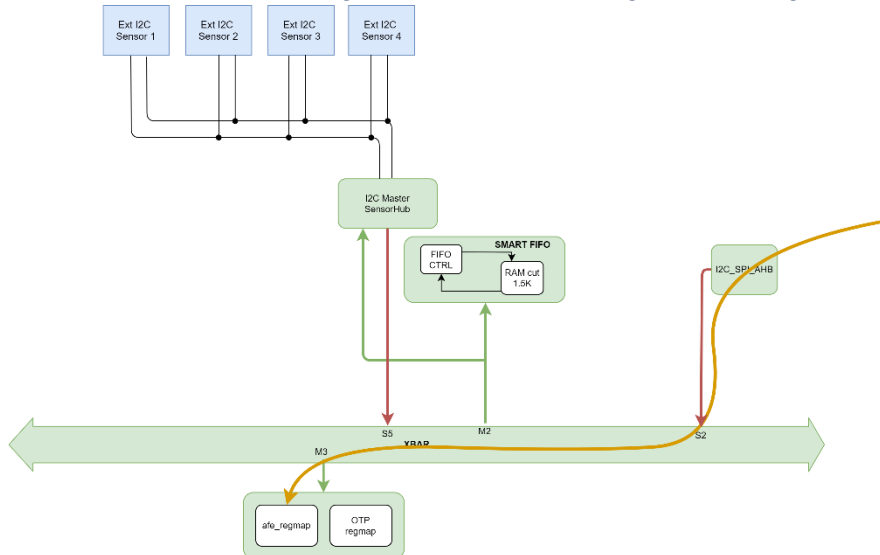
Port name	Direction	Type	TOP CONNECTION
HTRANS	input	[1:0]	
HSIZES	input	[2:0]	
HWRITES	input		
HREADY	input		
HWDATAS	input	[31:0]	
HREADYOUTS	output		
HRESPS	output		
HRDATAS	output	[31:0]	
ahb_master_on	output		
HRDATAM	input	[31:0]	
HREADYM	input		
HRESPM	input	[1:0]	
HGRANTM	input		
HSELM	output		
HADDRM	output	[31:0]	
HTRANS	output	[1:0]	
HWRITEM	output		
HSIZEM	output	[2:0]	
HBURSTM	output	[2:0]	
HPROTM	output	[3:0]	
HWDATAM	output	[31:0]	
HBUSREQM	output		
HLOCKM	output		

Table 26: Sensor Hub I/O List

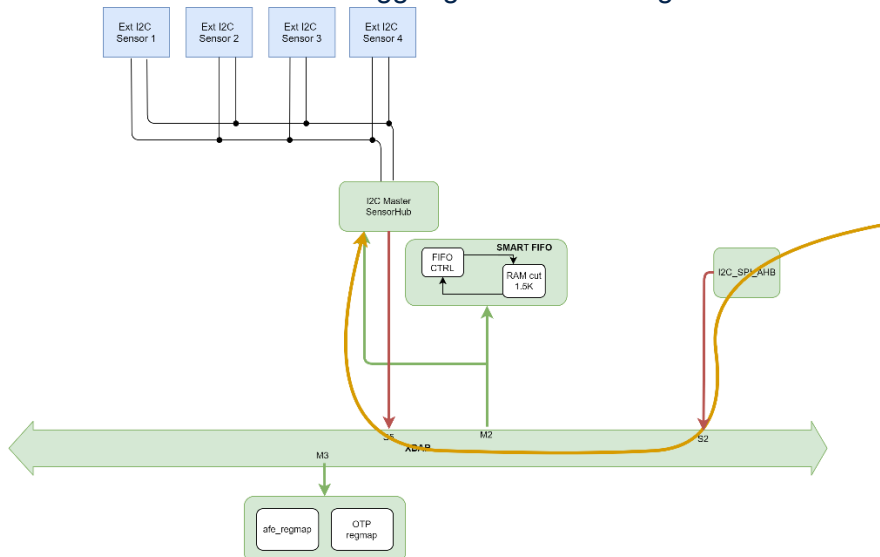
[DDS_STBIO1_XXXX] Description

Sensor hub related data flow is composed by the following phases :

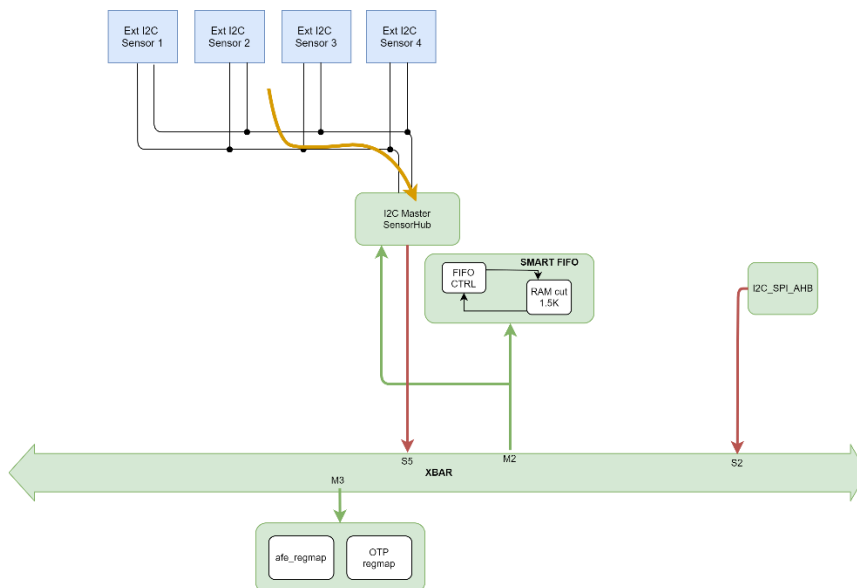
1. Initial configuration of STBIO1 registers through I2C/SPI interface



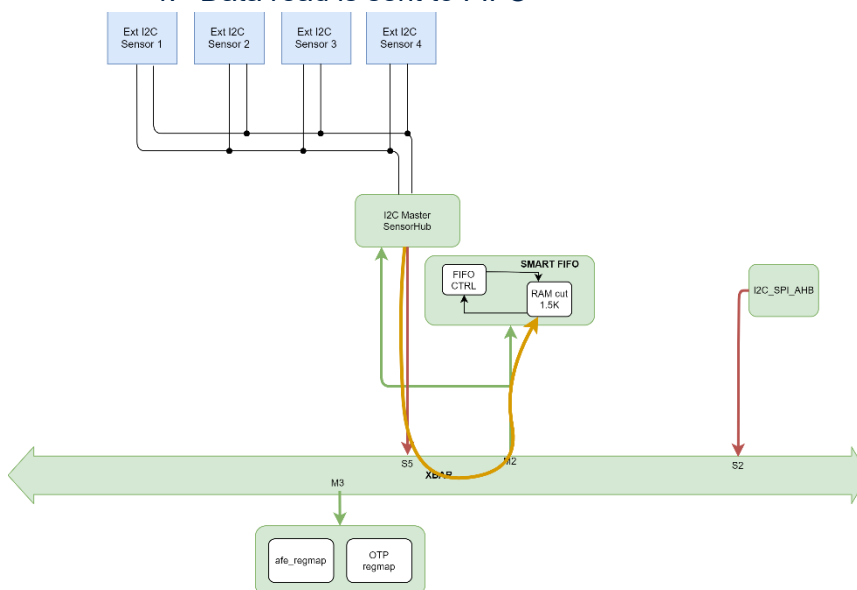
2. I2C master trigger generation through I2C/SPI interface



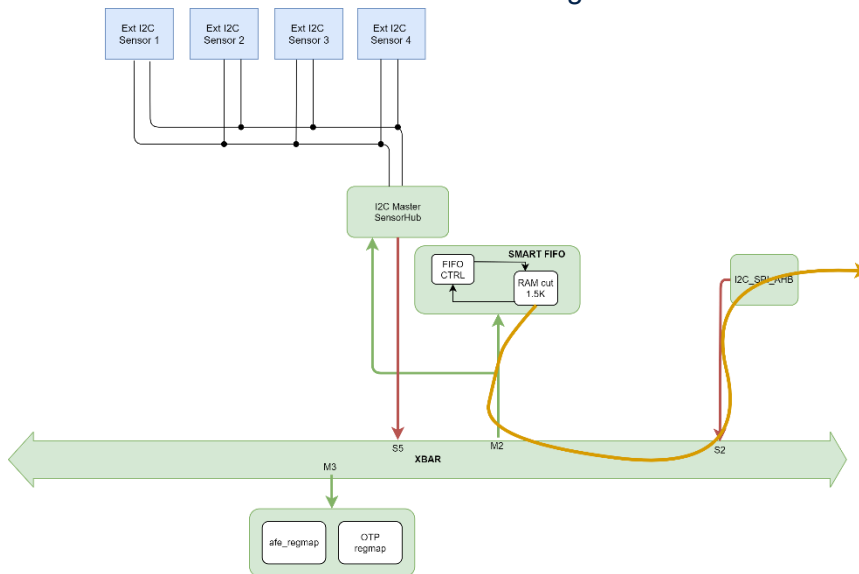
3. External I2C sensors are read by sensor hub



4. Data read is sent to FIFO



5. Data read can be read through I2C/SPI interface



[End]

[DDS_STBIO1_XXXX] Description

Sensor hub configuration sequence at system level to be added

[End]

12.2.3.1. SensorHub Peculiar Requirements

[DDS_STBIO1_9000] Description

Sensor hub IP is an IP used to read up-to 4 external I2C devices.

Sensor hub IP is compliant with FM+ I2C standard.

Both read and write operation are possible:

- Write:
 - o 1 byte on one I2C target
- Read:
 - o up-to 12byte on each I2C target
 - o up-to 4 I2C target

[End]

[DDS_STBIO1_9317] Requirement

If the sensor hub IP is not enabled, the SCL and SDA pad shall not toggle.

[Vpriority High] [TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9011] Requirement

Input scl_master_in of sensor hub shall be connected to input buffer from sensor hub's SCL PAD.

[Vpriority High] [TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9012] Requirement

Input sda_master_in of sensor hub shall be connected to input buffer from sensor hub's SDA PAD.

[Vpriority High] [TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9013] Requirement

Output scl_master_out of sensor hub shall be connected to enable of output buffer of sensor hub's SCL PAD (in case it is a bi-directional PAD).

[Vpriority High] [TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9014] Requirement

Output sda_master_out of sensor hub shall be connected to enable of output buffer of sensor hub's SDA PAD.

[Vpriority High] [TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9017] Requirement

The output port sleep_master shall be set to 1 when an I2C operation is not on-going.

[Vpriority High] [TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9004] Requirement

Input sys_clk of sensor hub shall be connected to 8MHz clock to generate I2C communications compliant with FM+ I2C standard.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9018] Requirement

The trigger of sensor hub shall be possible in both ways, HW via dvalid_masterblaze_ext signal or SW using DVALID_REGISTER.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9024] Requirement

Only I2C target addressed through SLAVE0_DEV_ADD register shall be the destination of an I2C Write Operation.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9027] Requirement

The bit 0 of SLAVE1_DEV_ADD, SLAVE2_DEV_ADD, SLAVE3_DEV_ADD registers shall be ignored: ONLY read operations towards I2C target 1, 2 and 3 shall be performed.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9029] Requirement

AUX_SENS_ON possible values shall be 0,1,2,3. All the values shall be tested since MasterBlaze supports up-to 4 I2C targets connected to I2C network.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9030] Requirement

SLAVEx_NUMOP bit-fields of SLAVEx_CONFIG registers shall define how many bytes per slave are read as per effect of an I2C operation (x from 1 to 12).

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9031] Requirement

All the values from 1 to 12 shall be valid for SLAVEx_NUMOP bit-fields of SLAVEx_CONFIG registers.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9032] Requirement

SLAVE0_NUMOP bitfield shall be ignored if an I2C Write operation is issued to the I2C Slave 0.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9033] Requirement

All the SENSOR_HUB_x registers shall be reset every time a new I2C operation starts.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9035] Requirement

When Quokka MasterBlaze μ P runs (sleep_master signal = 0), any dvalid triggered (HW or SW) shall be ignored.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9034] Requirement

The NACK_REGISTER register is reset every time a new I2C operation starts.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9260] Requirement

Each data stored in FIFO shall be composed by 4 words (16bytes).

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9261] Requirement

3 words (12bytes) of the data stored in FIFO shall contain the data from each I2C slave.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9262] Requirement

24-bits of the data stored in FIFO shall be reserved at '0'.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9263] Requirement

The last 8-bits of the data stored in FIFO shall contain the TAG field.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9300] Requirement

The AHB Write to registers has no effect if the IP masterblaze processor is busy (I2C operation on-going); the registers value shall not be updated.

[TO: IPOS_Senshub_MB_i2c]

[End]

[DDS_STBIO1_9500] Requirement

In case also other sources that writes into FIFO are active, then sensor hub shall work always in multi-fifo mode (fifo_mode input of sensor hub set to '1').

[End]

12.2.4. FIFO_CTRL integration

Generic name	Type	Value	TOP CONNECTION
AW		13	

Table 27: Fifo Parameter

Port name	Direction	Type	Description	TOP CONNECTION
TEST_MODE	input		ScanMode	u_pad_mux. scan_mode
HCLK	input		AHB Master Handler Clock 16MHz clock	u_pmu.clk_fifo
HRESETn	input			u_pmu.resetn_fifo
HSEL	input		AHB Slave Port	u_afa_xbar.hsel_M2
HADDR	input	[AW-1:0]	AHB Slave Port	u_afa_xbar.haddr_M2
HMASTER	input	[3:0]	AHB Slave Port	u_afa_xbar.hmaster_M2
HTRANS	input	[1:0]	AHB Slave Port	u_afa_xbar.htrans_M2
HSIZE	input	[2:0]	AHB Slave Port	u_afa_xbar.hsize_M2
HWRITE	input		AHB Slave Port	u_afa_xbar.
HWDATA	input	[31:0]	AHB Slave Port	u_afa_xbar.
HREADY	input		AHB Slave Port	u_afa_xbar.
HREADYOUT	output		AHB Slave Port	u_afa_xbar.
HRDATA	output	[31:0]	AHB Slave Port	u_afa_xbar.
HRESP	output		AHB Slave Port	u_afa_xbar.
O_FIFO_RAM_A	output	[ADDR_WIDTH-1:0]		u_ca_fifo_bist_c.ags_1
O_FIFO_RAM_D	output	[23:0]		u_ca_fifo_bist_c.csns_1
O_FIFO_RAM_CSN	output			u_ca_fifo_bist_c.dgs_1
O_FIFO_RAM_M	output	[23:0]		u_ca_fifo_bist_c.wens_1
O_FIFO_RAM_WEN	output			u_ca_fifo_bist_c.mgs_1
I_FIFO_RAM_Q	input	[23:0]		u_ca_fifo_bist_c.dgrs_1

Table 28: FIFO Controller I/O List

12.2.4.1. FIFO_CTRL Peculiar Requirements

[DDS_STBIO1_00000102] Requirement

The parameter MEM_LOC shall define the first Memory Location of the FIFO RAM: 0xC4000.

[TO: IPOS_FIFO_CTRL_00000102]

[End]

[DDS_STBIO1_00000202] Requirement

The FIFO register map first memory location shall be defined as MEM_LOC + 0x1000: 0xC5000.

[TO: IPOS_FIFO_CTRL_00000202]

[End]

[DDS_STBIO1_89456557] Requirement:

The FIFO Controller shall be able to address 1.5 kbyte divided into 32-bit address.

[TO: IPOS_FIFO_CTRL_89456557]

[End]

[DDS_STBIO1_89459082] Requirement:

The FIFO Controller shall work at 16 MHz clock frequency.

[TO: IPOS_FIFO_CTRL_89459082]

[End]

[DDS_STBIO1_89455261] Requirement:

The FIFO Controller shall have an AHB Slave Interface connected to the AHB BUS and a Memory Master Interface connected to the Physical Memory.

[TO: IPOS_FIFO_CTRL_89455261]

[End]

[DDS_STBIO1_89455449] Requirement:

The FIFO Controller shall have two working modes: Multiple FIFO and Unique FIFO. These two modes are mutually exclusive.

[TO: IPOS_FIFO_CTRL_89455449]

[End]

[DDS_STBIO1_89455232] Requirement:

As default, the FIFO shall work in Unique FIFO Mode.

[TO: IPOS_FIFO_CTRL_89455232]

[End]

[DDS_STBIO1_89456975] Requirement:

The user shall select the Multiple FIFO mode setting to 1 the first bit into the FIFO_CFG_MODE and 0 for the Unique FIFO mode.

[TO: IPOS_FIFO_CTRL_89456975]

[End]

[DDS_STBIO1_89457435] Requirement:

The FIFO Controller shall have a 6 i_depth input of 8 bit to be used in Multiple FIFO Mode to define the depth of each subFIFOs.

[TO: IPOS_FIFO_CTRL_89457435]

[End]

[DDS_STBIO1_89457790] Requirement:

The FIFO 6 memory locations to access each sub-FIFO in Multiple mode are obtained adding multiples of 256 to the memory base address.

Example: FIFO Memory Base Address = 0xC4000, then:

FIFO 0 memory location	0xC4000
FIFO 1 memory location	0xC4100
FIFO 2 memory location	0xC4200
FIFO 3 memory location	0xC4300
FIFO 4 memory location	0xC4400
FIFO 5 memory location	0xC4500

Table 29: SUB FIFO Memory Address

[TO: IPOS_FIFO_CTRL_89457790]

[End]

[DDS_STBIO1_00000008] Requirement:

If input signal HMASTER is set to 0 or 1, the FIFO Controller shall enter Memory mode.

[TO: IPOS_FIFO_CTRL_00000008]

[End]

[DDS_STBIO1_00000199] Requirement

DATA TYPE	BIT: 23 - 20	BIT: 19 - 16	BIT: 15 - 0
PPG FRAME 0	0000	20 BIT DATA	
PPG FRAME 1	0001	20 BIT DATA	
PPG FRAME 2	0010	20 BIT DATA	
PPG FRAME 3	0011	20 BIT DATA	
PPG FRAME 4	0100	20 BIT DATA	
PPG FRAME 5	0101	20 BIT DATA	
PPG FRAME 6	0110	20 BIT DATA	
PPG FRAME 7	0111	20 BIT DATA	
PPG FRAME 8	1000	20 BIT DATA	
PPG FRAME 9	1001	20 BIT DATA	
PPG FRAME 10	1010	20 BIT DATA	
PPG FRAME 11	1011	20 BIT DATA	
TIME SLOT DATA	1111	0000	16 BIT DATA
ECG0_AC	1100	0000	16 BIT DATA
ECG0_DC	1101	0000	16 BIT DATA
ECG1_AC	1100	0001	16 BIT DATA
ECG1_DC	1101	0001	16 BIT DATA
ECG2_AC	1100	0010	16 BIT DATA
ECG2_DC	1101	0010	16 BIT DATA
BIA_AC_P	1110	0001	16 BIT DATA
BIA_DC_P	1110	0011	16 BIT DATA
BIA_AC_Q	1110	0000	16 BIT DATA

BIA_DC_Q	1110	0010	16 BIT DATA
GSR	1111	1000	16 BIT DATA
DATA_QUOKKA_ELAB	1111	0001 to 0111	16 BIT DATA

Table 30: Data Tag

00000199[End]

[DDS_STBIO1_00001199] Requirement

When in Multiple Fifo, the 6 sub-FIFO offset shall coincide with the following table :

FIFO OFFSET	DATA TYPE
ECG FIFO OFFSET	ECG0_AC, ECG0_DC
BIA FIFO OFFSET	BIA_AC_P, BIA_DC_P, BIA_AC_Q, BIA_DC_Q, ECG1_AC, ECG2_DC
GSR FIFO OFFSET	GSR, ECG1_DC, ECG2_AC
PPG FIFO OFFSET	PPG

Table 31 : FIFO Offset Data Table

[END]

12.3. ISPU integration

Por name	Direction	Comment	TOP CONNECTION
clk	input	clk 16MHz from pmu	u_pmu_clk_stredl_sig
main_rst_n	input	reset in 16MHz domain	u_pmu_resetrn_stredl_sig
POR_n	input	reset in 16MHz domain	u_pmu_resetrn_stredl_sig
ispu_xbar_sw_resetrn	input	Software reset ISPU xbar	u_regmap_otp_ispu_xbar_sw_resetrn_sig
ispu_core_sw_resetrn	input	Software reset ISPU core	u_regmap_otp_ispu_core_sw_resetrn_sig
scan_mode	input	Scan mode	u_pad_mux_scan_mode_sig
scan_enable	input	Scan enable	u_pad_mux_scan_enable_sig
test_rst_n	input	Scan reset	u_pad_mux_scan_rstn_sig
bist_mode	input	Bist mode enable	u_pad_mux_bist_mode_sig
interrupt	input	Interrupt lines	u_ca_fifo_bist_c_oen_1, u_fifo_irq_FIFO_U_M_IRQ_STATU S_sig, u_regmap_otp_irq_SENTHUB_STS _sig, u_regmap_otp_irq_ISPU_MC_INTR _STS_sig, u_regmap_otp_sw_interrupt_sig, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1,

			u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1
clk_en	output	Enable clock at ISPU core	u_ispu_clk_en_sig
S_HSEL	input	AHB HSEL Slave Port	u_xbar_afe_hsel_M0_sig
S_HREADY	input	AHB HREADY Slave Port	u_xbar_afe_hreadymux_M0_sig
S_HREADYOUT	output	AHB HREADYOUT Slave Port	u_ispu_S_HREADYOUT_sig
S_HTRANS	input	AHB HTRANS Slave Port	u_xbar_afe_htrans_M0_sig
S_HADDR	input	AHB HADDR Slave Port	u_xbar_afe_haddr_M0_sig
S_HSIZE	input	AHB HSIZE Slave Port	u_xbar_afe_hsize_M0_sig[1:0]
S_HWRITE	input	AHB HWRITE Slave Port	u_xbar_afe_hwrite_M0_sig
S_HWDATA	input	AHB HWDATA Slave Port	u_xbar_afe_hwdata_M0_sig
S_HRDATA	output	AHB HRDATA Slave Port	u_ispu_S_HRDATA_sig
S_HRESP	output	AHB HRESP Slave Port	u_ispu_S_HRESP_sig

M_HSEL	output	AHB HSEL Master Port	u_ispu_M_HSEL_sig
M_HREADY	input	AHB HREADY Master Port	u_xbar_afe_hready_S0_sig
M_HTRANS	output	AHB HTRANS Master Port	u_ispu_M_HTRANS_sig
M_HADDR	output	AHB HADDR Master Port	u_ispu_M_HADDR_sig
M_HSIZE	output	AHB HSIZE Master Port	u_ispu_M_HSIZE_sig
M_HWRITE	output	AHB HWRITE Master Port	u_ispu_M_HWRITE_sig
M_HWDATA	output	AHB HWDATA Master Port	u_ispu_M_HWDATA_sig
M_HRDATA	input	AHB HRDATA Master Port	u_xbar_afe_hrdata_S0_sig
M_HRESP	input	AHB HRESP Master Port	u_xbar_afe_hresp_S0_sig
DBG_HREADY	input	AHB HREADY Debug Master Port	u_xbar_afe_hready_S4_sig
DBG_HTRANS	output	AHB HTRANS Debug Master Port	u_ispu_DBG_HTRANS_sig
DBG_HADDR	output	AHB HADDR Debug Master Port	u_ispu_DBG_HADDR_sig

DBG_HSIZE	output	AHB HSIZE Debug Master Port	u_ispu_DBG_HSIZE_sig
DBG_HWRITE	output	AHB HWRITE Debug Master Port	u_ispu_DBG_HWRITE_sig
DBG_HBURST	output	AHB HDBURST Debug Master Port	NO CONN
DBG_HWDATA	output	AHB HWDATA Debug Master Port	u_ispu_DBG_HWDATA_sig
DBG_HRDATA	input	AHB HRDATA Debug Master Port	u_xbar_afe_hrdata_S4_sig
DBG_HRESP	input	AHB HRESP Debug Master Port	u_xbar_afe_hresp_S4_sig
trst_n	input	JTAG Test Reset	u_pad_mux_ispu_trst_n_sig
tck	input	JTAG Test Clock	u_pad_mux_ispu_tck_sig
tms	input	JTAG Test Mode Select	u_pad_mux_ispu_tms_sig
tdi	input	JTAG Test Data In	u_pad_mux_ispu_tdi_sig
tdo	output	JTAG Test Data Out	u_ispu_tdo_sig
tdo_en	output	JTAG Test Data Out Enable	u_ispu_tdo_en_sig

ca_stredl_bist_c_rst_bistctrl_n	input	BIST Asynchronous Reset Pin for collar Serial Interface and Controller	u_pmu_tst_rst_bistctrl_n_sig
ca_stredl_bist_c_clk_bistctrl	input	BIST Low Frequency Serial Shift Clock for the controller	u_pmu_tst_clk_bistctrl_sig
ca_stredl_bist_c_bist_mode	input	BIST Global BIST Mode Selection Bus	u_ta_afe_bist_ctrl_bist_mode_ctrl_sig
ca_stredl_bist_c_march_shift	input	BIST March element serial data out enable to SP/ROM/RF/DP collars	u_ta_afe_bist_ctrl_march_shiftSP_sig
ca_stredl_bist_c_collar_config	input	BIST Mode selection pins for SP/ROM/RF/DP memory collars (modefunc, modestab, modecoll)	u_ta_afe_bist_ctrl_collar_configSP_sig
ca_stredl_bist_c_march_in	input	BIST March element serial data in	u_ta_afe_bist_ctrl_march_outSP_sig
ca_stredl_bist_c_march_run	input	BIST March run to SP/ROM/RF/DP collars	u_ta_afe_bist_ctrl_march_runSP_sig

ca_stredl_bist_c_memsafe	input	BIST Pin to globally deactivate the selected memories by making CSN high internally	u_ta_afe_bist_ctrl_memsafe_ctrl_sig
ca_stredl_bist_c_sif_reg_en	input	BIST Enable for various types of data transfer (Bitmap, memory selection, programmable algorithms) through serial interface.	u_ta_afe_bist_ctrl_sif_reg_en_ctrl_sig
ca_stredl_bist_c_sif_load_en	input	BIST Allows to control the data loading from collar registers to serial interface registers of the collar	u_ta_afe_bist_ctrl_sif_load_en_ctrl_sig
ca_stredl_bist_c_shift_en	input	BIST Enable for Serial Input Data from Tester/External World	u_ta_afe_bist_ctrl_shift_en_ctrl_sig
ca_stredl_bist_c_sif_update_en	input	BIST Allows to control the data loading from collar	u_ta_afe_bist_ctrl_sif_update_en_ctrl_sig

		registers to serial interface registers of the collar	
ca_stredl_bist_c_march_end	output	BIST March end signal from each SP / ROM / RF / DP collar type	u_ispu_ca_stredl_bist_c_march_end_sig
ca_stredl_bist_c_bbad_1	output	BIST Indicates bbad for a specific memory. There are as many bbad signals as the number of memories.	NO CONN
ca_stredl_bist_c_bfail_1	output	BIST Indicates bfail for a specific memory. There are as many bbad signals as the number of memories.	u_ispu_ca_stredl_bist_c_bfail_1_sig
ca_stredl_bist_c_repairable_1	output	BIST Indicates repairable for a specific memory. There are as many repairable signals as	NO CONN

		the number of memories. For a shared collar there would be as many repairable outputs as the number of memories	
ca_stredl_bist_c_bbad_2	output	BIST Indicates bbad for a specific memory. There are as many bbad signals as the number of memories.	NO CONN
ca_stredl_bist_c_bfail_2	output	BIST Indicates bfail for a specific memory. There are as many bbad signals as the number of memories.	u_ispu_ca_stredl_bist_c_bfail_2_sig
ca_stredl_bist_c_repairable_2	output	BIST Indicates repairable for a specific memory. There are as many repairable signals as	NO CONN

		the number of memories. For a shared collar there would be as many repairable outputs as the number of memories	
ca_stredl_bist_c_bbad_3	output	BIST Indicates bbad for a specific memory. There are as many bbad signals as the number of memories.	NO CONN
ca_stredl_bist_c_bfail_3	output	BIST Indicates bfail for a specific memory. There are as many bbad signals as the number of memories.	u_ispu_ca_stredl_bist_c_bfail_3_sig
ca_stredl_bist_c_repairable_3	output	BIST Indicates repairable for a specific memory. There are as many repairable signals as	NO CONN

		the number of memories. For a shared collar there would be as many repairable outputs as the number of memories	
ca_stredl_bist_c_bbad_4	output	BIST Indicates bbad for a specific memory. There are as many bbad signals as the number of memories.	NO CONN
ca_stredl_bist_c_bfail_4	output	BIST Indicates bfail for a specific memory. There are as many bbad signals as the number of memories.	u_ispu_ca_stredl_bist_c_bfail_4_sig
ca_stredl_bist_c_repairable_4	output	BIST Indicates repairable for a specific memory. There are as many repairable signals as	NO CONN

		the number of memories. For a shared collar there would be as many repairable outputs as the number of memories	
ca_stredl_bist_c_serial_out	output	BIST Serial Output Data from Tester / External World	u_ispu_ca_stredl_bist_c_serial_out_sig
ca_stredl_bist_c_bypass	input	BIST Memory Bypass Pin for RAM type memories. One top level port per collar.	u_pad_mux_scan_mode_sig
ca_stredl_bist_c_serial_in	input	BIST Serial Input Data from Tester / External World	u_ta_afe_bist_ctrl_serial_out_ctrl_sig
ca_stredl_bist_c_test_se	input	BIST Scan Enable Signal Used internally by the BIST to prevent memory operation during scan shifting (a dedicated	u_pad_mux_scan_enable_sig

		test_se (one test_se per collar) is provided at the top level	
ca_stredl_bist_c_tst_gated clock	input	BIST Clock Gating Signal for the entire BIST. Used to force clock gating cell transparency in ATPG mode .	u_pad_mux_scan_enable_sig
ca_stredl_bist_c_tst_reset_mux	input	BIST Bypass for reset synchronizer. allows to force reset synchronizing circuit transparency in ATPG mode.	u_pad_mux_scan_mode_sig
ca_stredl_bist_c_oen_1	input	BIST Memory Output Enable 1	u_ca_fifo_bist_c_oen_1
ca_stredl_bist_c_oen_2	input	BIST Memory Output Enable 2	u_ca_fifo_bist_c_oen_1
ca_stredl_bist_c_oen_3	input	BIST Memory Output Enable 3	u_ca_fifo_bist_c_oen_1
ca_stredl_bist_c_oen_4	input	BIST Memory	u_ca_fifo_bist_c_oen_1

		Output Enable 4	
ca_stredl_bist_c_mem_atpg_mode	input	BIST For ATPG purpose: Used to constraint few memory inputs and improve controllability and observability .	u_ca_fifo_bist_c_oen_1
ca_stredl_bist_c_tst_mem_selck_1	input	BIST To select the clock and memory ports in ATPG mode 0: Functional Clock / Functional Ports 1: Test Clock/ Test Ports	u_ca_fifo_bist_c_oen_1
ca_stredl_bist_c_tst_mem_selck_2	input	BIST To select the clock and memory ports in ATPG mode 0: Functional Clock / Functional Ports 1: Test Clock/ Test Ports	u_ca_fifo_bist_c_oen_1
ca_stredl_bist_c_tst_mem_selck_3	input	BIST To select the	u_ca_fifo_bist_c_oen_1

		clock and memory ports in ATPG mode 0: Functional Clock / Functional Ports 1: Test Clock/ Test Ports	
ca_stredl_bist_c_tst_mem_selck_4	input	BIST To select the clock and memory ports in ATPG mode 0: Functional Clock / Functional Ports 1: Test Clock/ Test Ports	u_ca_fifo_bist_c_oen_1
ca_stredl_bist_c_par_mem_sel	input	BIST Parallel Memory Selection Bus	{ 1'b0, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1, u_ca_fifo_bist_c_oen_1 }

Table 32: ISPU I/O List

12.3.1. Requirements

[DDS_STBIO1_0500] Requirement:] Requirement:

Interrupt lines list:

Interrupt	bit	Description
irq_FIFO_U_M_IRQ_STsATUS	29	FIFO status [DDS_STBIO1_0501]
irq_SENSHUB_STS	28	SENSOR HUB status [DDS_STBIO1_0502]
irq_ISPU_MC_INTR_STS	27	ECG, BIA or PPG samples ready for elaboration[DDS_STBIO1_0503]
sw_interrupt	26:11	16 sw interrupt used by ADSP or external user [DDS_STBIO1_0504]
irq_timer	1	Internal timer [DDS_STBIO1_0505]

[END]

[DDS_STBIO1_0506] Description:

The ISPU is a dedicated IP for implementing complex processing algorithms. By default, it is turned off. This is located inside a power island

[End]

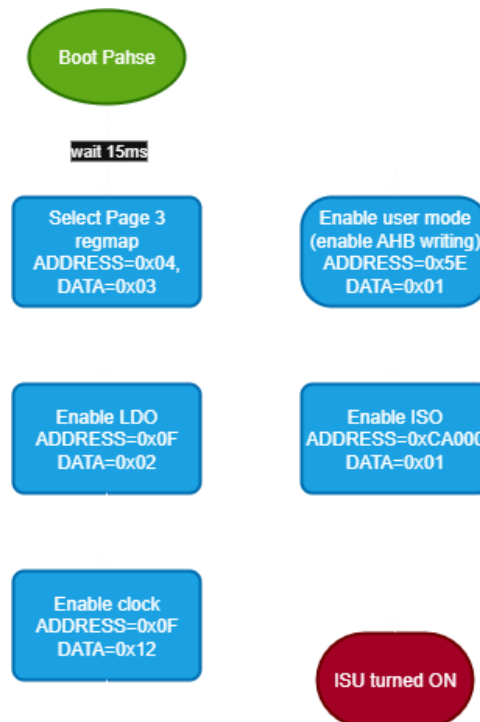


Figure 17: ISPU turn on phases

[DDS_STBIO1_0507] Requirement:

After the initial boot phase, to turn on the ISPU shall be:

- Select page 3 of the regmap (ADDRESS=0x04, DATA=0x03)
- Enable LDO 1V8 by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x02)
- Enable clock by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x12)
- Enable user mode by writing to regmap (ADDRESS=0x5E, DATA=0x01)
- Enable the power island by writing to regmap **POWER CONFIGURATION REGISTER** via AHB interface (ADDRESS=0xCA000, DATA=0x01)

[End]

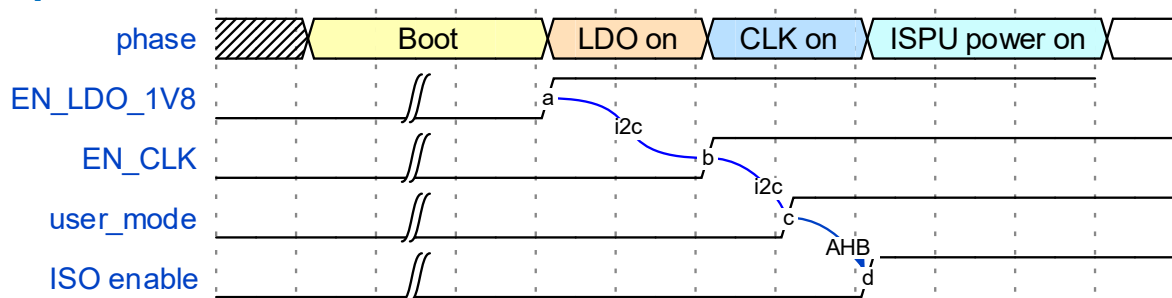


Figure 18: Timing turn on ISPU

[DDS_STBIO1_0508] Requirement:

To turn off the ISPU shall be:

- Disable the power island by writing to regmap **POWER CONFIGURATION REGISTER** via AHB interface (ADDRESS=0xCA000, DATA=0x00)

[End]

[DDS_STBIO1_0509] Requirement:

To put the ISPU into sleep mode, it is necessary to perform a read at address 0x80054.

[End]

[DDS_STBIO1_05010] Requirement:

To wake up the ISPU, it is necessary to trigger an interrupt.

[End]

[DDS_STBIO1_05011] Requirement:

When the ISPU is in sleep mode, it is possible to perform reads and writes from the data and code memory by triggering an AHB transition.

[End]

Stato corrente	TMS = 0	TMS = 1
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Test-Logic-Reset	Run-Test/Idle	Test-Logic-Reset
Run-Test/Idle	Run-Test/Idle	Select-DR-Scan
Select-DR-Scan	Capture-DR	Select-IR-Scan
Capture-DR	Shift-DR	Exit1-DR
Shift-DR	Shift-DR	Exit1-DR
Exit1-DR	Pause-DR	Update-DR
Pause-DR	Pause-DR	Exit2-DR
Exit2-DR	Shift-DR	Update-DR
Update-DR	Run-Test/Idle	Select-DR-Scan
Select-IR-Scan	Capture-IR	Test-Logic-Reset
Capture-IR	Shift-IR	Exit1-IR
Shift-IR	Shift-IR	Exit1-IR
Exit1-IR	Pause-IR	Update-IR
Pause-IR	Pause-IR	Exit2-IR
Exit2-IR	Shift-IR	Update-IR

Update-IR	Run-Test/Idle	Select-DR-Scan
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12.4. STBIO1 Register Map

12.4.1. STBIO1 Register Map

Here is described the list of registers in the register map. The AFE_BLOCK registers are accessible by I2C/SPI direct access and AHB interface, while the others are only by AHB. The I2C/SPI direct access mode is done by the HOST INTF port (I2C/SPI AHB) and the address of the AFE_BLOCK registers are arranged in 4 pages of 64 registers.

BASE START ADDRESS	ADDRESS BLOCK	REGISTER WIDTH	BASE END ADDRESS	RESET	REQ_ID
0xC8000	AFE_BLOCK	8	0xC80EB	tst_main_reset_n, regmap_sw_resetrn	DDS_STBIO1_3000
0xC80EC	RESERVED		0xC8FFF	tst_main_reset_n, regmap_sw_resetrn	DDS_STBIO1_3001
0xC9000	OTP REGISTER	8	0xC902C	tst_main_reset_n	DDS_STBIO1_3002
0xC9030	RESERVED		0xC9FFF	tst_main_reset_n, regmap_sw_resetrn	DDS_STBIO1_3003
0xCA000	SYSTEM_REGISTER	32	0xCA140	tst_main_reset_n, regmap_sw_resetrn	DDS_STBIO1_3004
0xCA144	RESERVED		0xCCFFF	tst_main_reset_n, regmap_sw_resetrn	DDS_STBIO1_3005
0xCD000	OTP (TRIMM REGS)	32	0xCD02F	tst_main_reset_n, regmap_sw_resetrn	DDS_STBIO1_3006

Table 33: Address Register Map

The device is configurable/readable by the I2C/AHB bridge. This IP is in charge of take an I2C/SPI access and (eventually) transform it into an AHB transaction, to allow the configuration/reading of the device using the ISPU.

The only registers that are accessible also by an **I2C/SPI direct access**, are the ones of the AFE BLOCK section.

To reach a very Low-Power consumption, the AHB interface is disabled as default.

The I2C/SPI direct access mode is done by the HOST INTF port (I2C/SPI AHB) and the address of the AFE_BLOCK registers are arranged in 4 pages of 64 registers (Table 24).

In the following table are the base address and the size of each one page.

BASE ADDRESS	SIZE	PAGE
0x00	0x3F	0
0x40	0x3F	1
0x80	0x3F	2
0xC0	0x3F	3

Table 34: Location AFE_BLOCK Registers on the I2C/SPI pages.

For direct access operation two I2C/SPI operations need for reading/writing an AFE_BLOCK register:

1. Page Selection -> I2C write operation for writing the 2 bits of the PAGE_SEL (address 0x40 of I2C/SPI AHB Bridge)
2. AFE_BLOCK register R/W -> Read/Write the AFE_BLOCK register with the address reported in one of 4 pages.
 - a. e.g.: for reading the DEVICE_STATUS_REG (Register Map address 0x0096) the **PAGE_SEL=0x02, I2C/SPI direct access address=0x16**

To enable AHB Access, it needs using I2C/SPI direct access mode, for:

- First turn-on the LDO1V8 and the Fast-Ring Oscillator (set EN_LDO1V8 and EN_CLKF of VIREF_LDO1V8_CLK16M, see address in the Table 34).
- Check the Fast Clock(16MHz), is stable (when the Fast-Ring Oscillator is stable, the CLK_READY bit in DEVICE_CONFIG_OPERATION (0x0000) register is set to 1).

Now it's possible to configure the registers that are mapped on AHB_Interface as explained in the following example:

- Run an I2C/SPI transaction, with the type of AHB access word/byte, at the address 0x59 (one of I2C/SPI Internal Configuration Registers).
- Run an I2C/SPI transaction, with the register address that you want to configure, at the address 0x50 (one of I2C/SPI Internal Configuration Registers), starting from the LSB. In other word, if you need to access at register VIREF_LDO1V8_CLK16M at AHB address 0x000C_00CF, you have to perform four I2C/SPI accesses to write:
 - At address 0x50 0xCF
 - At address 0x51 0x00
 - At address 0x52 0x0C
 - At address 0x53 → 0x00
- Run an I2C/SPI write, of the new data, at address 0x60 (one of I2C/SPI Internal Configuration Registers).
- Run an I2C/SPI read transition, with the number of byte that you read, at address 0x68 (one of I2C/SPI Internal Configuration Registers).

- Run an I2C/SPI transaction with the data that you want to read, at address loaded into the previous step, at the address 0x70 (one of I2C/SPI Internal Configuration Registers).

It's possible to remap the whole AFE register map with AHB addresses (column "AHB Start Address" of Table 34), including the AFE_BLOCK block: in this case each register is accessible with its own AHB address (AHB start address + offset) and the I2C/SPI access to these registers must follow the previous procedure. To do that:

- Set the **user_mode** at 1 (set first bit-field of CTRL_REG_7 (0x5E), one of I2C/SPI Internal Configuration Registers).

This procedure is useful when you are thinking to configure the STBIO1 by ISPU (with a code previously charged in code memory), leaving to host interface (I2C/SPI) the rule to collect the data from FIFO after AFE sampling and processing phases.

If the ISPU is not used to configure the device, the **user_mode** shall remain to 0.

Pay Attention: it is strictly recommended to set the **user_mode** at startup, because when the **user_mode changes** value (from 0 to 1 or 1 to 0) **it's mandatory to rewrite the AFE_BLOCK registers** that are already configured.

12.4.2. STBIO1 Register Map Peculiar Requirements

[DDS_STBIO1_4000] Requirement:

Write enable for each OTP REGISTER shall be 0, in functional mode, if bit 31 of OTP_PRG10 is set 1.

[Vpriority High] [End]

12.4.2.1. AFE_BLOCK Register Summary

Offset	Register Name	Register Title	Direct Access	PAGE SEL
0x0000	DEVICE_CONFIG_OPERATION	DEVICE_CONFIG_OPERATION register	0x00	0x00
0x0001	DEVICE_CONFIG_CHANNEL	DEVICE_CONFIG_CHANNEL register	0x01	
0x0002	SET_ECG_FREQ_l	SET_ECG_FREQ_l register	0x02	
0x0003	SET_ECG_FREQ_h	SET_ECG_FREQ_h register	0x03	
0x0004	SET_BIA_FREQ_l	SET_BIA_FREQ_l register	0x04	
0x0005	SET_BIA_FREQ_h	SET_BIA_FREQ_h register	0x05	
0x0006	SET_GSR_FREQ_l	SET_GSR_FREQ_l register	0x06	
0x0007	SET_GSR_FREQ_h	SET_GSR_FREQ_h register	0x07	
0x0008	SET_PPG_FREQ_l	SET_PPG_FREQ_l register	0x08	
0x0009	SET_PPG_FREQ_h	SET_PPG_FREQ_h register	0x09	
0x000A	ADC_GENERAL_CONFIG	ADC_GENERAL_CONFIG register	0x0A	
0x000B	ADC_CAL_CONFIG	ADC_CAL_CONFIG register	0x0B	
0x000C	ADC_CONFIG_ECG	ADC_CONFIG_ECG register	0x0C	
0x000D	ADC_CONFIG_PPG	ADC_CONFIG_PPG register	0x0D	
0x000E	LDO1V2	LDO1V2 register	0x0E	

0x000F	DEVICE_ID_REG	DEVICE ID REGISTER	0x0F
0x0010	RLD_0	RLD_0 register	0x10
0x0011	RLD_1_DCCK_0	RLD_1_DCCK_0 register	0x11
0x0012	DCCK_1	DCCK_1 register	0x12
0x0013	CURR_INJ	CURR_INJ register	0x13
0x0014	CONTROL_CIP_CIN	CONTROL_CIP_CIN register	0x14
0x0015	IMP_OVERCURR	IMP_OVERCURR register	0x15
0x0016	BIO_BUF_AND_BIO_ASYNC_RST	BIO_BUF_AND_BIO_ASYNC_RST register	0x16
0x0017	IN_MUX	IN_MUX register	0x17
0x0018	ECG_CTRL	ECG_CTRL register	0x18
0x0019	IMP_CTRL	IMP_CTRL register	0x19
0x001A	GSR_CTRL	GSR_CTRL register	0x1A
0x001B	EN_CHOP_FREQ_MOD_DEMOD	EN_CHOP_FREQ_MOD_DEMOD register	0x1B
0x001C	PPG_ALC_CONFIG_PARAM_1	PPG_ALC_CONFIG_PARAM_1 register	0x1C
0x001D	PPG_ALC_CONFIG_PARAM_2	PPG_ALC_CONFIG_PARAM_2 register	0x1D
0x001E	DIGITAL_RAMP_CONFIG_PARAM_1	DIGITAL_RAMP_CONFIG_PARAM_1 register	0x1E
0x001F	DIGITAL_RAMP_CONFIG_PARAM_2	DIGITAL_RAMP_CONFIG_PARAM_2 register	0x1F
0x0020	DIGITAL_RAMP_CONFIG_PARAM_3	DIGITAL_RAMP_CONFIG_PARAM_3 register	0x20
0x0021	GENERAL_PPG_PARAMETERS_1	GENERAL_PPG_PARAMETERS_1 register	0x21
0x0023	GENERAL_PPG_PARAMETERS_2	GENERAL_PPG_PARAMETERS_2 register	0x23
0x0024	GENERAL_PPG_PARAMETERS_3	GENERAL_PPG_PARAMETERS_3 register	0x24
0x0025	FRAME_1_PARAMETERS_1	FRAME_1_PARAMETERS_1 register	0x25
0x0026	FRAME_2_PARAMETERS_1	FRAME_2_PARAMETERS_1 register	0x26
0x0027	FRAME_3_PARAMETERS_1	FRAME_3_PARAMETERS_1 register	0x27
0x0028	FRAME_4_PARAMETERS_1	FRAME_4_PARAMETERS_1 register	0x28
0x0029	FRAME_5_PARAMETERS_1	FRAME_5_PARAMETERS_1 register	0x29
0x002A	FRAME_6_PARAMETERS_1	FRAME_6_PARAMETERS_1 register	0x2A
0x002B	FRAME_7_PARAMETERS_1	FRAME_7_PARAMETERS_1 register	0x2B
0x002C	FRAME_8_PARAMETERS_1	FRAME_8_PARAMETERS_1 register	0x2C
0x002D	FRAME_9_PARAMETERS_1	FRAME_9_PARAMETERS_1 register	0x2D
0x002E	FRAME_10_PARAMETERS_1	FRAME_10_PARAMETERS_1 register	0x2E
0x002F	FRAME_11_PARAMETERS_1	FRAME_11_PARAMETERS_1 register	0x2F
0x0030	FRAME_12_PARAMETERS_1	FRAME_12_PARAMETERS_1 register	0x30
0x0031	FRAME_1_PARAMETERS_2	FRAME_1_PARAMETERS_2 register	0x31

0x0032	FRAME_2_PARAMETERS_2	FRAME_2_PARAMETERS_2 register	0x32	
0x0033	FRAME_3_PARAMETERS_2	FRAME_3_PARAMETERS_2 register	0x33	
0x0034	FRAME_4_PARAMETERS_2	FRAME_4_PARAMETERS_2 register	0x34	
0x0035	FRAME_5_PARAMETERS_2	FRAME_5_PARAMETERS_2 register	0x35	
0x0036	FRAME_6_PARAMETERS_2	FRAME_6_PARAMETERS_2 register	0x36	
0x0037	FRAME_7_PARAMETERS_2	FRAME_7_PARAMETERS_2 register	0x37	
0x0038	FRAME_8_PARAMETERS_2	FRAME_8_PARAMETERS_2 register	0x38	
0x0039	FRAME_9_PARAMETERS_2	FRAME_9_PARAMETERS_2 register	0x39	
0x003A	FRAME_10_PARAMETERS_2	FRAME_10_PARAMETERS_2 register	0x3A	
0x003B	FRAME_11_PARAMETERS_2	FRAME_11_PARAMETERS_2 register	0x3B	
0x003C	FRAME_12_PARAMETERS_2	FRAME_12_PARAMETERS_2 register	0x3C	
0x003D	FRAME_1_PARAMETERS_3	FRAME_1_PARAMETERS_3 register	0x3D	
0x003E	FRAME_2_PARAMETERS_3	FRAME_2_PARAMETERS_3 register	0x3E	
0x003F	FRAME_3_PARAMETERS_3	FRAME_3_PARAMETERS_3 register	0x3F	0x01
0x0040	FRAME_4_PARAMETERS_3	FRAME_4_PARAMETERS_3 register	0x00	
0x0041	FRAME_5_PARAMETERS_3	FRAME_5_PARAMETERS_3 register	0x01	
0x0042	FRAME_6_PARAMETERS_3	FRAME_6_PARAMETERS_3 register	0x02	
0x0043	FRAME_7_PARAMETERS_3	FRAME_7_PARAMETERS_3 register	0x03	
0x0044	FRAME_8_PARAMETERS_3	FRAME_8_PARAMETERS_3 register	0x04	
0x0045	FRAME_9_PARAMETERS_3	FRAME_9_PARAMETERS_3 register	0x05	
0x0046	FRAME_10_PARAMETERS_3	FRAME_10_PARAMETERS_3 register	0x06	
0x0047	FRAME_11_PARAMETERS_3	FRAME_11_PARAMETERS_3 register	0x07	
0x0048	FRAME_12_PARAMETERS_3	FRAME_12_PARAMETERS_3 register	0x08	
0x0049	FRAME_1_PARAMETERS_4	FRAME_1_PARAMETERS_4 register	0x09	
0x004A	FRAME_2_PARAMETERS_4	FRAME_2_PARAMETERS_4 register	0x0A	
0x004B	FRAME_3_PARAMETERS_4	FRAME_3_PARAMETERS_4 register	0x0B	
0x004C	FRAME_4_PARAMETERS_4	FRAME_4_PARAMETERS_4 register	0x0C	
0x004D	FRAME_5_PARAMETERS_4	FRAME_5_PARAMETERS_4 register	0x0D	
0x004E	FRAME_6_PARAMETERS_4	FRAME_6_PARAMETERS_4 register	0x0E	
0x004F	FRAME_7_PARAMETERS_4	FRAME_7_PARAMETERS_4 register	0x0F	
0x0050	FRAME_8_PARAMETERS_4	FRAME_8_PARAMETERS_4 register	0x10	
0x0051	FRAME_9_PARAMETERS_4	FRAME_9_PARAMETERS_4 register	0x11	
0x0052	FRAME_10_PARAMETERS_4	FRAME_10_PARAMETERS_4 register	0x12	
0x0053	FRAME_11_PARAMETERS_4	FRAME_11_PARAMETERS_4 register	0x13	
0x0054	FRAME_12_PARAMETERS_4	FRAME_12_PARAMETERS_4 register	0x14	
0x0055	FRAME_1_PARAMETERS_5	FRAME_1_PARAMETERS_5 register	0x15	
0x0056	FRAME_2_PARAMETERS_5	FRAME_2_PARAMETERS_5 register	0x16	

0x0057	FRAME_3_PARAMETERS_5	FRAME_3_PARAMETERS_5 register	0x17	
0x0058	FRAME_4_PARAMETERS_5	FRAME_4_PARAMETERS_5 register	0x18	
0x0059	FRAME_5_PARAMETERS_5	FRAME_5_PARAMETERS_5 register	0x19	
0x005A	FRAME_6_PARAMETERS_5	FRAME_6_PARAMETERS_5 register	0x1A	
0x005B	FRAME_7_PARAMETERS_5	FRAME_7_PARAMETERS_5 register	0x1B	
0x005C	FRAME_8_PARAMETERS_5	FRAME_8_PARAMETERS_5 register	0x1C	
0x005D	FRAME_9_PARAMETERS_5	FRAME_9_PARAMETERS_5 register	0x1D	
0x005E	FRAME_10_PARAMETERS_5	FRAME_10_PARAMETERS_5 register	0x1E	
0x005F	FRAME_11_PARAMETERS_5	FRAME_11_PARAMETERS_5 register	0x1F	
0x0060	FRAME_12_PARAMETERS_5	FRAME_12_PARAMETERS_5 register	0x20	
0x0061	FRAME_1_PARAMETERS_6	FRAME_1_PARAMETERS_6 register	0x21	
0x0062	FRAME_2_PARAMETERS_6	FRAME_2_PARAMETERS_6 register	0x22	
0x0063	FRAME_3_PARAMETERS_6	FRAME_3_PARAMETERS_6 register	0x23	
0x0064	FRAME_4_PARAMETERS_6	FRAME_4_PARAMETERS_6 register	0x24	
0x0065	FRAME_5_PARAMETERS_6	FRAME_5_PARAMETERS_6 register	0x25	
0x0066	FRAME_6_PARAMETERS_6	FRAME_6_PARAMETERS_6 register	0x26	
0x0067	FRAME_7_PARAMETERS_6	FRAME_7_PARAMETERS_6 register	0x27	
0x0068	FRAME_8_PARAMETERS_6	FRAME_8_PARAMETERS_6 register	0x28	
0x0069	FRAME_9_PARAMETERS_6	FRAME_9_PARAMETERS_6 register	0x29	
0x006A	FRAME_10_PARAMETERS_6	FRAME_10_PARAMETERS_6 register	0x2A	
0x006B	FRAME_11_PARAMETERS_6	FRAME_11_PARAMETERS_6 register	0x2B	
0x006C	FRAME_12_PARAMETERS_6	FRAME_12_PARAMETERS_6 register	0x2C	
0x006D	FRAME_1_PARAMETERS_7	FRAME_1_PARAMETERS_7 register	0x2D	
0x006E	FRAME_2_PARAMETERS_7	FRAME_2_PARAMETERS_7 register	0x2E	
0x006F	FRAME_3_PARAMETERS_7	FRAME_3_PARAMETERS_7 register	0x2F	
0x0070	FRAME_4_PARAMETERS_7	FRAME_4_PARAMETERS_7 register	0x30	
0x0071	FRAME_5_PARAMETERS_7	FRAME_5_PARAMETERS_7 register	0x31	
0x0072	FRAME_6_PARAMETERS_7	FRAME_6_PARAMETERS_7 register	0x32	
0x0073	FRAME_7_PARAMETERS_7	FRAME_7_PARAMETERS_7 register	0x33	
0x0074	FRAME_8_PARAMETERS_7	FRAME_8_PARAMETERS_7 register	0x34	
0x0075	FRAME_9_PARAMETERS_7	FRAME_9_PARAMETERS_7 register	0x35	
0x0076	FRAME_10_PARAMETERS_7	FRAME_10_PARAMETERS_7 register	0x36	
0x0077	FRAME_11_PARAMETERS_7	FRAME_11_PARAMETERS_7 register	0x37	
0x0078	FRAME_12_PARAMETERS_7	FRAME_12_PARAMETERS_7 register	0x38	
0x0079	SAVE_FRAME_DATA_RAW	SAVE_FRAME_DATA_RAW register	0x39	
0x0080	DIGITAL_RAMP_DEBUG_REG	DIGITAL_RAMP_DEBUG_REG register	0x00	0x02

0x0086	PPG_DEBUG_REG_0	PPG_DEBUG_REG_0 register	0x06	
0x0087	PPG_DEBUG_REG_1	PPG_DEBUG_REG_1 register	0x07	
0x0088	PPG_DEBUG_REG_2	PPG_DEBUG_REG_2 register	0x08	
0x0089	ADC_GENERAL_DEBUG_0	ADC_GENERAL_DEBUG_0 register	0x09	
0x008A	ADC_GENERAL_DEBUG_1	ADC_GENERAL_DEBUG_1 register	0x0A	
0x008B	ADC_ERROR_CAP_9_DEBUG	ADC_ERROR_CAP_9_DEBUG register	0x0B	
0x008C	ADC_ERROR_CAP_8_DEBUG	ADC_ERROR_CAP_8_DEBUG register	0x0C	
0x008D	ADC_ERROR_CAP_7_DEBUG	ADC_ERROR_CAP_7_DEBUG register	0x0D	
0x008E	ADC_ERROR_CAP_6_DEBUG	ADC_ERROR_CAP_6_DEBUG register	0x0E	
0x008F	ADC_ERROR_CAP_5_DEBUG	ADC_ERROR_CAP_5_DEBUG register	0x0F	
0x0090	ADC_ERROR_CAP_4_DEBUG	ADC_ERROR_CAP_4_DEBUG register	0x10	
0x0091	ADC_ERROR_CAP_3_DEBUG	ADC_ERROR_CAP_3_DEBUG register	0x11	
0x0092	ADC_ERROR_CAP_2_DEBUG	ADC_ERROR_CAP_2_DEBUG register	0x12	
0x0093	ADC_ERROR_CAP_1_DEBUG	ADC_ERROR_CAP_1_DEBUG register	0x13	
0x0094	ADC_ERROR_CAP_0_DEBUG	ADC_ERROR_CAP_0_DEBUG register	0x14	
0x0095	DEVICE_STATUS_REG	DEVICE_STATUS_REG register	0x15	
0x0096	ELAB_STATUS_REG	ELAB_STATUS_REG register	0x16	
0x0097	ANA_TEST_0	ANA_TEST_0 register	0x17	
0x0098	ANA_TEST_1	ANA_TEST_1 register	0x18	
0x0099	ANA_TEST_2	ANA_TEST_2 register	0x19	
0x009A	ANA_TEST_3	ANA_TEST_3 register	0x1A	
0x009B	ANA_TEST_4	ANA_TEST_4 register	0x1B	
0x009C	ADC_DATA_OUT_DEBUG_LOW	ADC_DATA_OUT_DEBUG_l register	0x1C	
0x009D	ADC_DATA_OUT_DEBUG_HIGH	ADC_DATA_OUT_DEBUG_h register	0x1D	
0x009E	DEBUG_GSR_REG	DEBUG GSR REGISTER	0x1E	
0x009F	B_IDAC_DEBUG_LOW	B_IDAC_DEBUG_l register	0x1F	
0x00A0	B_IDAC_DEBUG_HIGH	B_IDAC_DEBUG_h register	0x20	
0x00A1	SPARE_0_REG	AFE SPARE 0 REGISTER	0x21	
0x00A2	SPARE_1_REG	AFE SPARE 1 REGISTER	0x22	
0x00A3	TEST_GPIO	TEST_GPIO REGISTER	0x23	
0x00A4	PPG_STATUS	PPG_STATUS REGISTER	0x24	
0x00CF	VIREF_LDO1V8_CLK16M	VIREF_LDO1V8_CLK16M register	0x0F	0x03
0x00D2	GPIO_CTRL_OUT	GPIO_CONTROL_OUTPUT register	0x12	
0x00D3	GPIO1_CTRL_IN	GPIO1_CONTROL_INPUT register	0x13	
0x00D4	GPIO2_CTRL_IN	GPIO2_CONTROL_INPUT register	0x14	
0x00D5	GPIO3_CTRL_IN	GPIO3_CONTROL_INPUT register	0x15	

0x00D6	GPIO4_CTRL_IN	GPIO4_CONTROL_INPUT register	0x16
0x00D7	PAD_CTRL	PAD_CTRL register	0x17
0x00D8	PAD_CTRL_GPIO	PAD_CTRL_GPIO register	0x18
0x00D9	OFFSET_CALIBRATION_DATA1_LOW	OFFSET_CALIBRATION_DATA1_l register	0x19
0x00DA	OFFSET_CALIBRATION_DATA1_HIGH	OFFSET_CALIBRATION_DATA1_h register	0x1A
0x00DB	OFFSET_CALIBRATION_DATA2_LOW	OFFSET_CALIBRATION_DATA2_l register	0x1B
0x00DC	OFFSET_CALIBRATION_DATA2_HIGH	OFFSET_CALIBRATION_DATA2_h register	0x1C
0x00DD	OFFSET_CALIBRATION_DATA3_LOW	OFFSET_CALIBRATION_DATA3_l register	0x1D
0x00DE	OFFSET_CALIBRATION_DATA3_HIGH	OFFSET_CALIBRATION_DATA3_h register	0x1E
0x00DF	OFFSET_CALIBRATION_DATA4_LOW	OFFSET_CALIBRATION_DATA4_l register	0x1F
0x00E0	OFFSET_CALIBRATION_DATA4_HIGH	OFFSET_CALIBRATION_DATA4_h register	0x20
0x00E1	OFFSET_CALIBRATION_DATA5_LOW	OFFSET_CALIBRATION_DATA5_l register	0x21
0x00E2	OFFSET_CALIBRATION_DATA5_HIGH	OFFSET_CALIBRATION_DATA5_h register	0x22
0x00E3	OFFSET_CALIBRATION_DATA6_LOW	OFFSET_CALIBRATION_DATA6_l register	0x23
0x00E4	OFFSET_CALIBRATION_DATA6_HIGH	OFFSET_CALIBRATION_DATA6_h register	0x24
0x00E5	OFFSET_CALIBRATION_DATA7_LOW	OFFSET_CALIBRATION_DATA7_l register	0x25
0x00E6	OFFSET_CALIBRATION_DATA7_HIGH	OFFSET_CALIBRATION_DATA7_h register	0x26
0x00E7	OFFSET_CALIBRATION_DATA8_LOW	OFFSET_CALIBRATION_DATA8_l register	0x27
0x00E8	OFFSET_CALIBRATION_DATA8_HIGH	OFFSET_CALIBRATION_DATA8_h register	0x28
0x00E9	OFFSET_CALIBRATION_DATA9_LOW	OFFSET_CALIBRATION_DATA9_l register	0x29
0x00EA	OFFSET_CALIBRATION_DATA9_HIGH	OFFSET_CALIBRATION_DATA9_h register	0x2A
0x00EB	RESET_ASYNC_BIO_CH	RESET_ASYNC_BIO_CH REGISTER	0x2B

Table 35: AFE_BLOCK register list.

In the following are reported the configuration internal registers of HOST_IF I2C/SPI AHB. For more information and details regarding to IP functionalities and register it's need to reference the IP documentation.

Register Name	Address Offset	Reset Value	Access	Bits	Field name
PAGE_SEL	0x40	0x00	RW	7:2	Unused
				1:0	In case of external 4 pages registers expansion these bits are used to choose the page
ERROR_BYTE	0x41	0x00	RW	7:0	Byte send to the I2C/SPI master in case of error
FLUSH_FIFO	0x4B	0x00	RW	7:2	Unused
				1	Force flush of write fifo. Write 0 and then 1 to activate the function
				0	Force flush of read fifo. Write 0 and then 1 to activate the function
FIFO_STATUS	0x4C	0x00	RO	7:4	Unused
				3	Read fifo full
				2	Read fifo not empty
				1	Write fifo full
				0	Write fifo not empty
ERR_ITF	0x4E	0x00	RC	7:6	Unused
				5	Unexpected read fifo empty
				4	Unexpected write fifo full
				3	Number of bytes write operation already done, read operation is expected
				2	Read operation done without writing nr bytes before
				1	Write fifo flushed since not empty before write operation

				0	Read fifo flushed since not empty before read operation
ERR_AHB	0x4F	0x00	RC	7:5	Unused
				4	Time out reached from single write for BYTE_NR to following read
				3	Unused
				2	AHB error on read
				1	AHB error on write
				0	Unused
SAM_ADDRESS_BYTE 0	0x50	0x00	RW	7:0	First (LSB) byte of the AHB address
SAM_ADDRESS_BYTE 1	0x51	0x00	RW	7:0	Second byte of the AHB address
SAM_ADDRESS_BYTE 2	0x52	0x00	RW	7:0	Third byte of the AHB address
SAM_ADDRESS_BYTE 3	0x53	0x00	RW	7:0	Fourth (MSB) byte of the AHB address
CTRL_REG_0	0x57	0x00	RW	7:1	Unused
				0	Increment address flag. If '1' the AHB address is incremented after each access. If '0' the address is not incremented (fifo access)
CTRL_REG_1	0x58	0x00	RW	7:1	Unused
				0	I2C disable. If '1' the i2c slave is switched off. Every request from master gets all '0'
CTRL_REG_2	0x59	0x00	RW	7:1	Unused
				0	AHB access. If '1' the access is by word, by byte otherwise

CTRL_REG_3	0x5A	0x00	RW	7:0	Unused. Available for future expansion
CTRL_REG_4	0x5A	0x00	RW	7:0	Unused. Available for future expansion
CTRL_REG_5	0x5A	0x00	RW	7:0	Unused. Available for future expansion
CTRL_REG_6	0x5A	0x00	RW	7:0	Unused. Available for future expansion
CTRL_REG_7	0x5A	0x00	RW	7:0	Connected to 8 bit output o_ctrl_reg_7. They can be used during integration
IP_VERSION	0x5F	0x00	RO	7:0	Byte to specify IP version. To be used at integration time by means of IP parameter "ip_version"

Table 36: I2C/SPI internal registers.

13. Soft Reset Procedures

13.1. Main Controller Soft Reset

This procedure reset all the Main Controller FSMs and related IPs, reperforming the BOOT Phase without reset the Regmap.

[DDS_STBIO1_1114] Requirement:

To perform a SOFT Reset procedure, the user shall perform the following steps:

1. If Mode Operation has been set to 1, shall be reset to 0
2. Set the Soft Reset Main Controller Bit (SW_RESETN_REG[0]) to 0 for at least 1 ms
3. Set the Soft Reset Main Controller Bit (SW_RESETN_REG[0]) to 1
4. Wait for the DEVICE_STATUS_REGISTER to reach the IDLE Value

[END]

14. Measurement Cycle Examples

The Device Activity can be summarized by the following Flowchart:

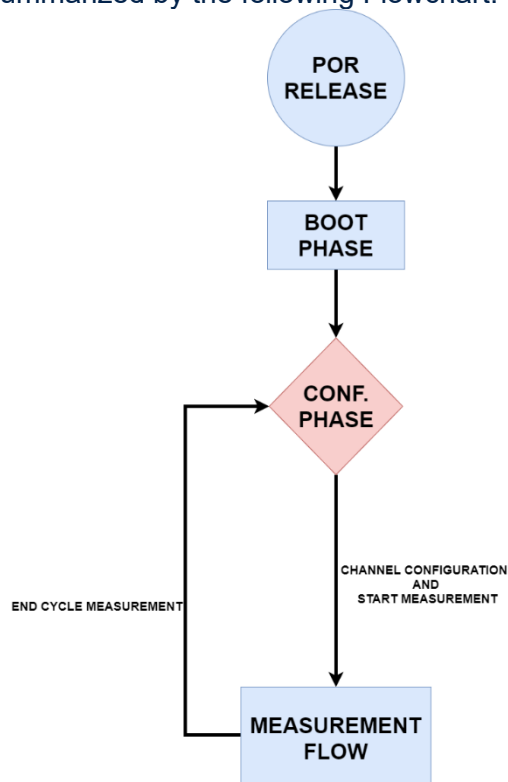


Figure 19: Device Activity Flowchart

In the next paragraphs, each phase will be explicated.

14.1. BOOT Phase

[DDS_STBIO1_0114] Requirement:

At the release of the POR signal from the analog domain, the PMU shall manage the turn-on of the LDO1V8 and starts clocks Start-Up Sequence (reported in PMU IPOS) [TO: IPOS_PMU] [END]

[DDS_STBIO1_0115] Requirement:

After At least 2 ms from the Release of the POR, the reset of the digital domain shall be de-asserted (an overlook on this part is described in the PMU IPOS) and the device shall start the Boot Phase [TO: IPOS_PMU] [END]

[DDS_STBIO1_0116] Requirement:

To check the end of the BOOT phase, the user shall read the main_controller_status_register (DEVICE STATUS REG register [2:0]) and follow these steps:

- Read the DEVICE STATUS REG [2:0]
- Check if the Value is 1 (BOOT phase still in process)
- If the value is 2, then the BOOT phase is ended and the device is in IDLE. [TO: IPOS_MAIN_CTRL] [END]

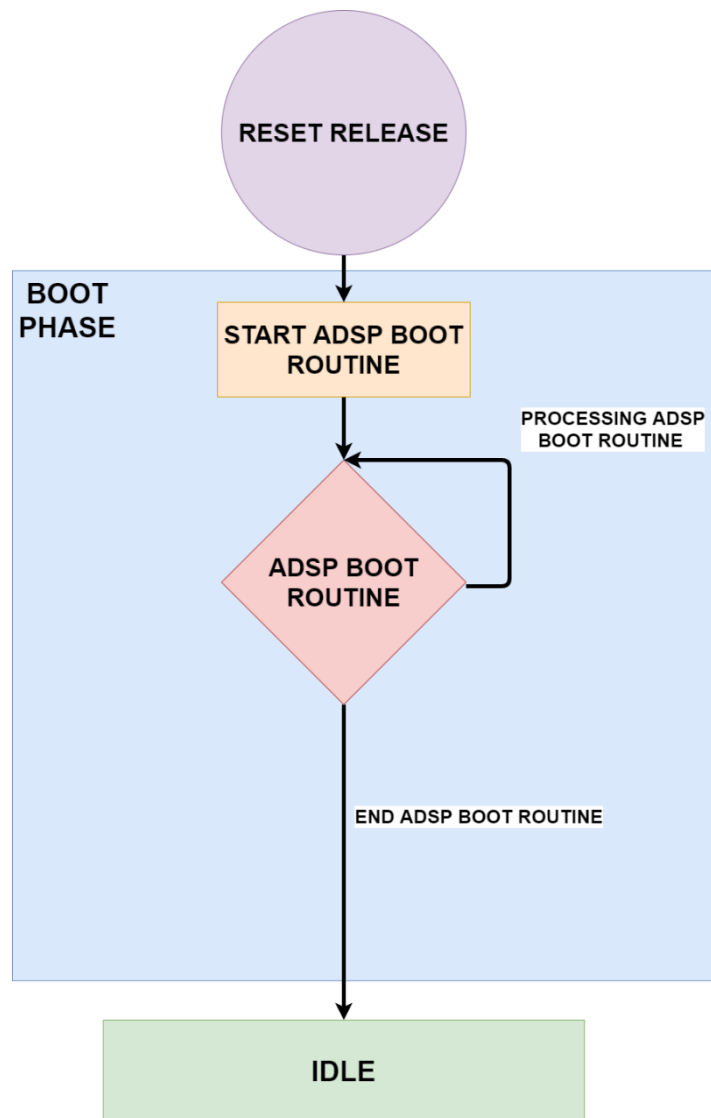


Figure 20: Boot Phase Flow

14.2. Configuration Phase

[DDS_STBIO1_0117] Requirement:

During the Configuration Phase (IDLE STATE of the Device_FSM), the user shall be able to:

- Configure Bio-Channel or Optical Channel, the ADC Calibration and then Start a measurement cycle.
- Put the device in Debug-Mode. [TO: IPOS_MAIN_CTRL] [END]

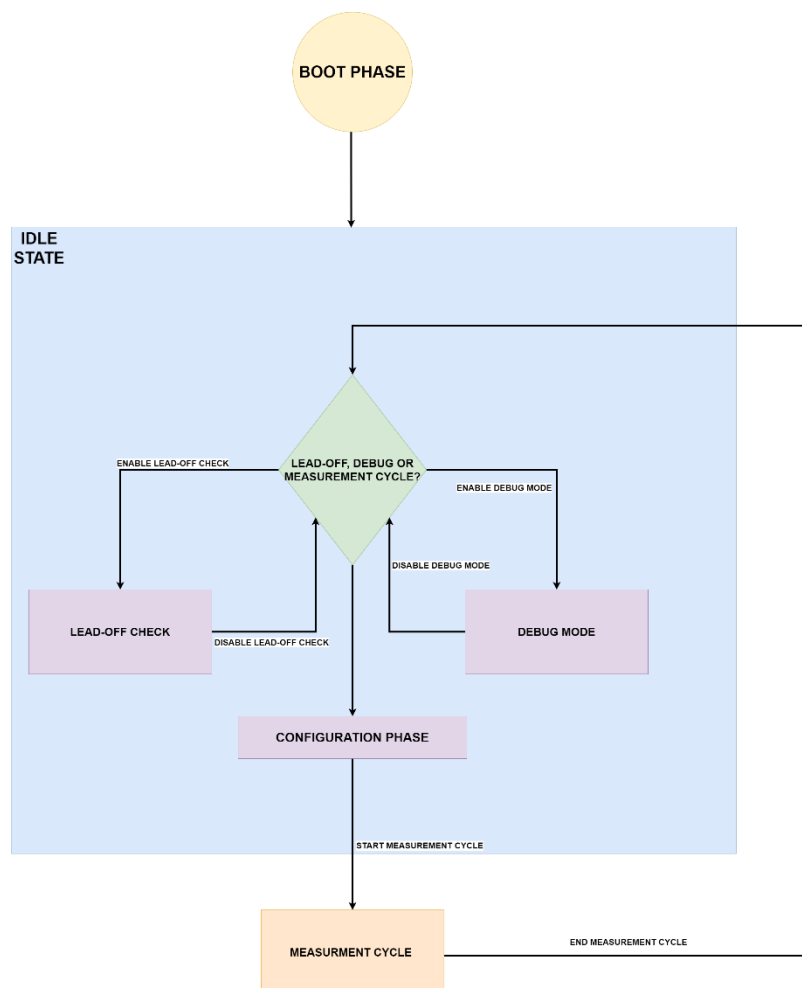


Figure 21: Configuration Phase Diagram

The start and the end of a Measurement Cycle is selected by the writing of the dedicated register as described in following examples.

14.3. Measurement Cycle

In this paragraph, different examples of Measurement Cycle will be analyzed to give the Application User an overview of the Device Workflow. All measurements will be performed in Data Storage Mode

14.3.1. ECG Only

This configuration generates each time slot, two averaged ECG data, which are ECG AC and ECG DC, respectively.

1. TIME SLOT LENGTH:
[DDS_STBIO1_0118] Requirement:
 - The user shall configure time slot duration in DEVICE_CONFIG_OPERATION register [4:2] [TO: IPOS_MAIN_CTRL] [END]
2. SELECT CHANNEL:
[DDS_STBIO1_0119] Requirement:
 - The user shall select active channel, in this case only ECG0_SEL and DS_SEL (Data Storage Mode) in DEVICE_CONFIG_CHANNEL register writing 0x21. [TO: IPOS_MAIN_CTRL] [END]
3. SELECT DIVISION INDEX FOR CONFIGURED CHANNELS:
[DDS_STBIO1_0120] Requirement:
 - The user shall select a division index for ECG, configuring M_ECG_l and M_ECG_h in SET_ECG_FREQ_l register and SET_ECG_FREQ_h register. [TO: IPOS_MAIN_CTRL] [END]
4. CONFIGURE ADC SAMPLING PERIOD:
[DDS_STBIO1_0121] Requirement:
 - The user shall set the number of clocks for sampling ECG data from ADC in ADC_CONFIG_ECG register. [TO: IPOS_MAIN_CTRL] [END]
5. GENERAL CONFIG:
[DDS_STBIO1_0122] Requirement:
 - The User shall set at 1 HC_EN in ECG_CTRL register. [TO: IPOS_MAIN_CTRL] [END]
6. SELECT OPERATIVE MODE:
[DDS_STBIO1_0123] Requirement:
 - The user shall start measurement writing register DEVICE_CONFIG_OPERATION [0] at 1. [TO: IPOS_MAIN_CTRL] [END]
7. CHECK DATA:
[DDS_STBIO1_0124] Requirement:
 - To check data in unique FIFO (default config) the user shall check the first MSB which will have to be equal to the table in DATA TAG Table. [TO: IPOS_MAIN_CTRL] [END]

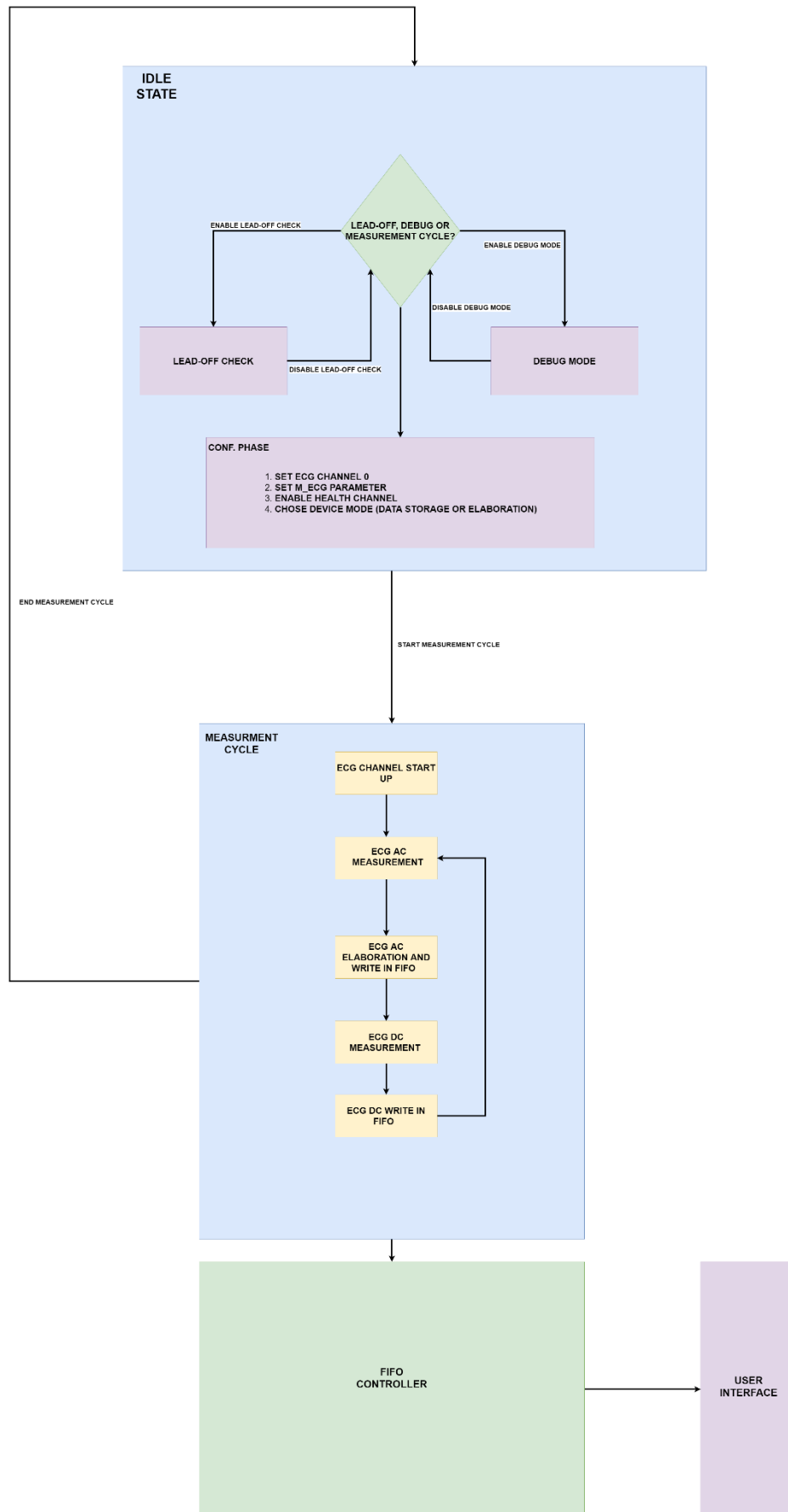


Figure 22: ECG ONLY FLOWCHART

14.3.2. ECG and BIA

This configuration generates each time slot, two averaged ECG data, which are ECG AC and ECG DC, respectively and four averaged BIA data which are BIA_AC_P, BIA_DC_P, BIA_AC_Q and BIA_DC_Q.

1. TIME SLOT LENGTH:
[DDS_STBIO1_0125] Requirement:
 - The user shall configure the time slot duration in DEVICE CONFIG OPERATION register [4:2]. [TO: IPOS_MAIN_CTRL] [END]
2. SELECT CHANNEL:
[DDS_STBIO1_0126] Requirement:
 - The user shall select active channel, in this case ECG0_SEL, BIA_SEL and DS_SEL (Data Storage Mode) in DEVICE CONFIG CHANNEL writing 0x23. [TO: IPOS_MAIN_CTRL] [END]
3. SELECT DIVISION INDEX FOR CONFIGURED CHANNELS:
[DDS_STBIO1_0127] Requirement:
 - The user shall select a division index for ECG, configuring M_ECG_l and M_ECG_h in SET ECG_FREQ_l register and SET ECG_FREQ_h register. [TO: IPOS_MAIN_CTRL] [END]
 [DDS_STBIO1_0128] Requirement:
 - The user shall select a division index for ECG, configuring M_BIA_l and M_BIA_h in SET BIA_FREQ_l register and SET BIA_FREQ_h register. [TO: IPOS_MAIN_CTRL] [END]
4. CONFIGURE ADC SAMPLING PERIOD:
[DDS_STBIO1_0129] Requirement:
 - The user shall set number of clocks for sampling ECG and BIA data from ADC in ADC CONFIG ECG register. [TO: IPOS_MAIN_CTRL] [END]
5. GENERAL CONFIG:
[DDS_STBIO1_0130] Requirement:
 - The user shall set at 1 HC_EN in ECG_CTRL register. [TO: IPOS_MAIN_CTRL] [END]
 [DDS_STBIO1_0140] Requirement:
 - The user shall set at 1 IMP_EN in IMP_CTRL register. [TO: IPOS_MAIN_CTRL] [END]
6. SELECT OPERATIVE MODE:
[DDS_STBIO1_0141] Requirement:
 - The user shall start measurement writing register DEVICE CONFIG OPERATION [0] at 1. [TO: IPOS_MAIN_CTRL] [END]
7. CHECK DATA:
[DDS_STBIO1_0142] Requirement:
 - To check data in unique FIFO (default config) the user shall check the first MSB which will have to be equal to the one described in the DATA TAG Table. [TO: IPOS_MAIN_CTRL] [END]

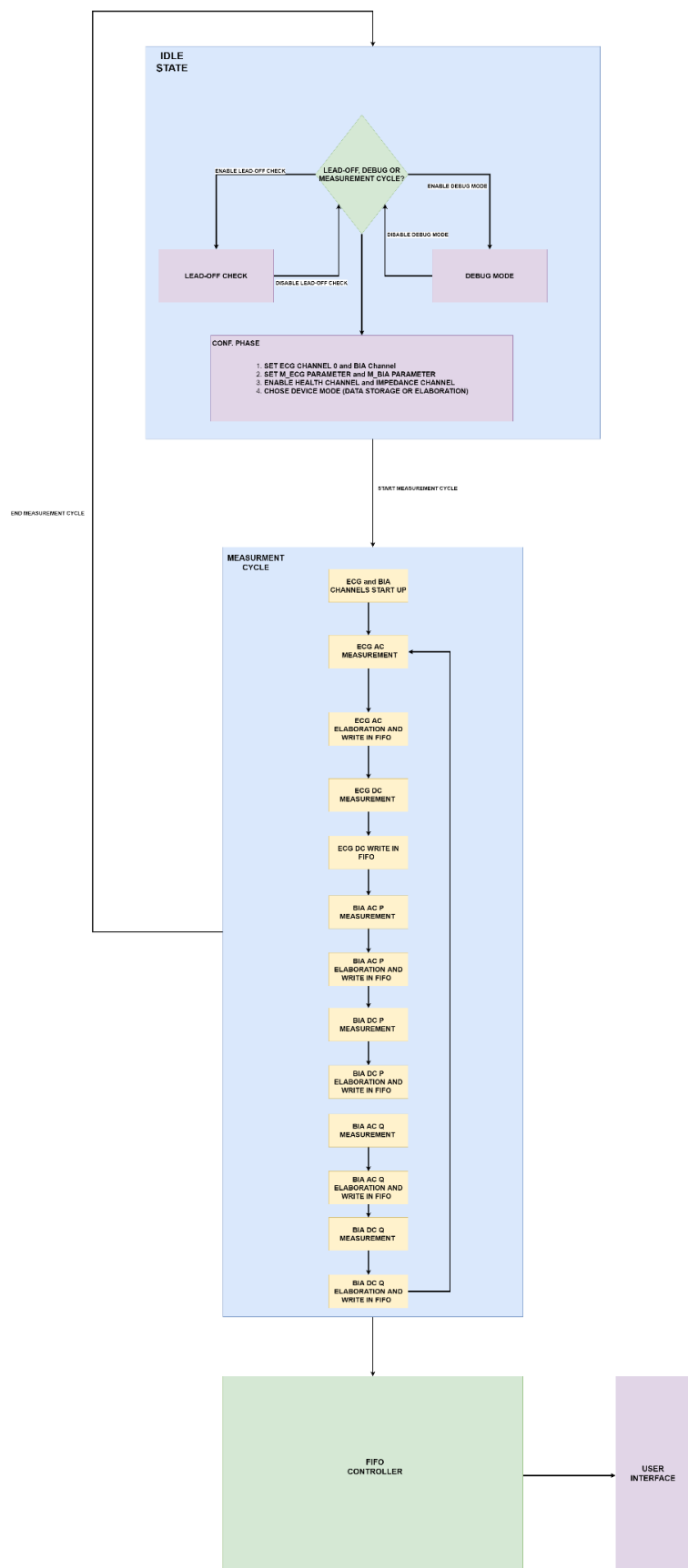


Figure 23: ECG and BIA Measurement Flowchart

14.3.3. PPG Only

This configuration generates each time slot, a number of PPG DATA according to user configuration.

1. TIME SLOT LENGTH:
[DDS_STBIO1_0143] Requirement:
 - The User shall configure time slot duration in DEVICE CONFIG OPERATION register [4:2].
2. SELECT CHANNEL:
[DDS_STBIO1_0144] Requirement:
 - The user shall select active channel, in this case only PPG_SEL and DS_SEL (data Storage) in DEVICE CONFIG CHANNEL register shall be set to 0x30.
3. SELECT DIVISION INDEX FOR CONFIGURED CHANNELS:
[DDS_STBIO1_0145] Requirement:
 - The user shall configure a division index PPG, writing M_PPG_l and M_PPG_h in SET PPG_FREQ_l register and SET PPG_FREQ_h register.
4. CONFIGURE ADC SAMPLING PERIOD:
[DDS_STBIO1_0146] Requirement:
 - The user shall configure the number of clocks for sampling PPG data from ADC in ADC CONFIG PPG register.
5. PPG FRAMES CONFIG:
[DDS_STBIO1_0147] Requirement:
The user shall configure each frame following these steps:
 - 5.1 ALC PARAMETERS:
 - T1 in PPG_ALC_CONFIG_PARAM_1 register.
 - T2, GAIN_START_SET, GAIN_MAX_SET in PPG_ALC_CONFIG_PARAM_2 register.
 - 5.2 DIGITAL RAMP PARAMETERS:
 - N_MAX, T_DELTA in DIGITAL_RAMP_CONFIG_PARAM_1 register.
 - DELTA, T_DELTA1 in DIGITAL_RAMP_CONFIG_PARAM_2 register.
 - DELTA1 in DIGITAL_RAMP_CONFIG_PARAM_3 register.
 - 5.3 GENERAL PPG PARAMETERS:
 - N_PPG_FRAME, WAVE_VELOCITY_MODE, SEL_PPG_ELAB in GENERAL_PPG_PARAMETERS_1 register.
 - T_TX_STARTUP in GENERAL_PPG_PARAMETERS_2 register.
 - T_RX_SET, PPG_START_UP_TIME_l in GENERAL_PPG_PARAMETERS_3 register.
 - PPG_START_UP_TIME_h in GENERAL_PPG_PARAMETERS_4 register.
 - 5.4 PARAMETERS FOR EACH FRAME CONFIGURED BY USER:
 - RX_CH_FRAME_{n}, FRAME_{n}_SAMPLES, FRAME_{n}_REPETITION in FRAME_n_PARAMETERS_1 register.
 - BTIA_GAIN_FRAME_{n}, TX_LEDSET_FRAME_l_{n} in FRAME_n_PARAMETERS_2 register.
 - TX_LEDSET_FRAME_h_{n}, TX_LEDOFF_FRAME_l_{n} in FRAME_n_PARAMETERS_3 register.
 - TX_LEDOFF_FRAME_h_{n} in FRAME_n_PARAMETERS_4 register.
 - IDAC_DC_PPG_FRAME_{n} in FRAME_n_PARAMETERS_5 register.
 - EN_TX_i in FRAME_n_PARAMETERS_6 register.

- CURR_SEL_TX_FRAME{n} in FRAME_n_PARAMETERS_7 register.
- TX1_OUTAB_SEL_FRAME_{n}, TX2_OUTAB_SEL_FRAME_{n}, TX3_OUTAB_SEL_FRAME_{n}, TX4_OUTAB_SEL_FRAME_{n} in FRAME_n_PARAMETERS_10 register.

6. SELECT OPERATIVE MODE:

[DDS_STBIO1_0148] Requirement:

- The user shall write DEVICE_CONFIG_OPERATION register [0] at 1.

7. CHECK DATA:

[DDS_STBIO1_0149] Requirement:

- To check data in unique FIFO (default config) the user shall check the first MSB. PPG DATA has 20 bits of data and 4 of TAG. The 4 bits, from 0 to 11, are used to identify the frame (as described in the table in DATA TAG table).

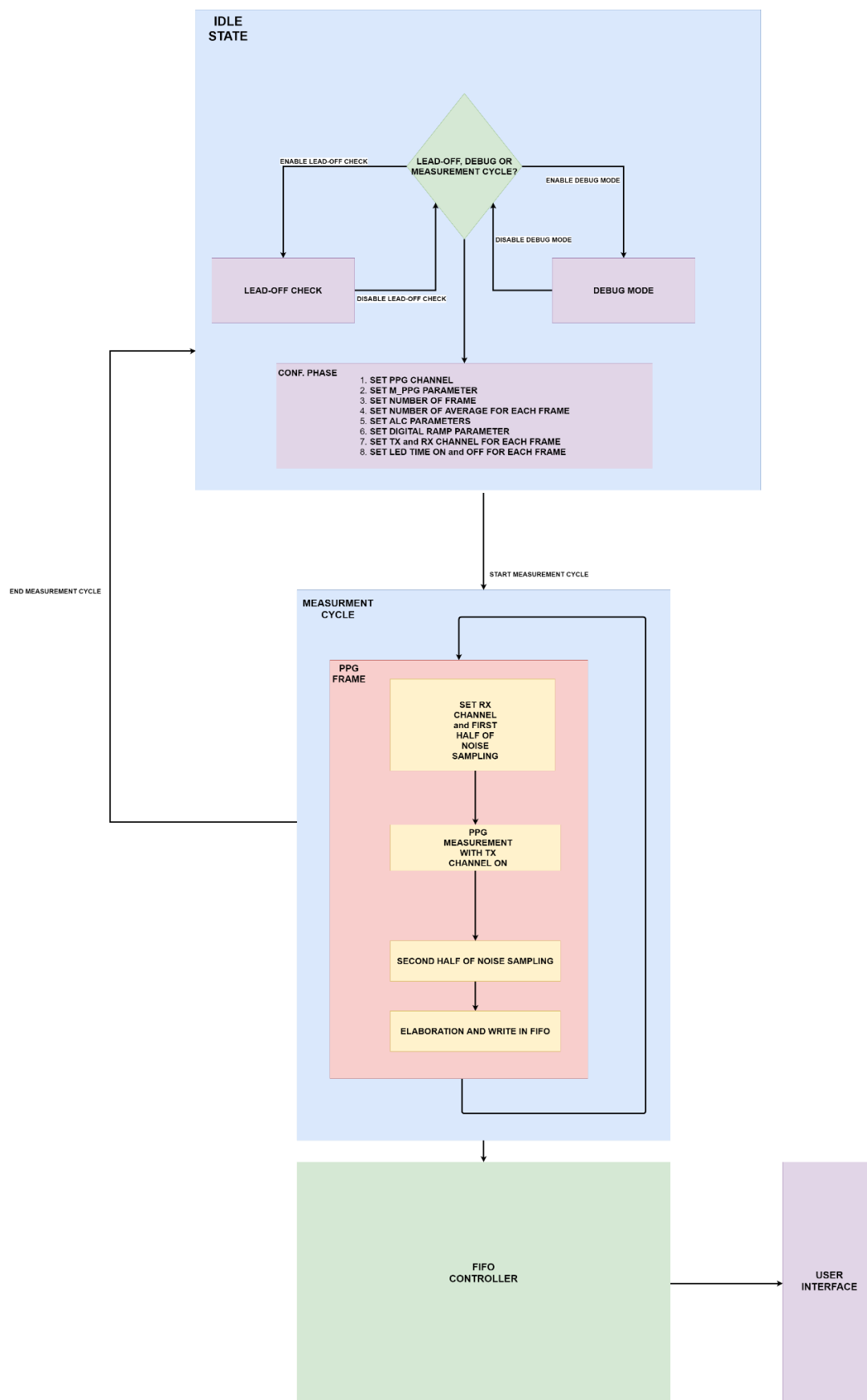


Figure 24: PPG Measurement Flowchart

16. BIST

16.1. Reference documents

Current Version	Document	Author	Change Description
1.0-17.00	/prj/ams_DELIVERIES/H9A_STBIO1/memorycuts/H9A_STBIO1- -GenericMBIST_CONFIG--BIST_STBIO1/1.0-17.00/ /doc/ref_manual/generic_mbist_um.pdf	STMicroelectronics TRnD	User manual, release notes and KPS for the GenericMBIST
1.0-17.00	/prj/ams_DELIVERIES/H9A_STBIO1/memorycuts/H9A_STBIO1- -GenericMBIST_CONFIG--BIST_STBIO1/1.0-17.00/ /doc/user_manual/generic_mbist_rm.pdf	STMicroelectronics TRnD	Reference manual for the GenericMBIST

16.2. Memory BIST Architecture

The BIST solution consists of memory-wrapper “collar” and common master “controller” and an IEEE 1149.1 compliant Test Access Port (TAP).

16.2.1. MBIST Collar

The BIST collar is a wrapper which is placed closest to memory. It executes the instructions received from controller in the form of march elements to test the memory. It runs at the speed of the memory under test.

After testing it reports the execution status to the controller and manages all the BIST status, diagnosis, repair information and debug data. The collar also provides support for testing multiple memories in parallel or in sequential order.

16.2.2. MBIST Controller

The controller acts as an interface between the collars and the user. All user BIST control is done through the controller. The controller controls various testing and reporting activities of the memory test. It allows various mode selections and programs the pre-stored synthesized algorithms instructions into collars attached to it.

Main functionalities of controller are:

- Mode management (RunBIST mode, Retention mode etc.)
- Program the configurable collar with default algorithms, march element by march element and initiate the activities on memory through collars by launching the march run.
- Compute the Global status based on the individual status from the collar.

TAP

- It includes a TAP controller, a 7-bit instruction register, three test data registers, and the bypass register.
- This responds to the control sequences supplied through the test access port (TAP) and generates the control signals required for BIST operation

16.2.3. Top Level Approach

Figure 66: Top Level Controller Configuration

In this configuration, there is only one controller for the entire chip. All associated collars are spread into the design (at top level and within subblocks)

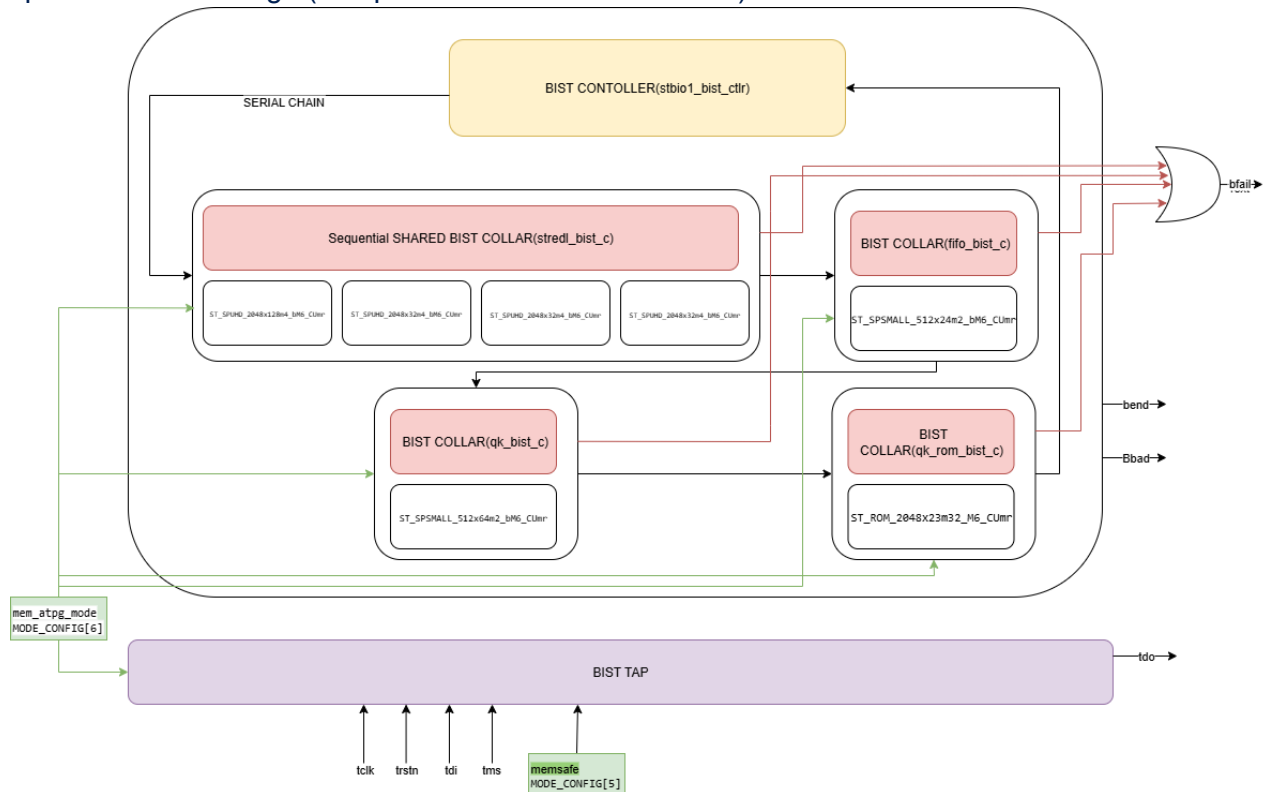


Figure 25 Top Level Controller Configuration

OUTPUT

- Bend: Indicates that all the activities of the test run of all Memories in the current BIST mode are complete
- Bfail: Indicates bfail for a specific memory. There are as many bfail signals as the number of memories. Bfail also goes high at the first cycle of each march element run. This pulse should be used as a reference in clk_m domain to calculate the number of cycles after which bfail went high in case of a failure in a specific march sequence run.
- Tdo: TAP serial output Updated on falling edge of tck

INPUT FROM PAD

- Tck: TAP clock
- Tdi: TAP serial input Sampled on rising edge of tck
- Tms: TAP test mode select Sampled on rising edge of tck
- Trstn: TAP reset

INPUT FROM REGISTER

- Memsafe (ADDRESS=0x3E, BIT=0x05): to globally deactivate the memory by making CSN high internally
- Mem_atpg_mode (ADDRESS=0x3E, BIT=0x06): Used to constraint few memory inputs and improve controllability and observability.

ANALOG PADS:

- sense internal V1V2 (GPIO1)
- force = V1V2 SUPPLY (GPIO2)

17. Digital DFT

17.1. Scan mode

[DDS_STBIO1_0699] Requirement:

After the initial boot phase, only with SPI interface, to turn on the Scan mode shall be:

- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x03)
- Enable LDO 1V8 by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x02)
- Enable clock by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x12)
- Enable debug mode by writing to regmap (ADDRESS=0x3E, DATA=0x01)

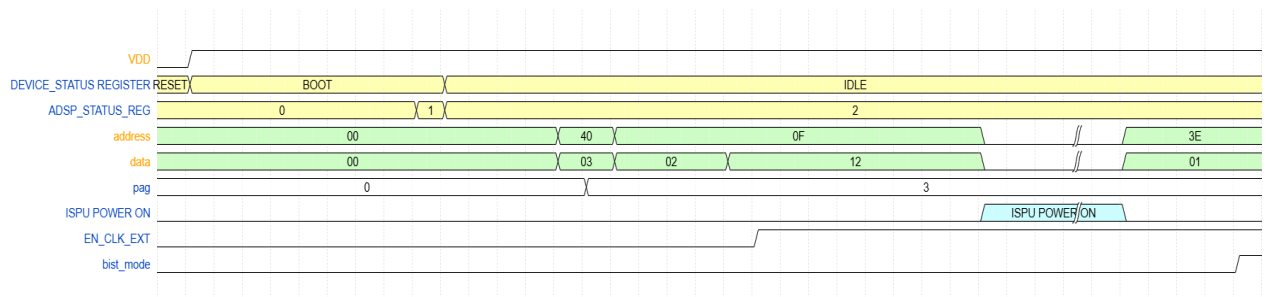


Figure 26 scan mode activation timing

[End]

17.2. Debug mode

[DDS_STBIO1_0700] Requirement:

After the initial boot phase, to turn on the Debug mode shall be:

- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x03)
- Enable LDO 1V8 by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x02)
- Enable clock by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x12)
- Enable debug mode by writing to regmap (ADDRESS=0x3E, DATA=0x08)

[End]

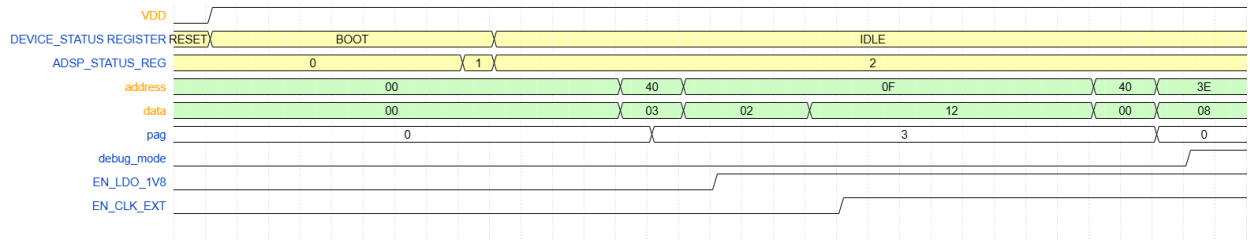


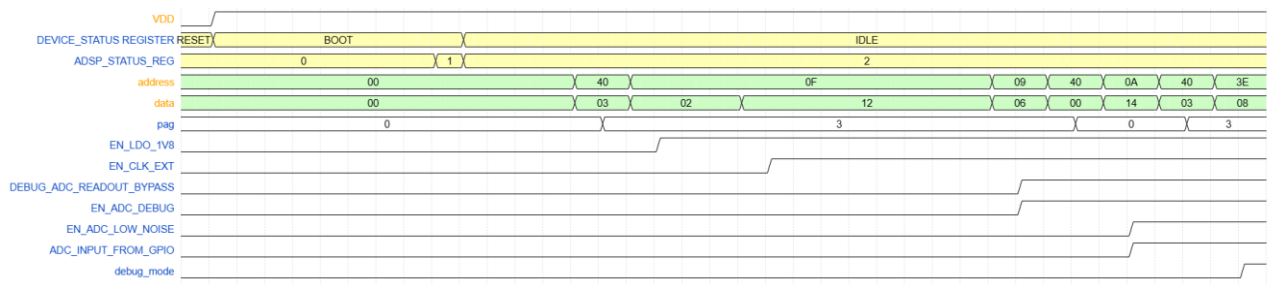
Figure 27 debug mode activation timing

17.3. ADC TEST LOW NOISE

[DDS_STBIO1_0701] Requirement:

To turn on the ADC TEST LOW NOISE mode shall be:

- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x03)
- Enable LDO 1V8 by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0xF, DATA=0x02)
- Enable clock by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x12)
- Enable ADC in debug mode (ADDRESS=0x09, DATA=0x06)
- Select page 2 of the regmap (ADDRESS=0x40, DATA=0x02)
- Enable ADC TEST LOW NOISE and input of ADC from GPIO (ADDRESS=0x0A, DATA=0x14)
- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x02)
- Enable debug mode by writing to regmap (ADDRESS=0x3E, DATA=0x08)



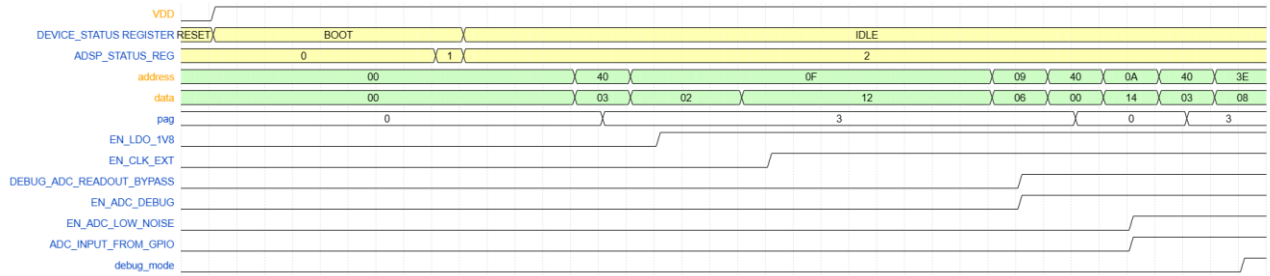
[End]

[DDS_STBIO1_0702] Requirement:

To turn on the ADC TEST LOW NOISE mode shall be:

- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x03)
- Enable LDO 1V8 by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0xF, DATA=0x02)
- Enable clock by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x12)
- Enable ADC in debug mode (ADDRESS=0x09, DATA=0x06)
- Select page 2 of the regmap (ADDRESS=0x40, DATA=0x02)
- Enable ADC TEST LOW NOISE and input of ADC from GPIO (ADDRESS=0x0A, DATA=0x14)
- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x02)

- Enable debug mode by writing to regmap (ADDRESS=0x3E, DATA=0x08)



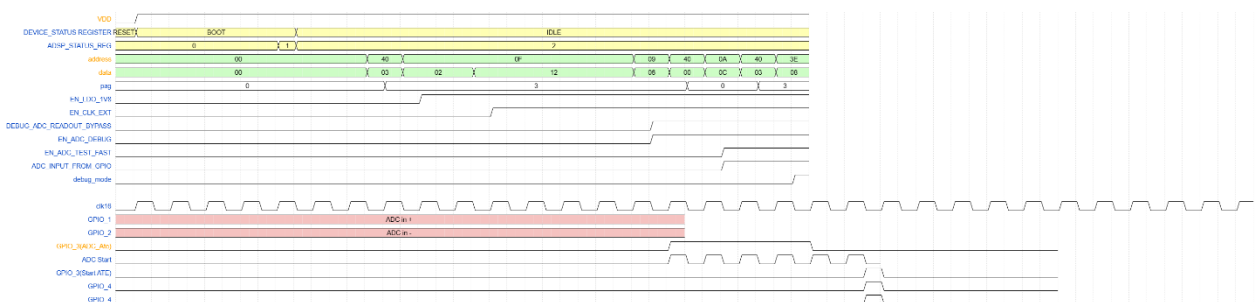
[End]

17.4. ADC TEST FAST

[DDS_STBIO1_0507] Requirement:

To turn on the ADC TEST LOW NOISE mode shall be:

- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x03)
- Enable LDO 1V8 by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0xF, DATA=0x02)
- Enable clock by writing to regmap **VIREF_LDO1V8_CLK16M register** (ADDRESS=0x0F, DATA=0x12)
- Enable ADC in debug mode (ADDRESS=0x09, DATA=0x06)
- Select page 2 of the regmap (ADDRESS=0x40, DATA=0x02)
- Enable ADC TEST LOW NOISE and input of ADC from GPIO (ADDRESS=0x0A, DATA=0x0C)
- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x02)
- Enable debug mode by writing to regmap (ADDRESS=0x3E, DATA=0x08)



[End]

17.5. BIST MODE

[DDS_STBIO1_0507] Requirement:

After the initial boot phase, to turn on the Debug mode shall be:

- Select page 3 of the regmap (ADDRESS=0x40, DATA=0x03)

- Enable LDO 1V8 by writing to regmap **VIREF_LDO1V8_CLK16M** register (ADDRESS=0x0F, DATA=0x02)
- Enable clock by writing to regmap **VIREF_LDO1V8_CLK16M** register (ADDRESS=0x0F, DATA=0x12)
- Enable the power island [\[DDS_STBIO1_0507\]](#)
- Enable bist mode by writing to regmap (ADDRESS=0x3E, DATA=0x04)

[End]

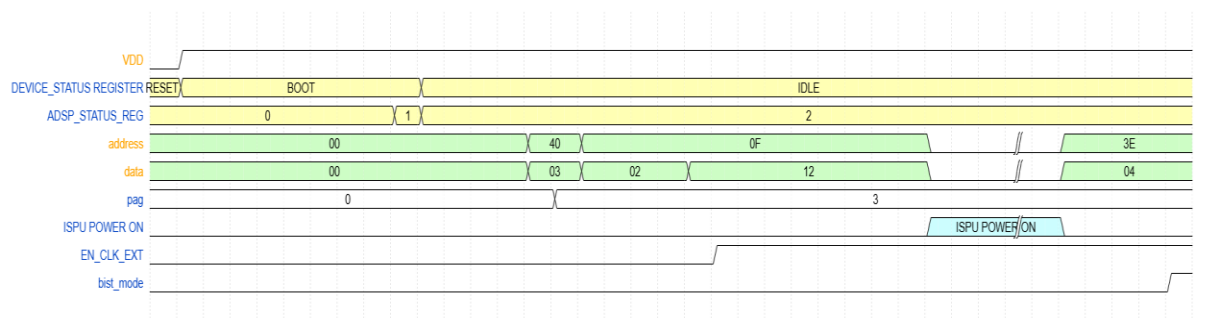


Figure 28 bist mode activation timing